Gross External Damages of Water Pollution in the United States

David A. Keiser*[1], Daniel J. Phaneuf[2], Catherine L. Kling[3], Yongjie Ji[4], Tihitina Andarge[5], Mohammad Mainul Hoque[3], Nilesh Shinde[6], and Yau-Huo (Jimmy) Shr[7]

[1]UMass Amherst and NBER; [2]University of Wisconsin–Madison; [3]Cornell University; [4]Iowa State University; [5]UMass Amherst; [6]Indiana University; [7]National Taiwan University

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Abstract

This paper provides the first estimates of the Gross External Damages (GED) of water pollution in the U.S. We focus on nutrient pollution, which is responsible for most surface water impairments. To measure marginal damages, or implicit prices, we model the transport of pollution through 2.5 million river and lake segments. Marginal damages are largest when emissions occur near population centers and in headwater streams. A significant portion of damages are exported downstream. Though agriculture is responsible for two-thirds of nutrient pollution, municipal wastewater treatment contributes more to GED. Total measured damages range from $35B to $100B per year.

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1 Introduction

Gross Domestic Product is one of the most common measures of economic activity. Its acronym, GDP, features prominently in mainstream media and is memorized by scores of introductory economics students each year. It serves as one of the most important indicators for setting domestic and global policies. Yet, important shortcomings exist. One widely known limitation is that GDP performs poorly in measuring (or omits entirely) the value of essential environmental goods and services. It also fails to account for the external damages to those goods and service from market-based economic activity.

The concept of augmenting national economic accounts to include measures of the environment is not new. At least fifty years ago, Nordhaus and Tobin (1973) illustrated how one could adjust common measures of economic growth for important non-market goods and services. For decades since, economists have proposed approaches to incorporate "nature's numbers" into estimates of economic activity (NRC 1999). Implementation of these efforts has begun in earnest on the international stage. For example, the World Bank estimates that the value of renewable natural capital per capita has fallen by 20 percent since 1995 (The World Bank 2024). Commitments by the U.S. government to estimate these values have significantly lagged behind those of other countries, although the Biden White House reignited U.S.-based efforts by directing a fifteen year, multi-agency plan to finally "put nature on the nation's balance sheet" (The White House 2023; Bagstad et al. 2021). In the absence of governmental accounts, some researchers have produced national-level statistics of air and climate pollution damages in the U.S. (Muller et al. 2011; Tschofen et al. 2019; Cropper and Park 2025). A critical roadblock has been the absence of other estimates of nature's "prices" (Arrow et al. 2012; Fenichel et al. 2016). This paper provides the first estimates of the marginal damages (i.e., implicit prices) of U.S. water pollution and uses them to estimate Gross External Damages (GED), the GDP-equivalent measure of the damages from water pollution.

A central innovation of this paper is a new national integrated assessment model (IAM) of water pollution. This model, which we term HydroEcon, links sources of pollution to physical and economic damages throughout the nation. To develop this model, we first propose a framework for constructing the social cost of water pollution while incorporating the spatial variability of damages. This first step guides our interpretation of how to aggregate damages across space and uses of water resources. Like air, but unlike carbon, the damages from water pollution depend critically on the location of emissions (Keiser and Muller 2017).

In moving from theory to measurement, we focus on nutrient pollution, as it "is one of America's most widespread, costly, and challenging environmental problems" (USEPA 2024). Two recent assessments conducted by the U.S. Environmental Protection Agency (USEPA) found that 50% of the nation's lakes and over 40% of the nation's rivers and streams were in poor condition due, in part, to elevated nutrient levels (USEPA 2018, 2022a). Nutrients (nitrogen and phosphorus) were the most widespread stressor identified. Agricultural and urban run-off, as well as municipal wastewater treatment plants are leading contributors. The effects of excess nutrients are marked by thick, odorous, and potentially harmful algal mats, as well as decreased clarity and ecological integrity.

To construct our IAM, we model the flow and fate of water pollution throughout the conterminous U.S. using the National Hydrography Dataset Plus Version 2 (NHDPlusV2) and the U.S. Geological Survey's (USGS) SPAtially-Referenced Regression On Watershed attributes (SPARROW) model. The NHDPlusV2 provides a comprehensive mapping of rivers, streams, lakes, and other waterbodies in the U.S.[1] Our model uses approximately 2.5 million river and stream segments and over 350,000 lakes that we observe from this database. We then link these water features with SPARROW to model baseline and simulated changes in nutrient pollution levels. SPARROW is a state-of-the-art hydrologic model that the USGS has refined over two and a half decades (Smith et al. 1997). It provides highly granular information on long-run average concentrations of nutrient pollution, its flow and fate, and sources of pollution. Throughout the paper, we often summarize data at the watershed level. We define a watershed in this paper as an 8-digit hydrologic unit code (HUC8). There are approximately 2,100 8-digit HUCs in the U.S., or roughly two-thirds of the number of U.S. counties.

Next, we link our hydrologic system and pollution to people. We focus on five main areas of impact: water-based recreation, property values, drinking water treatment, human health, and climate change. Prior literature has focused on recreation, housing, and drinking water treatment as the most likely human uses affected by water pollution in the U.S. (Freeman 1982; Dodds et al. 2009; Keeler et al. 2012; Keiser 2019). An emerging literature suggests that nutrient pollution also has important impacts on human health and climate (Jones 2019; Downing et al. 2021). To measure human health impacts in particular, we focus on the incidence and costs of low birthweight. Though nutrients can impact health in several ways, including through the increased risk of cancer and birth defects, we focus on low

[1]This mapping effort has been supported by the U.S. Environmental Protection Agency and USGS since the early 2000s and serves as the leading source for information on hydrologic networks for economic analyses (e.g., Keiser and Shapiro (2019a)).

birthweight given recent advances that allow us to link nutrients to harmful algal toxins and ultimately to low birthweight. There is also a corresponding economics literature that allows us to monetize this outcome.

To estimate localized damages, our model utilizes county-level recreation data from the U.S. Department of Agriculture's (USDA) National Survey on Recreation and the Environment, the count of housing units and assessed values from Core Logic's parcel-level data file, drinking water intake and service location data from USEPA, and county-level birth data from the University of Wisconsin Population Health Institute's County Health Rankings & Roadmaps (CHR&R) program, which relies on data from the National Center for Health Statistics. To estimate climate damages, we rely on water resource features that are important for determining climate impacts (i.e., lake size and pollution concentrations). We then develop valuation functions for each of the five main categories. These functions provide estimates of the economic damages that arise from changes in pollution concentrations. We estimate some of these empirical relationships within the paper and synthesize decades of research to establish others.

We proceed by simulating how baseline surface water quality concentrations change with a marginal change in pollution emissions at all of our 2.5 million river segments.[2] We conduct this simulation one river segment at a time. We track how marginal changes in loads impact local and downstream water quality. We then link these changes in water quality to changes in economic behavior and value. Using this procedure, we recover marginal damages, or implicit prices, of pollution for each watershed. We combine this information with data on loads by nutrient pollution sources to estimate GED. This step allows us to assign marginal damages and the respective GED contributions from agricultural, urban, and municipal sources. While we estimate separate damages for phosphorus and nitrogen pollution depending on the use category that they likely impact, we report most of our statistics as marginal damages of nutrient pollution. These nutrients are highly correlated, and it is often difficult to tease out their separate impacts on aquatic systems and economic uses.

We highlight five key sets of results. First, we provide new insights on how water pollution, and nutrient pollution in particular, affects key uses of water resources. IAMs for water pollution have been especially difficult to develop given the complex links between water quality and the array of uses it impacts (Keeler et al. 2016). Our work includes new analyses and synthesizes a large body of research to establish these connections. We perform a new

[2] As we explain in Section 3, we restrict this procedure to the 2.5 million river network segments that fall entirely within a HUC8 to simplify estimates of watershed-specific marginal damages.

national-scale recreation demand analysis and show that the results of this exercise, along with the study it builds on (Keiser 2019), aligns closely with published estimates of the elasticity of recreation with respect to water quality from localized studies. We also show how one can derive recreation trip elasticities from random utility models, a common modeling framework in the recreation demand literature. Further, we find a robust consistency in the estimates of the value of water quality with respect to home values and establish the link between water clarity and nutrients needed to connect these impacts to pollution sources. While the extant literature is less developed, our work also opens the door to incorporating additional human health impacts into national scale estimates of the damages of nutrients.

Second, our results suggest meaningful impacts of nutrient pollution across the U.S. Social costs are higher near population centers. In this regard, damages from water pollution follow similar patterns as other local pollutants, such as air pollution. This finding is not entirely surprising, given that many local pollutants scale with population. For instance, average marginal damages in the Northeast and West are highest with values of $539 and $532 per kg of nutrient pollution respectively.[3] In the South and Midwest, average marginal damages are significantly lower ($131/kg and $70/kg respectively) than those in other areas of the country, reflecting to some extent lower population numbers. GED reflects both prices and quantity. Though marginal damages in the Northeast are significantly higher than in the Midwest, GED is similar in these two regions. This finding is due to lower levels of emissions in the Northeast and much higher levels of emissions in the Midwest.

Third, our work provides new insights on the unique nature of water pollution. An important aspect of nutrient pollution is that it flows downstream over large distances. We find that localized damages account for approximately 83% of damages, whereas the remaining 17% arises from downstream impacts.[4] We also find that approximately 20% of damages are exported to downstream states, highlighting the importance of cross-boundary pollution. We find other notable spatial patterns in damages. Twenty percent of watersheds in the U.S. are responsible for approximately 80% of GED. These findings highlight particular locations where abatement efforts may have the highest societal returns. We also find that marginal damages are highest when pollution is emitted in smaller, headwater streams.

Fourth, we find that water-based recreation reflects the largest share of economic damages. Marginal damages for recreation are an order of magnitude larger ($149.4/kg) than those of other categories, including housing ($8.0/kg), drinking water treatment ($0.03/kg),

[3]Unless noted, all dollar values reported are adjusted to 2022 U.S.$ using the U.S. Bureau of Labor Statistics Consumer Price Index (CPI) Inflation Calculator (USBLS 2025).

[4]We refer to localized damages as those that occur within the same watershed as the source of pollution.

human health ($1.5/kg), and climate change ($11.7/kg).[5] Prior work has suggested a large role for water-based recreation in overall water pollution damages. We do, however, find that health and climate impacts are also important channels of damages and should serve as areas of future research. There is more uncertainty regarding the extent and magnitude of these impacts than the traditional categories of recreation and housing. For example, our baseline damage estimates conservatively assume that low birthweight impacts from algal toxins accrue within 1 mile of a lake. Current research does not provide clear guidance as to how far these effects extend and through what specific channel they accrue (i.e., through drinking water, aerosolization, or direct surface water contact). Extending the health impacts, even out to 10 miles, significantly increases damage estimates for this category. There are also other health impacts that we currently do not monetize and their inclusion could have a meaningful impact on damage estimates.

Finally, we highlight a surprising result with respect to the damages from agriculture. Non-point sources of nutrient pollution represent the overwhelming share of total pollution loads in U.S. waterways. In our hydrologic model, they represent approximately two-thirds of the major anthropogenic sources of pollution.[6] Agriculture is the largest contributor. This statistic is not new nor controversial, as non-point sources of pollution, and agriculture in particular, are widely known as the leading sources of nutrient pollution today. However, our analysis highlights the importance of considering the economic value of this pollution. We find that non-point sources contribute 50% of GED. Agriculture's contribution to damages (38%), while significant, is substantially less than its contribution to overall pollution levels (65%). Point sources, primarily wastewater treatment facilities, represent approximately 21% of pollution loads, but account for about half of GED (47%) due to their close proximity to people.

This study contributes to several strands of literature. First, we add to a growing literature related to environmental statistics and natural capital accounting. Our work most closely resembles research efforts that have used IAMs to measure the GED of climate and

[5]Two-thirds of the U.S. population participates in water-based recreation (Keiser 2019), which is one reason why damages from this category are large. However, we estimate that only around 740 drinking water treatment plants sourcing from surface water face potential nutrient pollution issues and less than 5% of single family homes are within $\frac{1}{2}$ km of a lake. As a global pollution issue, climate damages are not dependent on local population characteristics, so they are our only category of damages that do not scale with local population numbers.

[6]For this and related statistics, we refer to emissions and damages relative to the three major anthropogenic categories of nutrient pollution that we observe (Agriculture, Urban Non-Point Sources, and Municipal Wastewater Treatment) and a small number of other anthropogenic sources.

local air pollution (Muller and Mendelsohn 2007, 2009; Muller et al. 2011; Tschofen et al. 2019).[7] Cropper and Park (2025) describe these advances, while noting that the development of similar metrics for water pollution damages has lagged behind that of air pollution. We believe our study provides the first national measures of GED for water pollution for any country. Other related work has focused on measuring the value of natural capital and ecosystem services for inclusion in national accounting frameworks (Fenichel and Abbott 2014; Fenichel et al. 2016; Ouyang et al. 2020; Fenichel 2024), which nests into a much larger literature that has sought to develop more inclusive measures of wealth to supplement national income statistics (Arrow et al. 2012; Polasky et al. 2015; The World Bank 2024). The foundation for this work has a long history. Muller et al. (2025) provide an overview and note that Irving Fisher wrote over 100 years ago about the importance of a more comprehensive definition of wealth, citing examples of the value of fisheries, public parks, and freedom. From their first widespread use in the early 20th century, many have noted the limitations of national income statistics as a measure of well-being (Abraham and Mackie 2006; Rockoff 2020).

Second, we build on work focused on the economics of water resources. Keiser and Shapiro (2019b) show that economic research on water has lagged research on air pollution and climate change. This finding stands in contrast to opinion polls that suggest that water pollution has been the leading environmental concern among Americans for several decades and public and private spending on its control has exceeded that of air pollution (Keiser and Shapiro 2019b). Recent work on water resources has examined both water quality and quantity issues ranging from the value of wetlands for flood control (Taylor and Druckenmiller 2022), to the impact and regulatory scope of the Clean Water Act (Keiser and Shapiro 2019a; Keiser et al. 2021; Greenhill et al. 2024), water markets (Ayres et al. 2021; Rafey 2023), and water quality regulations in developing countries (Greenstone and Hanna 2014; Kahn et al. 2015; He et al. 2020), among others. A separate literature explores drinking water (Bennear et al. 2009; Currie et al. 2013; Christensen et al. 2023; Keiser et al. 2023) and there is a long-standing interest in water, sanitation, and hygiene (WASH) programs in developing countries (Kremer et al. 2011; Guiteras et al. 2015).

Our work on water contributes most directly to research that seeks to measure the economic damages of surface water pollution. Most revealed preference studies focus on water-based recreation and housing market impacts and most studies are local or regional in scale (Bockstael et al. 1987; Leggett and Bockstael 2000; Phaneuf et al. 2000; Phaneuf et al. 2008;

[7] A related, but distinct, literature has directly estimated the damages from climate and air pollution for use in IAMs (e.g., Carleton et al. (2022)).

Egan et al. 2009; Olmstead 2010; Walsh et al. 2011; Guignet et al. 2022; Kuwayama et al. 2022). A small, but growing number of papers examine benefits of water quality improvements using national-scale data (Keiser 2019; Keiser and Shapiro 2019a; Mamun et al. 2023). We ground our model in a micro-foundation that allows us to utilize estimates from these and other individual revealed preference studies.

An alternative stated preference literature has provided complementary estimates of the value of water quality (Desvousges et al. 1987; Van Houtven et al. 2007; Bateman et al. 2023). These estimates have served as the foundation of USEPA-led IAMs to measure the benefits of federal statutes and regulatory actions (Carson and Mitchell 1993; Bingham et al. 2000; Griffiths et al. 2012; USEPA 2015). Related work has sought to measure the national damages from soil erosion and nutrient pollution, and several studies have been motivated by the USDA's need to measure the benefits of national conservation programs (Clark 1985; Ribaudo 1986; Hansen and Ribaudo 2008; Dodds et al. 2009).

Our study also informs a large literature that has specifically focused on the economics of nutrient pollution (Rabotyagov et al. 2014; Cohen and Keiser 2017; Garnache et al. 2016; Paudel and Crago 2021; Liu et al. 2023; Skidmore et al. 2023; Raff and Meyer 2022; Del Rossi et al. 2023; Meyer and Raff 2024). In addition to the economic valuation literature discussed above, this work identifies cost-effective interventions and measures the impacts of these interventions on water quality. Many, though not all, of these studies focus on agriculture. The damage estimates from our model will allow researchers to examine the efficiency, in addition to the cost-effectiveness, of nutrient policies.

A final area of literature highlights the importance of economic geography and the spatial nature of pollution. The textbook description of efficient pollution policies requires setting marginal damages equal to marginal abatement costs (Baumol and Oates 1988; Phaneuf and Requate 2016). Implementing this basic concept at a granular level across a large landscape (i.e., a country) has been challenging. Until recently, there have been few estimates of marginal damages of local pollutants across a large spatial extent (Muller and Mendelsohn 2007, 2009; Hsiang et al. 2019). Local estimates of air pollution damages have paved the way for a host of impactful insights ranging from the distributional consequences of environmental policy (Fowlie and Muller 2019) to the benefits of renewable energy (Cullen 2013; Fell et al. 2021), electric vehicle adoption (Holland et al. 2016) and the returns to energy efficiency programs (Fowlie et al. 2018), as well as the role of economic geography more broadly (Hollingsworth et al. 2024). We are unaware of nationwide, local estimates of marginal damages of water pollution, though Keeler et al. (2016) estimate the social costs of nitrogen damages in the state of Minnesota and Farrow et al. (2005) and Doyle et al. (2014)

demonstrate how marginal damages can inform water quality trading programs. This lack of damage estimates prevents similar analyses of water-related policies, as well as the ability to explore multi-media impacts (Keiser and Muller 2017). This paper fills this gap.

The paper proceeds as follows. We present our conceptual model of economic damages in Section 2. We then highlight important components of our hydrologic model (Section 3) and describe the damage functions for individual uses of water resources (Section 4). Section 5 summarizes estimates of spatially explicit marginal damages and GED. Section 6 concludes. A series of appendices describes our model components in more detail.

2 Conceptual Model

2.1 Conceptual Model of Damages

In our framework, damages from nutrient pollution arise locally from three types of behavioral interactions with surface water quality. These include amenities, recreation, and drinking water services. There is also an infant health externality. Damages also arise globally through a climate change mechanism. The following stylized model briefly illustrates these mechanisms and provides the conceptual basis for our empirical IAM. Additional details are described in Appendix A.

The landscape is divided into a continuum of N residential locations. There are waterbodies with varying water quality levels distributed in space across the landscape. The concentration of nutrient pollutants in a waterbody j is denoted c_j. There are H households that live in the landscape. Each household takes as given the common set of residential locations, waterbodies, and water quality levels. Households maximize utility by selecting a residential location and taking recreation trips to each of the J waterbodies.

To ground our framework in the canonical recreation demand and hedonic property value models, we add the following structure. First, a residential location n conveys three local attributes: a bundle of neighborhood characteristics ζ_n, a local waterbody indexed $j = n$ with local nutrient concentration c_n, and a water utility that sources drinking water from the local waterbody. Second, we assume that location n provides fixed travel distances to the $J-1$ nonlocal waterbodies, so that households in location n have exogenous travel cost TC_{nj} for $j \neq n$. Finally, we assume that location n provides a continuum of available distances to the local waterbody. In equilibrium, the annual rental cost for a home in location n is $P(d_{hn}, c_n, \zeta_n) + \tau_n(c_n)$, where d_{hn} is the distance that household h lives from local waterbody

n and $\tau_n(c_n)$ is an additively separable annual fixed tariff paid to the local water utility. Households select a residential location n, which endows them with the local waterbody, travel costs TC_{nj} for $j \neq n$, and τ_n. They also select the distance to the local waterbody and recreation visits to the local and nonlocal waterbodies.

This structure allows us to connect nutrient pollution to social costs through four local mechanisms. In Appendix A we start with a quasi-linear in income utility function U that includes amenity services from the local waterbody $A(d_{hn}, c_n)$ and recreation trips. Amenity services are decreasing in both distance to the local waterbody and nutrient pollution. Recreation trips to the nonlocal and local waterbodies are denoted $X_{hj}(c_j)$ for $j \neq n$ and $X_{hn}(c_n)$, respectively. Trips are decreasing in nutrient pollution. The utility function also includes an infant health term $B_h(c_n)$ that we proxy using birthweight. Birthweight is decreasing in exposure to local nutrient pollution. Finally, nutrient pollution can influence the cost of treating drinking water and hence the drinking water tariff so that $\tau_n'(c_n) > 0$. Under our assumptions, the marginal damage to household h living in location n (holding location fixed) from a change in water quality in the landscape is

$$MD_{hn}(c_1, ..., c_J) = \tau_n'(c_n) - \frac{\partial U}{\partial A}\frac{\partial A(d_{hn}, c_n)}{\partial c_n} - \sum_{j=1}^{J}\frac{\partial U}{\partial X_j}\frac{\partial X_{hj}(c_j)}{\partial c_j} - \frac{\partial U}{\partial B}B_h'(c_n) \tag{1}$$

2.2 Total Marginal Damages and Gross External Damages

Equation 1 derives the marginal damage for a single household from a generic marginal change in nutrient concentrations across the landscape. Our main interest is in the total marginal damage from a change in nutrient loadings (emissions) at specific points in space.[8] To derive this, we express nutrient concentrations as a stylized function of emissions $E_1, ..., E_J$ into the waterbodies so that $c_j = c_j(E_1, ..., E_J)$. Based on this, the change in concentration $\triangle c_j$ at waterbody j from a change in loadings $\triangle E_i$ into waterbody i is determined by the hydrological relationship

$$\triangle c_j = \frac{\partial c_j(E_1, ..., E_J)}{\partial E_i}\triangle E_i, \quad \forall j \tag{2}$$

Combining equations 1 and 2, the total marginal damage (TMD), i.e., the sum of all five marginal damage components, from a change in loadings $\triangle E_i$ into waterbody i is

[8] This follows from a similar approach outlined by Muller and Mendelsohn (2007, 2009) for air pollution.

$$TMD(\triangle E_i) = \sum_{n=1}^{N} H_n \tau_n'(c_n) \triangle c_n - \sum_{n=1}^{N}\sum_{h=1}^{H} \frac{\partial U}{\partial A}\frac{\partial A(d_{hn}, c_n)}{\partial c_n}\triangle c_n - \sum_{h=1}^{H}\sum_{j=1}^{J} \frac{\partial U}{\partial X_j}\frac{\partial X_{hj}(c_j)}{\partial c_j}\triangle c_j$$
$$- \sum_{h=1}^{H}\sum_{n=1}^{N} \frac{\partial U}{\partial B} B_h'(c_n)\triangle c_n + \sum_{j=1}^{J} GHG_j(\triangle c_j), \tag{3}$$

where H_n is the number of households living in location n and GHG_j denotes damages from greenhouse gas emissions arising from the change in nutrient concentration at waterbody j. We refer to the first term as the drinking water mechanism, the second as the amenity mechanism, the third as the recreation mechanism, the fourth as the birthweight mechanism, and the last term as the climate change mechanism. In later sections we discuss how these are empirically measured.

To estimate the Gross External Damages (GED) attributable to a sector s from its nutrient loadings into the J waterbodies we use

$$GED_s = \sum_{i=1}^{J} TMD(\triangle E_i) \cdot E_{si}, \tag{4}$$

where E_{si} is the baseline annual emissions from sector s into waterbody i.

3 Hydrology and Pollution

We rely on the National Hydrography Dataset Plus Version 2 (NHDPlusV2) and the USGS SPAtially-Referenced Regression On Watershed attributes (SPARROW) model for four key components. First, the NHDPlusV2 provides the base map for water resources in our model and details how water moves throughout the hydrologic network. In total, we use over 2.5 million river segments that are, on average, less than 2 km in length, over 350,000 lakes that have surface area greater than 0.01 km^2, and over 2,100 watersheds (see Appendix B for more details). Second, we link the NHDPlusV2 to the USGS SPARROW model so that we can observe baseline local concentration estimates of total nitrogen (TN) and total phosphorus (TP), our two key measures of nutrient pollution. Third, we use decay rates from the SPARROW model and river routing from the NHDPlusV2 to predict how local and downstream concentrations change as we add marginal loads of nutrients to a particular

river segment. Fourth, the USGS SPARROW model provides estimates of pollution loads (e.g., quantity of emissions) by major sources in each watershed.

The USGS SPARROW is often described as a hybrid hydrologic model that marries a biophysical model of water pollution routing and decay with a non-linear statistical model that estimates the contributions of pollution from various sources. It provides predictions of long-run average pollution loads, concentrations, and yields into receiving waters. The SPARROW modeling framework was introduced by Smith et al. (1997), and it has been continuously updated by US Geological Survey researchers. Early SPARROW applications provided large scale estimates of water quality conditions calibrated for base years 1987 (Smith et al. 1997), 1992 (Alexander et al. 2008), and 2002 (Robertson et al. 2014), with the most recent efforts calibrated to base year 2012 using data from 1999 to 2014 (e.g., Robertson and Saad (2021)). Through each iteration the framework has evolved to feature finer scale spatial resolutions, improved input source data and data pre-processing, and increasingly sophisticated calibration and diagnostic procedures (see, e.g., Robertson and Saad (2019), pp. 7-9).

We combine results from five regional SPARROW models that provide predictions of nutrient concentrations and loads for the conterminous U.S. These models include the Northeast, Southeast, Midwest, Southwest, and Pacific. Each of these regional models provides estimates of average conditions over the period 1999 to 2014 and reflects estimates of loads around 2012. The models predict between 86 and 93 percent of the variation in TP loads across the five regions. While each model has the same foundational structure, there are some differences across models in terms of the predicted contributing sources and the underlying data used to calibrate the models. For instance, each regional model estimates nutrient contributions for the primary categories of municipal wastewater facilities, agricultural runoff, and urban runoff. However, each model also includes some modest differences in other sources. For example, the Northeast estimates phosphorus contributions from mineral erosion, while each of the other four models describe a similar category of contributions from "natural sources." Appendix B describes these regional SPARROW models in more detail.[9]

Figure 1 displays baseline pollution concentrations (Panel A) and loads (Panel B) of TP across the U.S. Appendix Figure B.1 displays the corresponding patterns for TN. As

[9]The five individual models are the result of a coordinated effort to represent the conterminous U.S. using common methodology, underlying data sources, and computational techniques. Assembly and preprocessing of water quality and stream flow data used in the five regional models was completed in a single step as reported in Saad et al. (2019). The inputs and outputs from the models are fully comparable, while allowing regional heterogeneity to be reflected in the model-specific parameterizations.

expected, nutrient concentrations are highest in the Midwest due to agriculture's outsized role in contributing to nutrient pollution. Coastal areas also have high concentrations due to large urban centers and nutrient pollution from municipal waste.[10]

Figure 1, Panel C displays the river network and the corresponding variation in decay from the fate and transport model. Given a particular river segment, this figure maps the fraction of TP that is transported to the farthest point downstream. Dark blue represents less decay (more transported) and light blue represents more decay. The map shows, that as expected, there is less decay when pollution is added to the main channel of a major river (i.e., the Mississippi) than when added to a headwater stream of that main channel. We also see that the SPARROW models generally predict less decay in the river networks of the Pacific Northwest and more decay in the headwater streams in the Midwest and Southeast. We also perform some checks on these data to better understand the transport model. Panel D displays estimates from a regression of the fraction of TP transported downstream on various distance bins, while controlling for differences in decay at the HUC2 (major watershed) and state levels. This figure shows that decay increases significantly within the first 300-600 km. TP falls by about 20 percentage points within the first two distance bins. TP then continues to decrease as it moves downstream, with most TP decaying by 2,700 km (i.e., roughly the distance from a headwater stream in the Upper Mississippi to the Gulf of Mexico). Appendix Figure B.4 plots similar estimates of decay based on the local river flow and stream level. These results suggest a pattern in which TP decays more (less is transported) when added to a smaller headwater stream and less (more is transported) when added to a major river.

An important contribution of this paper is the development of new routines that allow us to quickly assess how marginal changes in nutrient pollution move within and across watersheds and the resulting economic damages associated with these changes. To measure damages, we simulate adding a marginal load of nutrients in each individual river segment, trace how those nutrients travel downstream, and identify how concentrations change in each affected watershed. We then repeat this procedure for the 2.5 million river and lake segments

[10]We exclude coastal segments from SPARROW when constructing our estimates of concentrations and loads due to our focus on inland surface waters. The main consequence of this decision is that we exclude approximately 17k metric tons of TP emitted by coastal municipal wastewater treatment plants. These loads do not factor into our estimate of GED.

[11]We restrict this algorithm to the 2.5M river segments that fall completely within a corresponding HUC8 to simplify reporting average marginal damages at the watershed level. This restriction does not impact total GED because GED is calculated using watershed-specific loads combined with the average marginal damages of each watershed. We use all 2.7M river and lake segments for routing purposes and to determine how local and downstream concentrations change as a result of pulses in these 2.5M segments.

to estimate segment-specific marginal damages.[11] This procedure is an operationalized version of equation 2 that relates a marginal change in nutrient load at a point in space to changes in local and downstream nutrient concentrations. Our algorithm for each segment i-specific pulse is as follows:

i. We first calculate an unweighted average of annual TN and TP loads for each watershed in the U.S.

ii. Next, we calculate "marginal" loads of TN and TP that are equal to 1% of the nationwide average watershed values calculated in step i.[12] These represent ΔE_i in equation 2.

iii. For each river or lake segment i, we add ΔE_i to the long-term average flux for that segment.[13] We refer to this as an emissions pulse. We do this one pulse at a time for each segment.

iv. For each segment i, we then compute the change in long-term average nutrient flux for all river and lake segments that are downstream from segment i following the pulse. We compute the resulting change in long-term average concentration from the change in flux at all downstream segments using the long-term average flow for the segment. The pulse decays as it moves downstream according to decay rates given by SPARROW.

v. To measure marginal damages that result from these changes in concentrations, we calculate the change in all watershed concentrations as a flow-weighted average of concentration changes at all river or lake segments contained in watersheds that are impacted by this pulse. This generates Δc_j in equation 2 for the reference watershed ($j = i$) and all downstream watersheds.

vi. We divide the resulting change in marginal damages by ΔE_i to obtain estimates of damages in terms of kg/yr of TN and TP.

We calculate watershed averages of these segment specific marginal damages. We then multiply watershed marginal damage estimates by the corresponding emissions load in that watershed to estimate GED.

[12] These values are 1,959 kg/yr for TP and 15,867 kg/yr for TN.

[13] We use loads to refer to nutrient emissions at a point in space and flux to refer to the physical amount of pollution passing through a river or lake segment. Long term average flux depends on upstream and local loads.

4 Damage Modules

Here, we describe our five damage modules. These modules link changes in water quality conditions to changes in economic value. They include five of the most likely categories of damages due to nutrient pollution, including impacts on recreation, housing, drinking water treatment, human health, and climate change.

4.1 Recreation

We estimate the lost recreation value from a change in water quality using three main steps.

First, we define a representative individual, the set of available destinations, and baseline water quality. The representative individual is defined at the census tract (ct) level. The set of available destinations for the individual – referred to as the choice set – includes lakes that are larger than 0.1 km^2 and within 100 miles of the center of the census tract. Figure 2 displays the location of these lakes and compares them to the set of all lakes greater than 0.01 km^2 used in some modules. In Appendix C.1, we provide support for these choices. We summarize our decision process here. Many freshwater recreation demand studies focus on lakes. In addition, recreation surveys suggest that the most frequently visited lakes are larger lakes ($>$ 1 km^2), though recreation occurs at lakes even smaller than our 0.1 km^2 threshold (Evans et al. 2011). Thus, we place a greater weight on larger lakes when constructing our representative measure of water quality. Some national surveys suggest that a substantial amount of recreation occurs within 25-30 miles of an individual's home (Keiser 2019; Keiser and Shapiro 2019a), but an analysis done by the USEPA in the 1990s suggests that choice sets for water-based recreation may extend out to and beyond a 100 mile radius (Viscusi et al. 2008). We choose 100 miles as our baseline given that this distance is a feasible driving distance for day trips, but we place greater weight on those lakes closer to home.

We use J_{ct} to denote the lakes in the choice set. For baseline water quality, we calculate a weighted average that reflects the size of the lakes within the choice set, their distance to the representative individual, and nutrient concentrations. Specifically, for nutrient concentration c_j at each of the $j = 1, \ldots, J_{ct}$ lakes, we assign water quality for the census tract according to

$$WQ_{ct} = \sum_{j=1}^{J_{ct}} \frac{\frac{a_j \cdot c_j}{d_{ct,j}}}{\sum_{j=1}^{J_{ct}} \frac{a_j}{d_{ct,j}}}, \tag{5}$$

where a_j is the area of the lake j and $d_{ct,j}$ is the distance from the census tract center to the lake. We use the concentration of TP to measure c_j in the recreation module.

Next, we construct a dose response function to relate TP to the number of water-based recreation days that the representative individual takes in a year. Let $X_{ct}(WQ^0_{ct})$ denote the baseline number of trips taken by a representative person living in tract ct, where the superscript 0 denotes baseline water quality. The change in trips from a marginal change in water quality is given by

$$\triangle X_{ct} = X_{ct}(WQ^0_{ct}) \cdot \epsilon_{WQ} \cdot (\frac{\%\triangle WQ_{ct}}{100}), \tag{6}$$

where ϵ_{WQ} is the elasticity of water recreation trips with respect to water quality, and $\%\triangle WQ_{ct}$ is the percent change in water quality defined as

$$\%\triangle WQ_{ct} = 100 \cdot (\frac{WQ^1_{ct} - WQ^0_{ct}}{WQ^0_{ct}}), \tag{7}$$

and superscript 1 denotes the new water quality level.

To compute equation 6 for each census tract, we use the National Survey of Recreation and the Environment (NSRE) to calibrate $X_{ct}(WQ^0_{ct})$. This survey, conducted by the USDA's Forest Service, represents one of the most comprehensive recreation surveys in the U.S. We use over 70,000 responses from surveys conducted between 1999 and 2009 to measure recreation visits that are for swimming, cold and warm-water fishing, boating (with a motor), and visits to noncoastal waterbodies for other recreation purposes. These years are the most recent available to us and align with the time period of our water quality data. We focus on these categories since they reflect the most likely water-based recreation uses impacted by pollution and were the focus of a recent study (Keiser 2019).[14] We use these data to calculate the average per capita annual water-based recreation trips at the county level, and assign $X_{ct}(WQ^0_{ct})$ based on the county that the census tract is in. Figure 3 and Appendix Figure C.1 plot our baseline recreation day and participation estimates respectively. Patterns from the NSRE generally follow expectations that water-based recreation activities occur more often in water-rich regions, including New England, the Upper Midwest, and parts of the Mountain West and Southeast. We observe fewer trips and lower participation rates in the arid Southwest.

[14]We assume most of these activities reflect freshwater recreation. Though we exclude activities that are likely saltwater, such as saltwater fishing, surfing, and beach visitations, it is possible that some of these data reflect coastal activities. We do not observe where these activities occur; we do observe the home county, participation rates, and the number of days for each activity.

To calibrate the elasticity of trips with respect to water quality, we perform two new analyses. First, we build on Keiser (2019). To the best of our knowledge, this study is the only analysis that directly addresses the endogeneity of water quality in recreation demand. The study links water-based recreation activities reported in the NSRE to local phosphorus pollution. It utilizes upstream concentrations of TP as an instrument for local TP concentrations to establish a plausibly causal link between pollution and recreation. One limitation of Keiser (2019) is that the analysis excluded a large portion of the Southwestern U.S. due to limited water quality data. In Appendix C.1, we present results from an analysis that utilizes the research design in Keiser (2019) and our more comprehensive water quality data. We find a meaningful and statistically significant relationship between TP and water-based recreation. Figure 3, Panel B shows that these new estimates, along with the original findings from Keiser (2019), suggest that the elasticity of water-based recreation with respect to TP ranges between 0.1 and 0.3.

We conducted a thorough review of other water-based recreation demand studies to understand how these elasticities compare to the literature. Most of these other studies examine recreation for a much smaller geographic area such as an individual state. Figure 3, Panel B displays estimates of recreation trip elasticities with respect to TP, Secchi disc depth (i.e., clarity), and other water quality parameters.[15] Our review finds a median elasticity of recreation trips with respect to TP of 0.18. As we discuss in the property value module below, we estimate that the elasticity of Secchi depth with respect to TP is approximately 0.5. This implies that the elasticity of recreation trips with respect to Secchi depth should be twice as large as the elasicity with respect to TP. We find that one-half of the median Secchi depth elasticity is 0.09. As Figure 3 shows, there is a remarkable consistency in elasticities across studies. Given our new analysis, the prior findings from Keiser (2019), and our review of the literature, we set our baseline elasticity, ϵ_{WQ}, equal to 0.15 and explore sensitivity analyses that allow for different elasticities. Appendix C.1 provides more details and shows how one can recover recreation trip elasticities from random utility maximization models.

We then derive an expression for the marginal value of water quality operating through the recreation mechanism using results from Chan and Wichman (2022). We define the consumer surplus from recreation trips at baseline and changed water quality conditions as

[15]Secchi disc depth measures water clarity by determining the distance at which a Secchi disc is no longer visible as it travels down the water column.

$$CS^0_{ct} = \int_{TC_0}^{\infty} X_{ct}(WQ^0_{ct})dTC \tag{8}$$

$$CS^1_{ct} = \int_{TC_0}^{\infty} X_{ct}(WQ^1_{ct})dTC, \tag{9}$$

where TC_0 is a measure of baseline travel cost. The marginal value of a change in water quality is the difference between the two consumer surplus measures - i.e., the change in recreation consumer surplus from the water quality change:

$$\triangle CS_{ct} = CS^1_{ct} - CS^0_{ct}. \tag{10}$$

To compute 10 with available data, we note that baseline consumer surplus can be rewritten as the product of the average per trip value (referred to as the recreation day value – RDV) and baseline trips:

$$CS^0_{ct} = \frac{CS^0_{ct}}{X(WQ^0_{ct})} \cdot X(WQ^0_{ct}) = RDV^0_{ct} \cdot X(WQ^0_{ct}). \tag{11}$$

If the recreation day value is approximately constant, the change in consumer surplus from a marginal change in water quality is:

$$\frac{\partial CS^0_{ct}}{\partial WQ_{ct}} = RDV_{ct} \cdot \frac{\partial X(WQ^0_{ct})}{\partial WQ_{ct}}. \tag{12}$$

Chan and Wichman (2022) show that equation 12, consisting of the product of a dose-response function and a unit value, performs well as an approximation to equation 10 in most applied contexts.

We emphasize that the average consumer surplus per trip is the appropriate unit value in equation 12, rather than the implicit price of a trip, because our quantity of interest is water pollution. The righthand side of equation 12 is the implicit price of nutrient pollution, as revealed through the recreation mechanism.

Finally, we ascribe a value to each water-based recreation day that is lost following a decrease in water quality. Specifically, if we assume a constant recreation day value for a trip ($RDV_{ct} = RDV$), the total lost recreation value can be estimated using

$$\triangle RV_{ct} = \triangle X_{ct} \cdot RDV. \tag{13}$$

We set RDV at \$78 per day based on a review of the Oregon State Recreation Use Value Database (2016). This database reports over 400 estimates of the value of water-based recreation activities from 55 studies across the U.S. conducted since 2000. \$78 per day reflects the median of the median estimate from each study. This estimate of the value of a water-based recreation day is almost identical to the value across all recreation activities (\$77). As a robustness exercise we also survey recent studies using the random utility maximization (RUM) framework to estimate water-based recreation demand. The median per trip value from this independent assessment is \$79, lending support to our baseline RDV. In Appendix C.1 we provide details on our use of the Oregon State Database and separate summary of the RUM-based recreation studies. To recover an aggregate value, we multiply the representative agent recreation costs from equation 13 by the adult population in the census tract to quantify the total change in recreation value for the census tract.

4.2 Property Values

Property value models have long been utilized to quantify the amenity value stemming from water quality changes at the local level (e.g., within a single lake, watershed, county, or state). To develop this damage module, we conducted a review of hedonic studies published between 2016 and 2024 and compared these findings to a meta-analysis by Guignet et al. (2022) that analyzed similar studies between 1979 and 2017. We detail this review in Appendix C.2. The published literature reinforces the importance of lakes in hedonic analyses, suggests that the capitalization of water quality falls with distance from a waterbody, and establishes a relatively consistent elasticity of housing prices with respect to water quality, especially water clarity. Our approach accounts for each of these key components.

Given the focus on Secchi depth in many hedonic studies, we first establish the relationship between TP and Secchi depth. We believe this is a reasonable linkage given that nutrients are a major factor in determining water clarity (Keeler et al. 2012; Keeler et al. 2016). In Appendix C.2, we present results from a regression analysis that utilizes USEPA National Lake Assessments (NLAs) conducted on several thousand lakes spanning fifteen years and four surveys (USEPA 2007, 2012, 2017, 2022a). These surveys report measured concentrations of TP and Secchi depth by lake. Our analysis suggests a robust and statistically

significant elasticity of Secchi depth with respect to TP of approximately 0.5. To account for the variability in this relationship across space, we estimate and utilize separate elasticities by the nine eco-regions used in the NLAs.

Next, we identify the elasticity of housing values with respect to water quality. Figure 3, Panel D summarizes our review of the literature. It shows that the elasticity of housing values with respect to Secchi depth is quite stable. It is approximately 0.18 for near waterfront homes. We find that half of this value, 0.09, is close in magnitude to the elasticity of housing values with respect to TP, 0.12, reported by a recent study that focused on nutrients (Walsh and Milon 2016). We also find that recent elasticity estimates are similar to those reported by Mamun et al. (2023). This study conducted a national-scale hedonic analysis using sales that occurred in proximity to over 1,600 freshwater lakes across the U.S. The authors find that a 1% increase in water clarity, measured by Secchi depth, on average leads to 0.18% and 0.04% increases in the value of single-family homes within 100 meters and between 100 and 300 meters from the lakeshore, respectively. Mamun et al. (2023) is the most comprehensive nation-wide hedonic analysis among those we reviewed. Given the consistency in elasticity estimates and comprehensiveness of Mamun et al. (2023), we utilize these elasticities for near and far from lakeshore homes.

We calculate the annualized marginal damage from a change in nutrient concentration at lake j according to

$$\Delta HV_j = \alpha \cdot \left(V_j^1 \cdot \eta_1 + V_j^2 \cdot \eta_2\right) \cdot \left(\frac{\nu_j \cdot \%\Delta c_j}{100}\right). \tag{14}$$

In this expression, $\eta_1 = 0.18$ and $\eta_2 = 0.04$ are the elasticities of home price with respect to water clarity for properties in the $<$ 100 m and 100-300 m distance from shoreline bands, respectively, and V_j^1 and V_j^2 denote the aggregate value of the housing stock near lake j in the two distance bands. In the second term, ν_j is the elasticity of water clarity with respect to TP. The elasticity for lake j takes one of nine values ranging from 0.39 to 0.50, based on the eco-region where the lake is located. The term $\%\Delta c_j$ is the percent change in phosphorus concentration in lake j. Their product is the percentage change in water clarity generated by the increase in TP concentration in lake j. Finally, α=0.0448 is an annualization rate that translates changes in the value of the housing stock into a yearly flow value. We rely on Bishop et al. (2025) for this value.

Finally, we calibrate our model using the value of assessed homes close to lakes. We begin by merging the waterbody shapefile of NHDPlusV2 with information from the Core Logic assessment database to calculate the number of properties and total assessed value of

single family houses within 100 meters and between 100 and 300 meters from the shore for all lakes greater than 0.1 km^2. We identify a total of 3.51 million single family residential houses located within 300 meters of 27,084 different lakes.[16] Figure 3, Panel C displays these values by county. The figure shows the highest valued homes near lakes are in New England, portions of the Upper Midwest, and the West coast. To link water quality to these homes, we assign each lake in a HUC8 the average watershed value for TP concentration from our SPARROW model. Appendix C.2 provides additional details of our property value module.

4.3 Drinking Water

Nutrient pollution generates external costs for drinking water facilities that draw raw water from surface water and groundwater and process it to supply potable water to consumers. To protect public health, the USEPA sets a limit of 10 mg/l for nitrate and 1 mg/l for nitrite in finished drinking water. When levels in raw water exceed these thresholds, community water systems (CWSs) must treat the water using methods such as altering intake sources, blending water, sourcing low-nitrate raw water, or applying nitrate removal technologies like reverse osmosis and ion exchange. However, the treatment involves additional costs in the form of capital, labor, energy and chemicals for the treatment process, which are ultimately passed through to households. In this section we describe how we construct an empirical representation of these costs, which corresponds to the first term in equation 3.

We sketch our basic steps here and provide additional details in Appendix C.3. First, drawing on the EPA's Safe Drinking Water Information Systems (SDWIS) data, we identify the set of CWSs that either source raw water from surface waters, groundwater under the influence of surface water, or purchase water from systems that source water from either of the first two categories.[17] There are approximately 50,000 CWSs in the US. We calculate 11,581 systems fall into one of these three categories. These water systems are of different sizes based on the population they serve. USEPA provides intake source location for 54 percent of them. These systems are spread across 1,244 HUC8 watersheds. We assign water quality for each CWS based on our SPARROW model prediction for TN concentration in its home HUC8 watershed.

[16] We use a smaller subset of lakes than in the recreation module because we do not observe residential parcels within 300 m of some lakes. The population of the approximately 200 counties with lakes greater than 0.1 km^2 but no Core Logic transactions represent around 1% of the U.S. population.

[17] Nitrogen contamination of groundwater can also generate treatment costs. However, the focus of this study is on surface water pollution.

Not all CWSs treat for nitrate and nitrite and the SDWIS does not report which systems do. To determine the systems that likely incur abatement costs, we use assumptions based on long-run average nitrogen levels, nitrogen violation history in the recent past (2018-2022), and ownership of treatment technologies that can remove nitrogen, though may also remove other contaminants. With these steps, we arrive at a set of 743 community water systems that we assume incur treatment costs. These systems are the basis for our calculations.

We rely on data and results from Mosheim and Ribaudo (2017) to estimate the drinking water treatment cost of an increase in nitrogen concentration.[18] The authors report variable cost elasticities with respect to nitrogen removal of ϵ_l=0.003 and ϵ_s=0.04 for large and small systems, respectively. The data includes values for the variable costs of CWS operations among systems in the sample. We use sample variation in variable costs (VC) across five geographic regions and large versus small systems to assign VC_n^f for each plant n that treats its raw water, where $f = l, s$ indicates if the system is large or small. We assume that a percentage increase in TN concentration in a system's source water implies an equal percentage increase in abatement requirements. We calculate the change in drinking water treatment costs for a system n from a change in nutrient pollution according to

$$\Delta DW_n^f = VC_n^f \cdot \left(\frac{\epsilon_f \cdot \%\Delta c_n}{100} \right), \tag{15}$$

where $\%\Delta c_n$ is the percentage change in nitrogen concentration in the HUC8 where the system sources its water from.

4.4 Human Health

Recent research has found that water pollution can affect infant health, even at levels that continue to exist in the U.S. For example, DiSalvo and Hill (2024) show that drinking water contamination at levels below regulatory thresholds increases the incidence of low birthweight and preterm births. For surface water, Dias et al. (2023) link glyphosate exposure to infant mortality and Flynn and Marcus (2023) find that investments in wastewater treatment technology following passage of the Clean Water Act reduced the incidence of low birthweight near waterbodies with improved quality. Likewise, Jones (2019) finds that exposure to freshwater harmful algal blooms increases the incidence of low infant birthweight. In this subsection, we build on this literature to quantify one element of the health costs of nutrient

[18] Nitrate and nitrite are forms of nitrogen. Total nitrogen, for which we observe predicted concentrations in this paper, is comprised of nitrate, nitrite, organic nitrogen, and ammonia. Mosheim and Ribaudo (2017) predict changes in costs as a function of total nitrogen.

pollution by examining the nationwide link between harmful algal blooms and the incidence of low infant birthweight.

In freshwater systems, excess nutrients fuel cyanobacteria (sometimes referred to as blue-green algae) growth, which can form dense blooms on the surface of aquatic resources (M. Beaulieu et al. 2013; Huisman et al. 2018). Some species of cyanobacteria generate cyanotoxins that are harmful to humans. Among these, microcystin is relatively well-studied and it is common in U.S. inland waters. In the 2022 NLA, 50 percent of sampled waterbodies contained detectable levels of microcystin, with concentrations that are strongly correlated with nutrient levels. Exposure to waterborne microcystin can occur through skin contact or ingestion (Lad et al. 2022); inhalation of aerosolized microcystin is also increasingly recognized as an exposure avenue (Lim et al. 2023).

There is evidence that exposure to microcystin in utero affects infant health. Zhao et al. (2020) found that microcystin exposure among pregnant mice caused fetal weight loss and inhibited placenta development. The reproductive toxicity of microcystin more generally is summarized by Zhang et al. (2021). Pilotto et al. (1999) present an early epidemiological study showing that first trimester exposure to a high frequency of cyanobacteria blooms was associated with a higher incidence of low birthweight. Finally, Jones (2019) exploits a natural experiment in a midwestern lake to show that the incidence of low birthweight in counties bordering the lake is causally related to changes in microcystin concentration.

We link changes in nutrient concentration to the incidence of low birthweight and the economic cost of low-weight births using five steps. We briefly describe these steps here, with additional details included in Appendix C.4. First, we use the findings in Jones (2019) to calibrate an elasticity of low birthweight incidence with respect to microcystin. In his context, the collapse of the zebra mussel population in Gull Lake, Michigan, in August 2010 caused a large and rapid decrease in microcystin concentration in the lake.[19] Levels remained low until the zebra mussel population recovered. Jones (2019) compares low-weight births in the two counties bordering the lake before and after the shock to control counties in the state. He finds that births in the treated counties were 1.4 percentage points less likely to be underweight after the shock. This corresponds to a 17.3 percent decrease in the incidence of low birthweight. The zebra mussel die-off caused average microcystin concentrations to fall from 0.20 µg/l to 0.04 µg/l in the lake – a decrease of 80 percent.[20] Based on these

[19]Filter feeding zebra mussels are a catalyst for cyanobacteria growth in some freshwater systems. See Sarnelle et al. (2005).

[20]In the 2022 NLA, the median concentration of microcystin among lakes where it was detected is 0.19 µg/l.

figures, we assign ϵ_{LB}=0.216 as the elasticity of low birthweight incidence with respect to microcystin.

Second, we empirically estimate the relationship between TP concentration and microcystin concentration in U.S. lakes, using the four waves of the National Lakes Assessment. As with the relationship between Secchi depth and TP, we allow for ecosystem heterogeneity by including ecoregion fixed effects and we recover ecoregion-specific elasticities. We use ϕ_j to denote the elasticity of microcystin with respect to TP for lake j. The elasticity takes one of nine values ranging from 0.24 to 0.62.

Third, we assume that lakes larger than 0.01 km^2 are potential sources of microcystin exposure for local populations. We define the exposed population based on distance to the nearest lake. For our baseline analysis we use a 1-mile distance band to define the exposure zone and a 10-mile buffer in a robustness check. We discuss motivation for these choices in Appendix C.4. For each lake j we use the USEPA's EnviroAtlas program data to measure the total population living in distance band k (1, 10 miles) from lake j.[21] These same data allow us to measure the fraction of the population living in county c that are within distance band k from their nearest lake j. We combine the population data with county-level health data assembled by the University of Wisconsin Population Health Institute. The County Health Rankings & Roadmaps (CHR&R) program publishes yearly statistics describing several health and well-being indicators, including the fraction of births less than 2,500 g.[22] We use the 2023 report, which summarizes natality data from the National Center for Health Statistics for the seven years spanning 2014 to 2020. We measure the county-level incidence of low birthweight as the proportion of all births that were low weight in the county during the seven years. We measure total annual births in each county as the average across the seven years.

We assign annual births in the exposure zone for each lake by assuming the proportion of county-level births near lake j is equal to the proportion of the county-level population living k miles from lake j. This allows us to measure B_{jk} as the baseline number of annual births that occur within k miles of lake j. Furthermore, we use the CHR&R data to measure the baseline proportion of low-weight births near lake j, based on the county the lake is in. We denote this using LB_j. The 2014-2020 average is 0.082 and ranges from 0.029 to 0.22.

Fourth, we predict how $\%\Delta c_j$, a percentage change in phosphorus concentration in lake j, generates additional low weight births in proximity to lake j. Specifically, we calculate

[21] See USEPA (2025).

[22] See University of Wisconsin Population Health Institute (2023).

the percentage change in the incidence of low birthweight near lake j using

$$\%\Delta LB_j = \epsilon_{LB} \cdot \phi_j \cdot \%\Delta c_j. \tag{16}$$

We then compute the number of additional low weight births k miles from lake j following the change in TP as

$$\Delta LWB_{jk} = \frac{\%\Delta LB_j}{100} \cdot LB_j \cdot B_{jk}. \tag{17}$$

Finally, we calculate the economic cost of a low weight birth by summing the present discounted value of lost lifetime earnings and the immediate medical costs generated by a low birthweight occurrence.[23] For the former, we use the findings from Bharadwaj et al. (2018), who study the long-term impact of birthweight on individual health and economic outcomes. Using a twins fixed effects design, they find that a 10 percent increase in birthweight causes a 1.25 percent increase in permanent labor income. Moving from an average birthweight of 3,400 g to the low birthweight threshold of 2,500 g represents a 26 percent decrease. Combining these, we assume that a low birthweight occurrence decreases the individual's lifetime earnings by 3.25 percent. In a recent study, Guvenen et al. (2022) found that the average present discounted value of lifetime earnings for an individual in the U.S. is \$1.43 million (2022 USD). Using this as a baseline, we assume that the present value of lost lifetime earnings for a low birthweight individual is \$46,500.

Medical costs are higher on average for low birthweight infants than for births without complication. Russell et al. (2007) quantify these differences and find that a low birthweight occurrence increases the median length of hospital stay by two days and median hospital cost by \$4,200 (2022 USD). Summing up the lost earnings and additional hospital costs, we assign VLB=\$50,700 as the economic cost of an additional low weight birth. The economic damages operating through the birthweight mechanism from $\%\Delta c_j$ in lake j is

$$\Delta BV_j = \Delta LWB_{jk} \cdot VLB. \tag{18}$$

[23]The human capital approach can either over or underestimate the true welfare loss associated with health damages (Phaneuf and Requate 2016). True welfare losses (compensating or equivalent variation) would reflect differences in preferences, the value of avoiding pain and suffering, broader measures of well-being, and would not apply only to those who ultimately enter the workforce. However, human-capital estimates are the best approximation currently available.

4.5 Climate Change

Nutrient concentrations in surface water contribute to eutrophication in lentic ecosystems (lakes and impoundments), and this generates emissions of greenhouse gases such as methane (CH_4), carbon dioxide (CO_2), and nitrous oxide (N_2O) (DelSontro et al. 2018; J. J. Beaulieu et al. 2019). Indeed, aquatic ecosystems are the largest contributor to global methane emissions (Rosentreter et al. 2021). In this section we describe how we estimate climate change damages from methane emissions that arise from increased nutrient pollution. Appendix C.5 describes the analogous equations for CO_2 and N_2O.

We use the methods described in Downing et al. (2021) to derive equations linking changes in phosphorus concentrations in lakes and reservoirs to annual changes in methane emissions and economics damages. We first use the waterbody shapefile from the NHDPlusV2 to identify 355,986 lakes larger than 0.01 km^2 across the U.S. An individual lake is our unit of analysis and so we link each lake to the SPARROW model output to assign a baseline water quality level for TP. The equations linking CH_4 and eutrophication are based on chlorophyll a ($chla$) concentration and so we convert TP concentration c_j in lake j to chlorophyll a concentration using McCauley et al. (1989):

$$\log_{10}(\text{chla}_j) = 0.725 \cdot \log_{10}(c_j) - 0.131. \tag{19}$$

To predict methane emissions from eutrophication, DelSontro et al. (2018) analyzed historical data from diverse lentic ecosystems to generate a prediction for CH_4 emissions based on chlorophyll a concentrations according to

$$\log_{10}(\text{CH}_{4j}) = 0.94 + 0.778 \log_{10}(\text{chla}_j), \tag{20}$$

where the units for CH_4 are mg per m^2 of surface area per day. We use equations 19 and 20 and the surface areas of lakes to predict how an increase in phosphorus concentration Δc_j generates additional annual methane emissions CH_{4j} from lake j.

To assign costs to the emission changes we use the social cost of one metric ton of methane in the current year, based on USEPA (2022b). Using a 2% discount rate[24] we assign $SCM = \$1799$ and calculate the annual social cost from a change in methane emissions in lake j based on

$$\Delta CC_j = \Delta CH_4(\Delta c_j) \cdot SCM.$$

[24] Although the Trump Administration has revoked the 2023 updated Circular A-4 and its recommendation of a 2% discount rate, we employ the 2% value as it is the best current estimate based on years of additional study and thorough vetting through the federal regulatory process.

5 Results

5.1 Damages by Location

Table 1 reports marginal damages and GED by U.S. Census regions. Panel A reports damages that account for pollution transport and fate. These estimates include both the watershed of origin ("local") as well as watersheds that are impacted downstream ("downstream impacts"). Panel B reports only damages that occur within the watershed in which the pollution is emitted. Column (1) reports average marginal damages by Census Region. We see that marginal damages are highest in the Northeast ($539/kg) and West ($532/kg) and considerably lower in the South ($131/kg) and Midwest ($70/kg).

Panel A of Figure 4 provides the visual representation of these results by watershed. Panel B displays corresponding GED estimates. These maps show areas where GED and marginal damages have similar patterns, as well as where they diverge. Marginal damages and GED are generally higher in population centers near the coasts and in water-rich areas, such as Appalachia, the Upper Midwest, and New England. Table 1, Column (2) summarizes GED by region. Here, GED is highest in the West and lowest in the Midwest. Although marginal damages are four to eight times lower in the South and Midwest relative to the West, GED are still substantial in these regions due to high levels of pollution loads and the fact that GED reflects both marginal damages and quantities. This is evident visually, as there are large portions of the Mississippi River Basin with low marginal damages, but high GED. On the other hand, while the Northeast has high marginal damages, it has lower levels of pollution, and thus relatively lower overall GED. Marginal damages are also high in the Southwestern U.S. due to the fact that baseline flow is lower in this area (see Appendix Figure B.3), but GED is not correspondingly high in this specific area due to lower overall loads. GED for the U.S. is approximately $53B annually.

Next, we examine the spatial concentration of total damages. In Panel C of Figure 4, we rank all watersheds in the U.S. according to their level of GED and normalize this ranking from 0 to 1. This value is captured by the x-axis. On the y-axis, we plot the corresponding cumulative share of GED. Thus, if pollution damages were evenly distributed across watersheds, we would see a 45-degree line. Instead, we show that damages are highly concentrated in a small fraction of watersheds. The top 20 percent of watersheds as ranked by GED are responsible for over 80% of GED. We also plot the location of these watersheds in Panel D. Many of these top watersheds fall near high population centers and a number of these trace out portions of the Mississippi River.

5.2 Cross-boundary Pollution

Next, we examine how the transport and fate of nutrient pollution impacts downstream communities. In Table 1, Panel B, we report the amount of marginal damages and GED that only accrue from within the watershed in which the pollution is generated. By comparing Panels A and B, we are able to analyze the importance of hydrologic boundaries in measuring damages. We find that the local component of marginal damages is approximately 83% on average. This figure varies from approximately 69% in the Midwest to 91% in the Northeast. Approximately 90% of damages are local in the West and 78% in the South. This suggests that a meaningful component of marginal damages arise due to their non-local impacts on water users. The share of local contributions to GED are identical to that of marginal damages at the national level (83%) and the variation across regions follows a similar pattern as marginal damages.

Table 1 also highlights the spillover of pollution across political boundaries. Columns (3) and (4) of Panel A show that approximately 80% of total GED accrues within state boundaries, meaning that a material amount of damages (20%) are born by downstream states. While the percent of national damages that are in-state are similar to the national percentage of damages that are confined to local watersheds, these percentages vary more widely across census regions. The percentage of damages that are in-state are 57% in the Northeast, 74% in the Midwest, 83% in the South, and 97% in the West.

5.3 Uses of Water Resources

In Table 2, we explore how different uses of water resources contribute to our estimates of damages. In Columns (1) and (2), we display estimates of damages that account for local and downstream impacts. In Columns (3) and (4), we examine the extent to which these damages are driven by local impacts. The most immediate finding from this table is that recreation comprises the largest share of damages. Approximately 88% of marginal and gross external damages can be attributed to impacts on water-based recreation. This finding is likely driven by the fact that such a large share of the U.S. population participates in some water-based recreation activity (approximately two-thirds). This finding is also consistent with the general notion that the largest share of water quality benefits likely accrue to water-based recreation users (Freeman 1982; Olmstead 2010; Keiser et al. 2019).

Human health, housing, and climate change impacts are also important components of damages, though contribute significantly less to marginal damages and GED. Their share of

GED are approximately 0.9%, 4.7% and 6.8%, respectively. While relatively small compared to recreation, overall damages are still substantial at $0.45B, $2.47B and $3.59B annually. Drinking water treatment impacts provide the smallest contribution to damages. One important caveat is that drinking water damages represent treatment costs and exclude any potentially important human health impacts that may arise from any residual or untreated pollution.

Figure 5 and Appendix Figure C.5 display the corresponding variation in gross external and marginal damages by watershed by use. Both marginal and gross external damages for recreation largely follow the same patterns as seen in Figure 4. This is unsurprising given the significant contribution of this category to overall damages. In terms of GED, overall damages for most use categories are higher on the coasts and in the eastern half of the U.S., though more evenly spread out for human health and climate change. GED from drinking water treatment impacts are generally larger in the Midwest and the northern half of the country.

5.4 Industry Contributions

Table 3 reports marginal damages and GED by source, as well as total emissions by source. Here, we report emissions for TP, as it is responsible for over 99% of damages. As is well-documented, agriculture is responsible for the largest share of emissions (Column 1). It is responsible for 65% of anthropogenic sources of nutrient pollution (200 thousand metric tons per year). Municipal wastewater contributes 21% of overall pollution loads (65 thousand metric tons) and urban run-off is estimated to contribute 12% (37 thousand metric tons). However, one of the most striking findings from this analysis is that while agriculture is responsible for a majority of pollution, municipal wastewater contributes the largest share of GED. It is responsible for about half of all damages ($24.8B/yr). While still significant, agriculture contributes approximately 38% of GED ($19.8B/yr) and urban non-point source pollution contributes 12% ($6.3B/yr). Column (2) suggests that this difference in pollution loads versus pollution damages is due to the spatial location of polluting activities. As shown in Figure 4, marginal damages are significantly lower in the Midwest, where agriculture is primarily located. Municipal point sources on the other hand are often located closer to population centers and thus contribute larger damages on the margin. These differences in marginal damages are large enough to influence the overall contribution of these sources to GED. Figure 6 displays maps of GED by source and by watershed.

5.5 Influence of Hydrologic and Geographic Features

In this section, we seek to better understand the variability in damages and how particular hydrologic factors may lead to unique patterns for water pollutants.

Figure 7 plots results from regressions of marginal damages at the river segment level on several hydrologic and geographic variables. These regressions explore each component in isolation, while controlling for state and HUC2 fixed effects. We cluster standard errors at the HUC8 level. There are 18 HUC2s in the contiguous U.S. and they serve as the largest watershed delineation. Each of our HUC8 watersheds nest within these larger regions. Thus, these plots examine how marginal damages vary while controlling for some of the cross-country variation we see in Figure 4.

Panel A shows that marginal damages are significantly smaller in larger bodies of water. Relative to very small streams (those with less than 1 cubic feet per second (cfs)), marginal damages decline by $100 to $200 per kg when flow exceeds 1,000 cfs. These rivers are often significant in size and well known. Examples include the North Platte River (1,000 cfs), the Potomac River (10,000 cfs), and the Ohio River (115,000 cfs).[25] These results suggest that dilution may significantly impact marginal damages, especially when river flows exceed 1,000 cfs. Panel B suggests a related finding. Here, stream level represents the number of network segments that are upstream from a terminal point (i.e., outlet). For example, the main stem of the Mississippi that discharges into the Gulf of Mexico is a level 1, whereas headwaters of the Mississippi may be several levels higher. The maximum level that we observe in our data is a level 14 stream. Most streams (less than 95%) are classified as less than a level 7 stream, but only 2% are level 1. Similar to our flow results, we find that marginal damages are lower for lower level streams and significantly higher for higher level streams. Level 10 and higher streams have marginal damages that are approximately $400 more than level 1 streams.

Panel C plots the relationship between the distance that pollution travels and marginal damages. Although pollutants that travel farther have the opportunity to impact more people, we surprisingly find very little relationship between marginal damages and transport distance. Finally, Panel D plots the relationship between marginal damages and the local population within 25 miles of the relevant HUC8. Marginal damages are significantly higher when local population numbers are larger. In particular, when emissions occur near areas with more than 1 million individuals, marginal damages are approximately $300 more than the areas with the fewest people.

[25] These flow estimates reflect average flow reported in the NHDPlusV2.

Thus far, we have considered these factors in isolation. We also regress marginal damages on several specifications that control for each of these components. Appendix Table C.12 displays these results using various fixed effects, from none (Column 1) to HUC2 and state fixed effects (Column 4). Column (4) reports results from a similar specification to these plots. We find that the relationship between marginal damages and stream flow is still negative, though not statistically significant. Marginal damages decline by approximately \$20 per 100,000 cfs. This magnitude is approximately an order of magnitude smaller than the estimates in Panel A of Figure 4. However, we continue to find a meaningful impact of stream level on marginal damages. The results imply that a stream level greater than 10 has marginal damages that are \$300 larger than those in level 1. Given that these results are conditional on stream flow, transport distance, and population, these results hint at some interesting, possible network impacts associated with these water bodies. We find that damages are positive, but not significant for transport distance. Local population continues to be statistically significant and meaningful. Areas with a local population over 1 million have marginal damages that are approximately \$200 larger than the smallest areas.

These results should not be interpreted as causal in nature. Rather, these relationships highlight variation in damages that may be unique to water and suggest the potential need for more stringent abatement efforts in higher level, headwater streams, as well as those near large population centers.

5.6 Sensitivity Analysis

We explore the sensitivity of our baseline marginal and GED estimates to a number of key parameters in our model. Table 4 reports these results.

First, we explore the sensitivity to our choice of a number of recreation parameters. This includes exploring the elasticity of recreation with respect to water quality, the distance used to define the recreation choice set, and the value of a recreation trip. We find that the elasticity and trip value are the two most important parameters. In moving from an elasticity of 0.1 to 0.3, damages triple as expected. Increasing the trip value to the mean value from the reported literature is roughly equivalent to increasing our elasticity parameter to 0.2. We do not find a meaningful difference in damages when we vary our choice set distance between 25 and 150 miles. This may reflect, in part, the fact that our recreation model down-weights sites that are further away.

Next, we examine the influence of our choices in the property values module. This includes extending the impacts of water quality out to 500 m using the same elasticity estimates as

in our baseline model, using an elasticity of home values with respect to Secchi depth of 0.25 for homes within 100 m from shore, and two different elasticity estimates for the relationship between Secchi depth and TP - one estimate that represents a nationwide average from a pooled regression analysis (Column (1) of Appendix Table C.7) and another that reflects a more demanding specification obtained by using a smaller subset of lakes (Column (4) of Appendix Table C.7). We find that increasing the elasticity of home values to water quality to 0.25 yields similar damage estimates as extending the base of homes that we assume are impacted by water quality. Housing damages are about 1.5 times larger. We also find that damages are about half of the baseline numbers when we reduce the parameter on the Secchi depth to TP elasticity; however, none of these changes meaningfully impact total GED for the country.

We examine the sensitivity of abatement costs for drinking water systems by varying two parameters. First, we lower the threshold for treating raw water to remove nitrogen from a concentration of 5 mg/l to 2.87 mg/l, which is the average value reported in Mosheim and Ribaudo (2017). Second, we double the variable-cost elasticity of nitrogen removal from source water from 0.004 to 0.008 for large systems, and from 0.03 to 0.06 for small systems. The lower treatment threshold more than doubles the number of systems requiring nitrogen abatement to comply with the USEPA MCL regulation. Estimated damages from additional abatement costs increase by roughly 1.5 times and nearly 2 times, respectively, under these two scenarios. Though these exercises increase drinking water damage estimates, they have little to no impact on our overall estimates of GED.

Next, we explore health results that extend low birthweight effects up to 10 miles from a lake and vary the elasticity between microcystin and TP. Similar to the Secchi depth analysis for hedonics, we explore a nationwide average elasticity (Column (1) of Appendix Table C.11) and one that utilizes a more demanding specification (Column (4) of Appendix Table C.11). These results show that the results are most sensitive to our assumption regarding the number of people impacted. Increasing our impacted population out to 10 miles increases health damages by a factor of 6.

Finally, we examine how climate change damages vary with the choice of discount rates and when we only consider methane (CH_4) rather than all three greenhouse gases. USEPA (2022b) reports that the social cost of one metric ton of GHG emission varies by discount rate and by the type of gas.[26] When we use a lower discount rate of 1.5% for all three GHGs,

[26]The social cost of one metric ton of CH_4 is \$2,478 at a 1.5% discount rate and \$1,390 at a 2.5% discount rate.The corresponding social costs for CO_2 and N_2O at a 1.5% discount rate are \$346 and \$90,454, respectively; at a 2.5% discount rate, they are \$122 and \$37,128.

the total climate change damages increase by 39% relative to the the baseline model. The damages shrink by 23% when use a discount rate of 2.5%. Next, we confirm that nearly all climate change damages (approximately 94%) are attributable to methane by excluding carbon dioxide (CO_2) and nitrous oxide (N_2O) from our calculations.

6 Conclusions

Our research shows that GED from nutrient pollution is between $35B and $100B per year, with a baseline estimate of $53B. This represents approximately 0.25% of GDP.[27] The closest comparison to this figure is work done to measure GED from air pollution in the U.S. Tschofen et al. (2019) find that GED from fine particulates ($PM_{2.5}$) is $790B to $1,010B per year over a similar time frame, implying that damages from $PM_{2.5}$ are up to twenty times larger than damages from the leading cause of surface water impairments.

In general, we find that damages are geographically heterogeneous and so spatially targeted interventions may have high returns. This finding is intuitive and follows similar patterns found in local air pollutants (Muller and Mendelsohn 2007, 2009). However, we also highlight unique features of GED for water pollutants. Damages are highly concentrated, with emissions into just 20% of watersheds generating 80% of total GED. We also find that meaningful shares of pollution damages are borne by individuals that live within hydrologic and political boundaries that are distinct from the sources of pollution. This finding provides support for a robust federal role in managing these pollutants. Finally, we note that marginal damages are higher when pollution is emitted in smaller, headwater streams and lower in larger streams.

Our work building HydroEcon has generated several insights about the economics of surface water quality. Our analysis of decades of water recreation demand and property value case studies reveals a remarkable consistency in estimated elasticities of trips and home prices with respect to water quality. These values are consistent whether we link recreation and housing directly to nutrients or to some other mediating variable, such as water clarity. We confirm that recreation is the primary damage generating mechanism, owing to the large number of people who visit freshwater locations. Amenity impacts are limited by the small fraction of residential homes that are near enough to waterbodies to capitalize water quality changes. Likewise, the small size of drinking water treatment costs arises from the comparatively small number of surface water sourcing plants that treat

[27] This figure reflects GDP in 2012, which is the baseline year for our emissions estimates.

nitrogen and the relatively low variable cost of doing so. We emphasize, however, that our exclusion of groundwater treatment costs may contribute to our low estimate. Finally, we lay the groundwork for linking nutrients to human health within a national IAM, though we are currently unable to measure all health benefits.

We see several important directions for future work. First, optimal pollution policy requires setting marginal damages equal to marginal abatement costs. One natural extension would be to consider these abatement options across space to find spatially-optimal levels of nutrient pollution. This information could inform a host of policy interventions, ranging from Clean Water Act programs that target wastewater treatment plants to USDA conservation programs, such as the Conservation Reserve Program and the Environmental Quality Incentives Program. The U.S. spends billions of dollars each year on abatement efforts. Knowledge of the corresponding benefits could inform optimal policy design.

Another fruitful area of research may be to consider how damages from particular sources vary for water versus air pollution. By combining this information on damages, policy makers and other stakeholders will have a more comprehensive picture of the benefits of controlling particular sources of pollution. In addition, our damage estimates reflect recreation, housing, drinking water treatment, infant birthweight, and climate change impacts. Unmeasured channels of benefits, such as the broader health impacts of exposure to nutrient pollution, could play an important role as future studies advance our understanding of how nutrient pollution impacts ecological and human health.

Finally, though nutrient pollution is a leading cause of impairments in U.S. waterways, there are tens of thousands of U.S. chemicals emitted into the environment each year. A natural next step would be to model and measure the economic damages of other key pollutants, such as bacteria and biochemical oxygen demand, among others.

References

Abraham, Katharine G., and Christopher Mackie. 2006. "A Framework for Nonmarket Accounting." *A New Architecture for the U.S. National Accounts.*

Alexander, Richard B., Richard A. Smith, Gregory E. Schwarz, Elizabeth W. Boyer, Jacqueline V. Nolan, and John W. Brakebill. 2008. "Differences in Phosphorus and Nitrogen Delivery to The Gulf of Mexico from the Mississippi River Basin." *Environmental Science & Technology* 42 (3): 822–830.

Arrow, Kenneth J., Partha Dasgupta, Larence H. Goulder, Kevin J. Mumford, and Kirsten Oleson. 2012. "Sustainability and the Measurement of Wealth." *Environment and Development Economics* 17(3):317–353.

Ayres, Andrew B., Kyle C. Meng, and Andrew J. Plantinga. 2021. "Do Environmental Markets Improve on Open Access? Evidence from California Groundwater Rights." *Journal of Political Economy* 129 (10): 2817–2860.

Bagstad, Kenneth J., Jane Carter Ingram, Carl D. Shapiro, Alessandra La Notte, Joachim Maes, Sara Vallecillo, C. Frank Casey, Pierre D. Glynn, Mehdi P. Heris, Justin A. Johnson, Chris Lauer, John Matuszak, Kirsten L.L. Olseon, Stephen M. Posner, Charles Rhodes, and Brian Voigt. 2021. "Lessons Learned from Development of Natural Capital Accounts in the United States and European Union." *Ecosystem Services* 52:101359.

Bateman, Ian J., Bonnie Keeler, Sheila M. Olmstead, and John Whitehead. 2023. "Perspectives on Valuing Water Quality Improvements Using Stated Preference Methods." *Proceedings of the National Academy of Sciences* 120(18):e2217456120.

Baumol, William J., and Wallace E. Oates. 1988. *The Theory of Environmental Policy.* Cambridge University Press.

Beaulieu, Jake J, Tonya DelSontro, and John A Downing. 2019. "Eutrophication Will Increase Methane Emissions from Lakes and Impoundments during the 21st Century." *Nature communications* 10 (1): 1375.

Beaulieu, Marieke, Frances Pick, and Irene Gregory-Eaves. 2013. "Nutrients and Water Temperature Are Significant Predictors of Cyanobacterial Biomass in a 1147 Lakes Data Set." *Limnology and Oceanography* 58 (5): 1736–1746.

Bennear, Lori S., Katrina K. Jessoe, and Sheila M. Olmstead. 2009. "Sampling Out: Regulatory Avoidance and the Total Coliform Rule." *Environmental Science and Technology* 43 (14): 5176–5182.

Bharadwaj, Prashant, Petter Lundborg, and Dan-Olof Rooth. 2018. "Birth Weight in the Long Run." *Journal of Human Resources* 53 (1): 189–231.

Bingham, Tayler H., Timothy R. Bondelid, Brooks M. Depro, Ruth C. Figueroa, A. Brett Hauber, Suzanne J. Unger, and George L. Van Houtven. 2000. *A Benefits Assessment of Water Pollution Control Programs Since 1972: Part 1, The Benefits of Point Source Controls for Conventional Pollutants in Rivers and Streams: Final Report.* Technical

report. Washington DC: Prepared by Research Triangle Institute for the Office of Water, Office of Policy, Economics, and Innovation, U.S. Environmental Protection Agency.

Bishop, Kelly C., Jakob Dowling, Nicolai V. Kuminoff, and Alvin Murphy. 2025. "Tax Policy and the Heterogeneous Costs of Homeownership." *Accepted at Journal of Urban Economics.*

Bockstael, Nancy E., Michael W. Hanemann, and Catherine L. Kling. 1987. "Estimating the Value of Water Quality Improvements in a Recreation Demand Framework." *Water Resources Research.*

Carleton, Tamma, Amir Jina, Michael Delgado, Michael Greenstone, Trevor Houser, Solomon Hsiang, Andrew Hultgren, Robert E Kopp, Kelly E. McCusker, Ishan Nath, James Rising, Ashwin Rode, Hee Kwon Seo, Arvid Viaene, Jiacan Yuan, and Alice Tianbo Zhang. 2022. "Valuing the Global Mortality Consequences of Climate Change Accounting for Adaptation Costs and Benefits." *Quarterly Journal of Economics* 137 (4): 2037–2105.

Carson, Richard T., and Robert Cameron Mitchell. 1993. "The Value of Clean Water: The Public's Willingness to Pay for Boatable, Fishable, and Swimmable Quality Water." *Water Resources Research* 29(7):2445–2454.

Chan, Nathan W., and Casey J. Wichman. 2022. "Valuing Nonmarket Impacts of Climate Change on Recreation: From Reduced Form to Welfare." *Environmental and Resource Economics* 81(1):179–213.

Christensen, Peter, David A. Keiser, and Gabriel E. Lade. 2023. "Economic Effects of Environmental Crises: Evidence from Flint, Michigan." *American Economic Journal: Economic Policy* 15 (1): 196–232.

Clark, Edwin H. 1985. "The Off-Site Costs of Soil Erosion." *Journal of Soil and Water Conservation* 40 (1): 19–22.

Cohen, Alex, and David A. Keiser. 2017. "The Effectiveness of Incomplete and Overlapping Pollution Regulation: Evidence from Bans on Phosphate in Automatic Dishwasher Detergent." *Journal of Public Economics* 150:53–74.

Cropper, Maureen L., and Yongjoon Park. 2025. "Measuring and Accounting for Environmental Public Goods: A National Accounts Perspective." Chap. Incorporating Air and Water Pollution into the National Income and Product Accounts, edited by Nicholas Z. Muller, Eli P. Fenichel, and Mary Bohman, 15–34. University of Chicago Press.

Cullen, Joseph. 2013. "Measuring the Environmental Benefits of Wind-Generated Electricity." *American Economic Journal: Economic Policy* 5(4):107–133.

Currie, Janet, Joshua Graff Zivin, Katherine Meckel, Matthew Neidell, and Wolfram Schlenker. 2013. "Something in the water: contaminated drinking water and infant health." *Canadian Journal of Economics* 46 (3): 791–810.

Del Rossi, Gemma, Mohammad Mainul Hoque, Yongjie Ji, and Catherine L Kling. 2023. "The Economics of Nutrient Pollution from Agriculture." *Annual Review of Resource Economics* 15 (1): 105–130.

DelSontro, Tonya, Jake J Beaulieu, and John A Downing. 2018. "Greenhouse Gas Emissions from Lakes and Impoundments: Upscaling in the Face of Global Change." *Limnology and Oceanography Letters* 3 (3): 64–75.

Desvousges, William H., V. Kerry Smith, and Ann Fisher. 1987. "Option Price Estimates for Water Quality Improvements: A Contingent Valuation Study for the Monongahela River." *Journal of Environmental Economics and Management* 14(3):248–267.

Dias, Mateus, Rudi Rocha, and Rodrigo R. Soares. 2023. "Down the River: Glyphosate Use in Agriculture and Birth Outcomes of Surrounding Populations." *The Review of Economic Studies* 90 (6): 2943–2981.

DiSalvo, Richard W., and Elaine L. Hill. 2024. "Drinking Water Contaminant Concentrations and Birth Outcomes." *Journal of Policy Analysis and Management* 43 (2): 368–399.

Dodds, Walter K., Wes W. Bouska, Jeffrey L. Eitzmann, Tyler J. Pilger, Kristen L. Pitts, Alyssa J. Riley, Joshua T. Schloesser, and Darren J. Thornbrugh. 2009. "Eutrophication of U.S. Freshwaters: Analysis of Potential Economic Damages." *Environmental Science and Technology* 43(1):12–19.

Downing, John A., Stephen Polasky, Sheila M. Olmstead, and Stephen C. Newbold. 2021. "Protecting Local Water Quality has Global Benefits." *Nature Communications* 12 (1): 2709.

Doyle, Martin W., Lauren A. Patterson, Yanyou Chen, Kurt E. Schnier, and Andrew J. Yates. 2014. "Optimizing the Scale of Markets for Water Quality Trading." *Water Resources Research* 50 (9): 7231–7244.

Egan, Kevin J., Joseph A. Herriges, Catherine L. Kling, and John A. Downing. 2009. "Valuing Water Quality as a Function of Water Quality Measures." *American Journal of Agricultural Economics* 91(1):106–123.

Evans, Keith S., Joseph A. Herriges, and Catherine L. Kling. 2011. *Iowa Lakes Survey 2009: A Report to the Iowa Department of Natural Resources.* Technical report.

Farrow, Scott R., Martin T. Schultz, Pinar Celikkol, and George L.Van Houtven. 2005. "Pollution Trading in Water Quality Limited Areas: Use of Benefits Assessment and Cost-Effective Trading Ratios." *Land Economics* 81 (2): 191–205.

Fell, Harrison, Daniel T. Kaffine, and Kevin Novan. 2021. "Emissions, Transmission, and the Environmental Value of Renewable Energy." *American Economic Journal: Economic Policy* 13(2):241–272.

Fenichel, Eli, Joshua Abbott, Jude Bayham, Whitney Boone, Erin Haacker, and Lisa Pfieffer. 2016. "Measuring the Value of Groundwater and other Forms of Natural Capital." *Proceedings of the National Academy of Sciences* 113(9):2382–2387.

Fenichel, Eli P. 2024. "A New Era of Economic Measurement for the Environment and Natural Capital." *Review of Environmental Economics and Policy* 18(2):321–330.

Fenichel, Eli P., and Joshua K. Abbott. 2014. "Natural Capital: From Metaphor to Measurement." *Journal of the Association of Environmental and Resource Economists* 1:1–27.

Flynn, Patrick, and Michelle Marcus. 2023. "A Watershed Moment: The Clean Water Act and Birth Weight." *Journal of Human Resources.*

Fowlie, Meredith, Michael Greenstone, and Catherine Wolfram. 2018. "Do Energy Efficiency Investments Deliver? Evidence from the Weatherization Assistance Program." *The Quarterly Journal of Economics* 133(3):1597–1644.

Fowlie, Meredith, and Nicholas Muller. 2019. "Market-based Emissions Regulation When Damages Vary Across Sources: What are the Gains from Differentiation?" *Journal of the Association of Environmental and Resource Economists* 6(3):593–632.

Freeman, Myrick A. 1982. *Air and Water Pollution Control.* John Wiley & Sons.

Garnache, Cloé, Scott M. Swinton, Joseph A. Herriges, Frank Lupi, and R. Jan Stevenson. 2016. "Solving the Phosphorus Pollution Puzzle: Synthesis and Directions for Future Research." *American Journal of Agricultural Economics* 98 (5): 1334–1359.

Greenhill, Simon, Hannah Druckenmiller, Sherrie Wang, David A. Keiser, Manuela Girotto, Jason K. Moore, Nobuhiro Yamaguchi, Alberto Todeschini, and Joseph S. Shapiro. 2024. "Machine Learning Predicts which Rivers, Streams, and Wetlands the Clean Water Act Regulates." *Science* 383 (6681): 406–412.

Greenstone, Michael, and Rema Hanna. 2014. "Environmental Regulations, Air and Water Pollution, and Infant Mortality in India." *American Economic Review* 104 (10): 3038–3072.

Griffiths, Charles, Heather Klemick, Matt Massey, Chris Moore, Steve Newbold, David Simpson, Patrick Walsh, and William Wheeler. 2012. "U.S. Environmental Protection Agency Valuation of Surface Water Quality Improvements." *Review of Environmental Economics and Policy* 6(1):130–146.

Guignet, Dennis, Matthew T. Heberling, Michael Papenfus, and Olivia Griot. 2022. "Property Values, Water Quality, and Benefit Transfer: A Nationwide Meta-Analysis." *Land Economics* 98(2):198–218.

Guiteras, Raymond, James Levinsohn, and Ahmed Mushfiq Mobarak. 2015. "Encouraging Sanitation Investment in the Developing World: A Cluster-randomized Trial." *Science* 348 (6237): 903–906.

Guvenen, Fatih, Greg Kaplan, Jae Song, and Justin Weidner. 2022. "Lifetime Earnings in the United States over Six Decades." *American Economic Journal: Applied Economics* 14 (4): 446–479.

Hansen, LeRoy, and Marc Ribaudo. 2008. *Economic Measures of Soil Conservation Benefits: Regional Values for Policy Assessment.* Technical report. USDA.

He, Guojun, Shaoda Wang, and Bing Zhang. 2020. "Watering Down Environmental Regulation in China." *Quarterly Journal of Economics* 135 (4): 2135–2185.

Holland, Stephen P., Erin T. Mansur, Nicholas Z. Muller, and Andrew J. Yates. 2016. "Are there Environmental Benefits from Driving Electric Vehicles? The Importance of Local Factors." *American Economic Review* 106(12):3700–3729.

Hollingsworth, Alex, Taylor Jaworski, Carl Kitchens, and Ivan Rudik. 2024. "Economic Geography and Air Pollution Regulation in the United States." Cornell Working Paper.

Hsiang, Solomon, Paulina Oliva, and Reed Walker. 2019. "The Distribution of Environmental Damages." *Review of Environmental Economics and Policy* 13(1):83–103.

Huisman, Jef, Geoffrey A. Codd, Hans W. Paerl, Bas W. Ibelings, Jolanda M. H. Verspagen, and Petra M. Visser. 2018. "Cyanobacterial Blooms." *Nature Reviews Microbiology* 16 (8): 471–483.

Jones, Benjamin A. 2019. "Infant Health Impacts of Freshwater Algal Blooms: Evidence from an Invasive Species Natural Experiment." *Journal of Environmental Economics and Management* 96:36–59.

Kahn, Matthew E., Pei Li, and Kaxuan Zhao. 2015. "Water Pollution Progress at Borders: The Role of Changes in China's Political Propotion Incentives." *American Economic Journal: Economic Policy* 7 (4): 223–242.

Keeler, Bonnie L., Jesse D. Gourevitch, Stephen Polasky, Forest Isbell, Chris W. Tessum, Jason D. Hill, and Julian D. Marshall. 2016. "The Social Costs of Nitrogen." *Science Advances* 2(10):e1600219.

Keeler, Bonnie L., Stephen Polasky, Kate A. Brauman, Kris A. Johnson, Jacques C. Finlay, Ann O'Neill, Kent Kovacs, and Brent Dalzell. 2012. "Linking Water Quality and Well-Being for Improved Assessment and Valuation of Ecosystem Services." *Proceedings of the National Academy of Sciences of the United States of America* 109(45):18619–18624.

Keiser, David A. 2019. "The Missing Benefits of Clean Water and the Role of Mismeasured Pollution Data." *Journal of the Association of Environmental and Resource Economists* 6(4):669–707.

Keiser, David A., Catherine L. Kling, and Joseph S. Shapiro. 2019. "The Low But Uncertain Measured Benefits of US Water Quality Policy." *Proceedings of the National Academy of Sciences* 116(12):5262–5269.

Keiser, David A., Bhash Mazumder, David Molitor, and Joseph S. Shapiro. 2023. "Water Works: Causes and Consequences of Safe Drinking Water in America." *Working Paper.*

Keiser, David A., and Nicholas Z. Muller. 2017. "Air and Water: Integrated Assessment Models for Multiple Media." *Annual Review of Resource Economics* 9:165–184.

Keiser, David A., Sheila M. Olmstead, Kevin J. Boyle, Victor B. Flatt, Bonnie L. Keeler, Catherine L. Kling, Daniel J. Phaneuf, Joseph S. Shapiro, and Jay P. Shimshack. 2021.

"A Water Rule that Turns a Blind Eye to Transboundary Pollution." *Science* 372 (6539): 241–243.

Keiser, David A., and Joseph S. Shapiro. 2019a. "Consequences of the Clean Water Act and the Demand for Water Quality." *Quarterly Journal of Economics* 134(1):349–396.

———. 2019b. "U.S. Water Pollution Regulation Over the Past Half Century: Burning Waters to Crystal Springs?" *Journal of Economic Perspectives* 33(4):51–75.

Kremer, Michael, Jessica Leino, Edward Miguel, and Alix Peterson Zwane. 2011. "Spring Cleaning: Rural Water Impacts, Valuation, and Property Rights Institutions." *Quarterly Journal of Economics* 126 (1): 145–205.

Kuwayama, Yusuke, Sheila Olmstead, and Jiameng Zheng. 2022. "A More Comprehensive Estimate of the Value of Water Quality." *Journal of Public Economics* 207:104600.

Lad, Apurva, Joshua D. Breidenbach, Robin C. Su, Jordan Murray, Rebecca Kuang, Alison Mascarenhas, John Najjar, Shivani Patel, Prajwal Hegde, Mirella Youssef, Jason Breuler, Andrew L. Kleinhenz, Andrew P. Ault, Judy A. Westrick, Nikolai N. Modyanov, David J. Kennedy, and Steven T. Haller. 2022. "As We Drink and Breathe: Adverse Health Effects of Microcystins and Other Harmful Algal Bloom Toxins in the Liver, Gut, Lungs and Beyond." *Life* 12 (3): 418.

Leggett, Christopher G., and Nancy E. Bockstael. 2000. "Evidence of the Effects of Water Quality on Residential Land Prices." *Journal of Environmental Economics and Management* 39:121–144.

Lim, Chris C., Jeonggyo Yoon, Kelly Reynolds, Lynn B. Gerald, Andrew P. Ault, Seulkee Heo, and Michelle L. Bell. 2023. "Harmful Algal Bloom Aerosols and Human Health." *eBioMedicine* 93:104604.

Liu, Pengfei, Yu Wang, and Wei Zhang. 2023. "The Influence of the Environmental Quality Incentives Program on Local Water Quality." *American Journal of Agricultural Economics* 105 (1): 27–51.

Mamun, Saleh, Adriana Castillo-Castillo, Kristen Swedberg, Jiarui Zhang, Kevin J. Boyle, Diego Cardoso, Catherine L. Kling, Christoph Nolte, Michael Papenfus, Daniel Phaneuf, and Stephen Polasky. 2023. "Valuing Water Quality in the United States Using a National Dataset on Property Values." *Proceedings of the National Academy of Sciences* 120 (15): e2210417120.

McCauley, Edward, John A Downing, and Susan Watson. 1989. "Sigmoid Relationships Between Nutrients and Chlorophyll among Lakes." *Canadian Journal of Fisheries and Aquatic Sciences* 46 (7): 1171–1175.

Meyer, Andrew, and Zach Raff. 2024. "The Effect of Water Pollution Regulation on Prices: Evidence from Wisconsin's Phosphorus Rule and Sewer Utility Bills." *Journal of Environmental Economics and Management* 124:102914.

Mosheim, Roberto, and Marc Ribaudo. 2017. "Costs of Nitrogen Runoff for Rural Water Utilities: A Shadow Cost Approach." *Land Economics* 93 (1): 12–39.

Muller, Nicholas Z., Eli P. Fencichel, and Mary Bohman. 2025. "Measuring and Accounting for Environmental Public Goods: A National Accounts Perspective." Chap. Introduction, 1–14. University of Chicago Press.

Muller, Nicholas Z., and Robert Mendelsohn. 2007. "Measuring the Damages of Air Pollution in the United States." *Journal of Environmental Economics and Management* 54(1):1–14.

———. 2009. "Efficient Pollution Regulation: Getting the Prices Right." *American Economic Review* 99(5):1714–1739.

Muller, Nicholas Z., Robert Mendelsohn, and William Nordhaus. 2011. "Environmental Accounting for Pollution in the United States Economy." *American Economic Review* 101(5):1649–1675.

Nordhaus, William D., and James Tobin. 1973. "The Measurement of Economic and Social Performance." Chap. Is Growth Obsolete?, edited by Milton Moss, 509–564. NBER.

NRC. 1999. *Nature's Numbers: Expanding the National Economic Accounts to Include the Environment.* National Academies Press.

Olmstead, Sheila M. 2010. "The Economics of Water Quality." *Review of Environmental Economics & Policy* 4(1):44–61.

Ouyang, Zhiyun, Changsu Song, Hua Zheng, Stephen Polasky, Yi Xiao, Ian J. Bateman, Jianguo Liu, Mary Ruckelshaus, Faqi Shi, Yang Xiao, Weihua Xu, Ziying Zou, and Grechen C. Daily. 2020. "Using Gross Ecosystem Product (GEP) to Value Nature in Decision Making." *Proceedings of the National Academy of Sciences* 117(25):14593–14601.

Paudel, Jayash, and Christine L. Crago. 2021. "Environmental Externalities from Agriculture: Evidence from Water Quality in the United States." *American Journal of Agricultural Economics* 103 (1): 185–210.

Phaneuf, Daniel J., Catherine L. Kling, and Joseph A. Herriges. 2000. "Estimation and Welfare Calculations in a Generalized Corner Solution Model with an Application to Recreation Demand." *The Review of Economics and Statistics* 82(1):83–92.

Phaneuf, Daniel J., and Till Requate. 2016. *A Course in Environmental Economics: Theory, Policy, and Practice.* Cambridge University Press.

Phaneuf, Daniel J., V. Kerrty Smith, Raymond B. Palmquist, and Jaren C. Pope. 2008. "Integrating Property Value and Local Recreation Models to Value Ecosystem Services in Urban Watersheds." *Land Economics* 84(3):361–381.

Pilotto, Louis S., Erich V. Kliewer, Robyn D. Davies, Michael D. Burch, and Robyn G. Attewell. 1999. "Cyanobacterial (Blue-Green Algae) Contamination in Drinking Water and Perinatal Outcomes." *Australian and New Zealand Journal of Public Health* 23 (2): 154–158.

Polasky, Stephen, Benjamin Bryant, Peter Hawthorne, Justin Johnson, Bonnie Keeler, and Derric Pennington. 2015. "Inclusive Wealth as a Metric of Sustainable Development." *Inclusive Wealth as a Metric of Sustainable Development* 40:445–466.

Rabotyagov, Sergey S., Todd D. Campbell, Michael White, Jeffrey G. Arnold, Jay Atwood, M. Lee Norfleet, Catherine L. Kling, Philip W. Gassman, Adriana Valcu, Jeffrey Richardson, R. Eugene Turner, and Nancy N. Rabalais. 2014. "Cost-effective Targeting of Conservation Investments to Reduce the Northern Gulf of Mexico Hypoxic Zone." *Proceedings of the National Academy of Sciences* 111(52):18530–18535.

Rafey, Will. 2023. "Droughts, Deluges, and (River) Diversions: Valuing Market-Based Water Reallocation." *American Economic Review* 113 (2): 430–471.

Raff, Zach, and Andrew Meyer. 2022. "CAFOs and Surface Water Quality: Evidence from Wisconsin." *American Journal of Agricultural Economics* 104 (1): 161–189.

Ribaudo, Marc O. 1986. *Reducing Soil Erosion: Offsite Benefits.* Technical report. USDA.

Robertson, Dale M., and David A. Saad. 2019. "Spatially Referenced Models of Streamflow and Nitrogen, Phosphorus, and Suspended Sediment Loads in Streams of the Midwestern United States." U.S. Geological Survey Scientific Investigations Report No. 2019-5114.

———. 2021. "Nitrogen and Phosphorus Sources and Delivery from the Mississippi/Atchafalaya River Basin: An Update Using 2012 SPARROW Models." *Journal of the American Water Resources Association* 57 (3): 406–429.

Robertson, Dale M., David A. Saad, and Gregory E. Schwarz. 2014. "Spatial Variability in Nutrient Transport by HUC 8, State, and Subbasin Based on Mississippi/Atchafalaya River Basin SPARROW Models." *Journal of the American Water Resources Association* 50 (4): 988–1009.

Rockoff, Hugh. 2020. "Off to a Good Start: The NBER and the Measurement of National Income." *NBER Working Paper Series.*

Rosentreter, Judith A, Alberto V Borges, Bridget R Deemer, Meredith A Holgerson, Shaoda Liu, Chunlin Song, John Melack, Peter A Raymond, Carlos M Duarte, George H Allen, et al. 2021. "Half of Global Methane Emissions Come from Highly Variable Aquatic Ecosystem Sources." *Nature Geoscience* 14 (4): 225–230.

Russell, Rebecca B., Nancy S. Green, Claudia A. Steiner, Susan Meikle, Jennifer L. Howse, Karalee Poschman, Todd Dias, Lisa Potetz, Michael J. Davidoff, Karla Damus, and Joann R. Petrini. 2007. "Cost of Hospitalization for Preterm and Low Birth Weight Infants in the United States." *Pediatrics* 120 (1): e1–e9.

Saad, David A., Gregory E. Schwarz, Denise M. Argue, David W. Anning, Scott A. Ator, Anne B. Hoos, Stephen D. Preston, Dale M. Robertson, and Daniel Wise. 2019. "Estimates of Long-Term Mean Daily Streamflow and Annual Nutrient and Suspended-sediment Loads Considered for Use in Regional SPARROW Models of the Conterminous United States, 2012 Base Year." U.S. Geological Survey Scientific Investigations Report No. 2019-5069.

Sarnelle, Orlando, Alan E. Wilson, Stephen K. Hamilton, Lesley B. Knoll, and David F. Raikow. 2005. "Complex Interactions Between the Zebra Mussel, *Dreissena polymorpha*, and the Harmful Phytoplankter, *Microcystis aeruginosa*." *Limnology and Oceanography* 50 (3): 896–904.

Skidmore, Marin, Tihitina Andarge, and Jeremy Foltz. 2023. "Effectiveness of Local Regulations on Nonpoint Source Pollution: Evidence from Wisconsin Dairy Farms." *American Journal of Agricultural Economics* 105 (5): 1333–1364.

Smith, Richard A., Gregory E. Schwarz, and Richard B. Alexander. 1997. "Regional Interpretation of Water-Quality Monitoring Data." *Water Resources Research* 33 (12): 2781–2798.

Taylor, Charles A., and Hannah Druckenmiller. 2022. "Wetlands, Flooding, and the Clean Water Act." *American Economic Review* 112 (4): 1334–1363.

The White House. 2023. *Fact Sheet: Biden-Harris Administration Releases National Strategy to Put Nature on the Nation's Balance Sheet.* Technical report.

The World Bank. 2024. "The Changing Wealth of Nations: Revisiting the Measurement of Comprehensive Wealth."

Tschofen, Peter, Ines Azevedo, and Nicholas Z. Muller. 2019. "Fine Particulate Matter Damages and Value Added in the US Economy." *Proceedings of the National Academy of Sciences* 116(40):19857–19862.

University of Wisconsin Population Health Institute. 2023. *County Health Rankings & Roadmaps.* https://www.countyhealthrankings.org/health-data/methodology-and-sources/data-documentation/national-data-documentation-2010-2023.

USBLS. 2025. *CPI Inflation Calculator.* https://www.bls.gov/data/inflation_calculator.htm.

USEPA. 2007. "National Lakes Assessment." http://water.epa.gov/type/lakes/lakessurvey_index.cfm.

———. 2012. "National Lakes Assessment." http://water.epa.gov/type/lakes/lakessurvey_index.cfm.

———. 2015. *Benefit and Cost Analysis for the Effluent Limitations Guidelines and Standards for the Steam Electric Power Generating Point Source Category.* Technical report. Environmental Protection Agency.

———. 2017. "National Lakes Assessment." http://water.epa.gov/type/lakes/lakessurvey_index.cfm.

———. 2018. "National Rivers and Streams Assessment." https://www.epa.gov/national-aquatic-resource-surveys/national-river-and-streams-assessment-2018-19-key-findings.

———. 2022a. "National Lakes Assessment." https://nationallakesassessment.epa.gov/webreport/.

USEPA. 2022b. "Report on the Social Cost of Greenhouse Gases: Estimates Incorporating Recent Scientific Advances."

———. 2024. "Nutrient Pollution." Accessed January 3, 2024. https://www.epa.gov/nutrientpollution.

———. 2025. *EnviroAtlas.* https://www.epa.gov/enviroatlas.

Van Houtven, George L., John Powers, and Subhrendu K. Pattanayak. 2007. "Valuing Water Quality Improvements in the United States Using Meta-Analysis: Is the Glass Half-Full or Half-Empty for National Policy Analysis?" *Resource and Energy Economics* 29:206–228.

Viscusi, W. Kip, Joel Huber, and Jason Bell. 2008. "The Economic Value of Water Quality." *Environmental and Resource Economics* 41:169–187.

Walsh, Patrick J., and J. Walter Milon. 2016. "Nutrient Standards, Water Quality Indicators, and Economic Benefits from Water Quality Regulations." *Environmental and Resource Economics* 64(4):643–661.

Walsh, Patrick J., J. Walter Milon, and David O. Scrogin. 2011. "The Spatial Extent of Water Quality Benefits in Urban Housing Markets." *Land Economics* 87(4):628–644.

Zhang, Shiyu, Xingde Du, Haohao Liu, Michael D. Losiewic, Xinghai Chen, Ya Ma, Rui Wang, Zhihui Tian, Linjia Shi, Hongxiang Guo, and Huizhen Zhang. 2021. "The Latest Advances in the Reproductive Toxicity of Microcystin-LR." *Environmental Research* 192:110254.

Zhao, Sujuan, Shengzheng Zhong, Fang Wang, Honghui Wang, Dexiang Xu, and Guangyu Li. 2020. "Microcystin-LR Exposure Decreased the Fetal Weight of Mice by Disturbance of Placental Development and ROS-Mediated Endoplasmic Reticulum Stress in the Placenta." *Environmental Pollution* 256:113362.

Table 1: Marginal and Gross External Damages by U.S. Census Region

	Marginal Damages ($/kg) (1)	Total GED ($B) (2)	In-State GED ($B) (3)	Downstream States GED ($B) (4)
Panel A: Local and Downstream Impacts				
Northeast	539.11	11.12	6.29	4.84
South	131.19	15.20	12.57	2.63
Midwest	69.61	9.78	7.23	2.54
West	531.82	16.38	15.94	0.44
U.S.	**170.55**	**52.48**	**42.03**	**10.45**
Panel B: Local Impacts				
Northeast	491.60	10.14	5.88	4.26
South	102.38	11.86	10.46	1.40
Midwest	47.73	6.71	5.30	1.41
West	477.53	14.71	14.52	0.18
U.S.	**141.10**	**43.42**	**36.16**	**7.26**

Notes: This table reports average marginal damages and Gross External Damages (GED) by U.S. Census regions. Panel A displays damages that allow for downstream fate and transport of pollution. Panel B reports the portion of damages that occur within the watershed in which a pollution source resides. Columns (1) and (2) report total average marginal damages and GED. Columns (3) and (4) report the portion of GED that occur within the state of origin versus downstream states.

Table 2: Marginal and Gross External Damages by Use Category

	Local and Downstream		Local	
	Marginal Damages ($/kg) (1)	GED ($B) (2)	Marginal Damages ($/kg) (3)	GED ($B) (4)
Use Category				
Recreation	149.40	45.92	127.0	39.06
Housing	8.03	2.47	5.21	1.60
Drinking Water Treatment	0.03	0.06	0.02	0.04
Human Health	1.47	0.45	1.21	0.37
Climate Change Impacts	11.67	3.59	7.62	2.34

Notes: This table reports average marginal damages and Gross External Damages (GED) by water use. Columns (1) and (2) report total average marginal damages and GED. Columns (3) and (4) report the portion of marginal damages and GED that occur within the watershed in which a pollution source resides.

Table 3: Marginal and Gross External Damages by Source

	Emissions (Metric Tons) (1)	Marginal Damages ($/kg) (2)	GED ($B) (3)
Non-Point Sources			
Agriculture	200,446	98.78	19.8
Urban	37,295	168.92	6.30
Point Sources			
Municipal Wastewater	65,336	380.19	24.84
Other Anthropogenic Sources			
Other	4,359	334.94	1.46

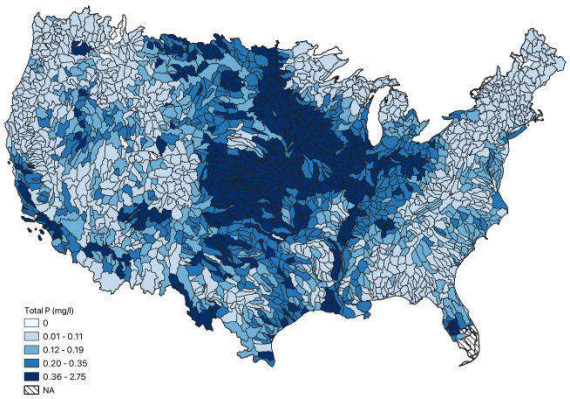
Notes: This table reports emissions of Total Phosphorus (TP) (Column 1), marginal damages (Column 2) and Gross External Damages (Column 3) by source type. Damages vary across sources based on location, not differences in pollutant types. We report emissions of TP since they are responsible for the largest share of nutrient damages in our model. Other phosphorus sources include emissions classified as mining and inter-basin transfers. We do not calculate damages for sources listed as natural, including those from channel erosion.

Table 4: Sensitivity Analysis for Marginal and Gross External Damages

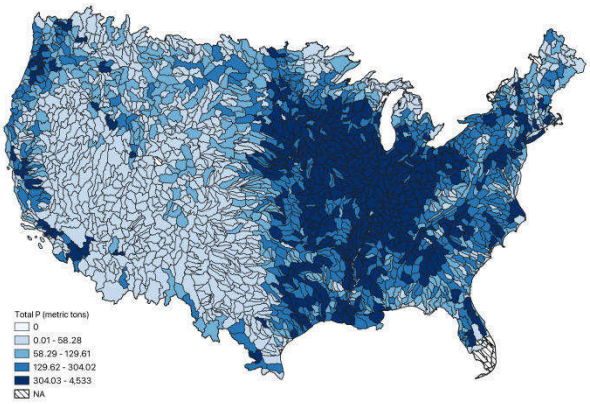
	Marginal Damages ($/kg) (1)	GED ($B) (2)
Recreation		
Baseline	149.4	45.9
Elasticity = 0.1	99.6	30.6
Elasticity = 0.2	199.1	61.2
Elasticity = 0.3	298.7	91.8
Choice Set Distance = 25 mi.	154.5	47.5
Choice Set Distance = 50 mi.	149.3	45.9
Choice Set Distance = 150 mi.	150.4	46.2
Trip Value = $108	206.8	63.6
Housing		
Baseline	8.0	2.5
Home Value-Secchi Depth Elasticity = 0.25	11.4	3.5
Secchi Depth-TP Elasticity = 0.47	8.3	2.6
Secchi Depth-TP Elasticity = 0.21	3.8	1.2
Homes within 500m	10.4	3.2
Drinking Water		
Baseline	0.03	0.06
Threshold TN concentration $\geq$ 2.87 mg/L	0.04	0.09
Doubled Variable Cost–TN Elasticity	0.05	0.11
Human Health		
Baseline	1.5	0.45
Microcystin-TP Elasticity = 0.45	1.9	0.6
Microcystin-TP Elasticity = 0.28	1.2	0.4
Impacted Population - 10 mi.	8.4	2.6
Climate Change		
Baseline	11.67	3.59
Discount Rate = 1.5%	16.25	5.0
Discount Rate = 2.5%	8.93	2.75
Methane (CH_4) Only	11.19	3.44
Total		
Baseline	170.5	52.5
Minimum	113.6	35.0
Maximum	334.8	103.0

Notes: This table displays baseline and sensitivity analyses for marginal damages and GED estimates by use category.

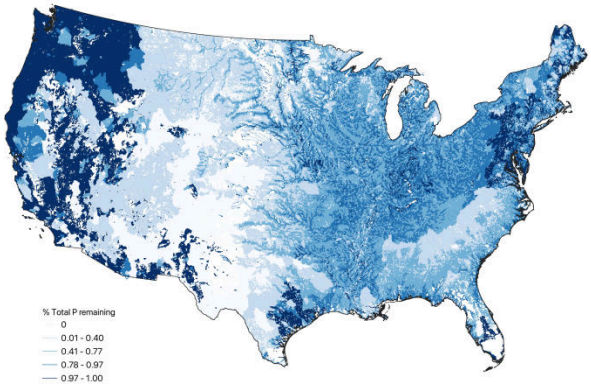
Figure 1: Baseline Water Quality Data



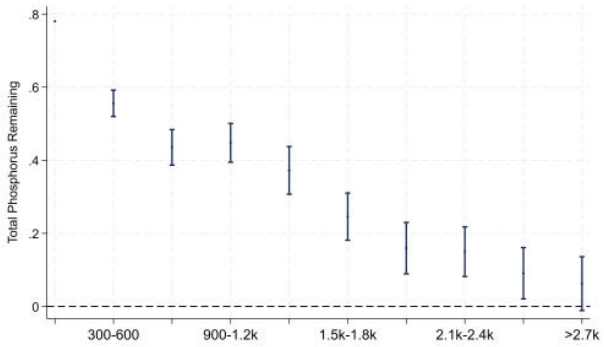
(a) Phosphorus Concentration (mg/l)



(b) Phosphorus Loads (metric tons/yr)



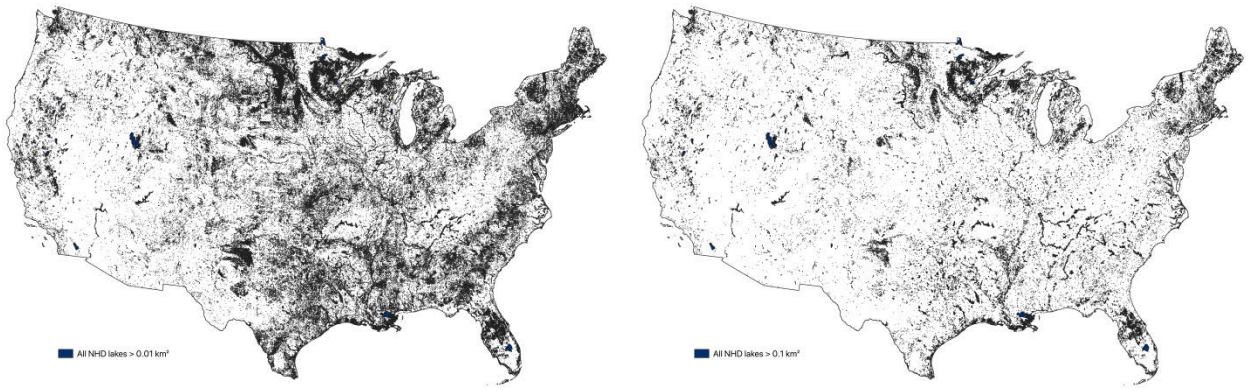
(c) Phosphorus Transported Downstream



(d) Transport of Phosphorus by Distance (km)

Notes: Panels a) and b) display baseline, long-run annual average concentrations and loads of Total Phosphorus (TP) by watershed. Darker shades of blue represent higher concentrations and greater loads. Panel c) displays the fraction of TP transported to the most downstream point from a particular river segment. Darker colors indicate less decay and lighter colors indicate more decay. Panel d) plots coefficient estimates from a regression of this fraction of TP remaining on distance to the terminal point, controlling for HUC2 and state fixed effects. Robust standard errors are clustered at the HUC8 level. Figures based on USGS SPARROW model estimates.

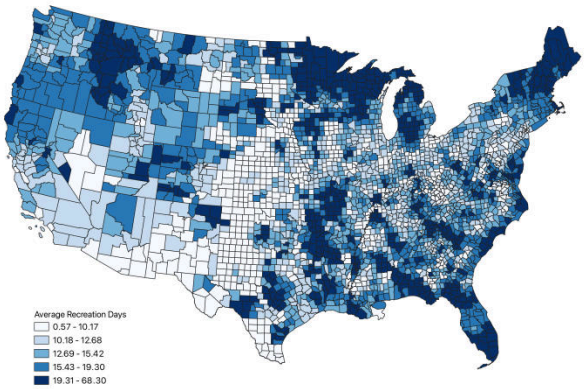
Figure 2: Location of Lakes



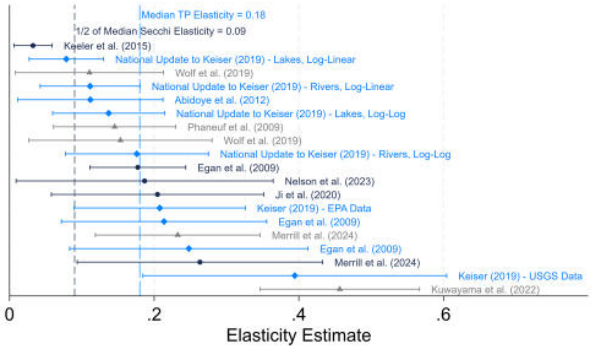
(a) Lakes > 0.01 km^2 (b) Lakes > 0.1 km^2

Notes: Panels a) and b) display lakes > 0.01 and 0.1 km^2 respectively. The baseline recreation and hedonic modules utilize lakes > 0.1 km^2. The baseline human health and climate change modules utilize lakes > 0.01 km^2.

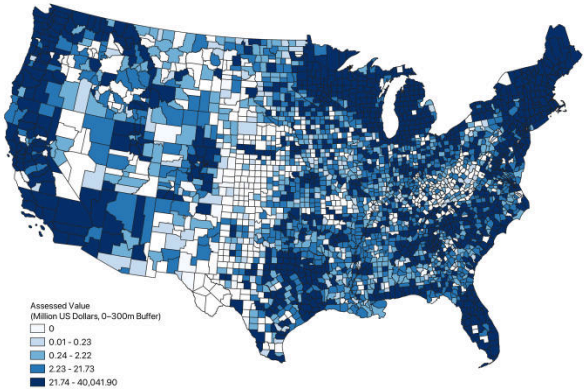
Figure 3: Recreation and Hedonic Module Components



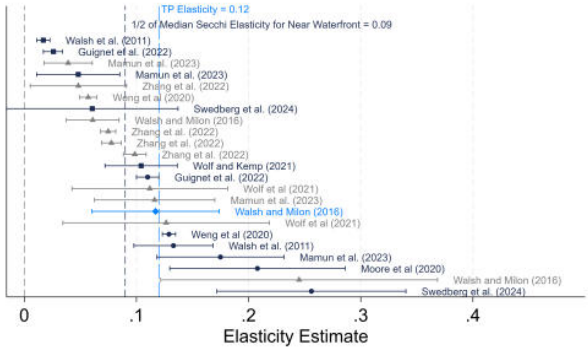
(a) Water-Based Recreation Days



(b) Recreation Trip Elasticity



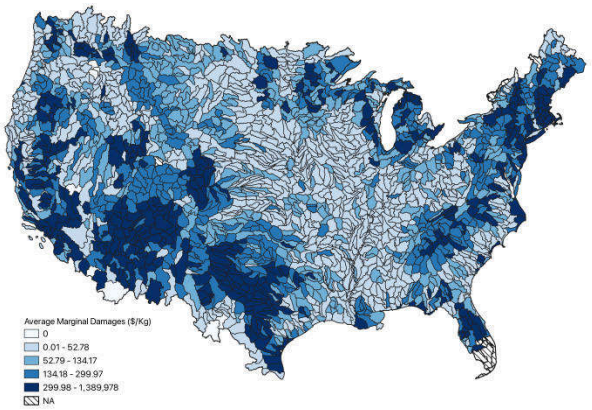
(c) Assessed Housing Values Near Lakes



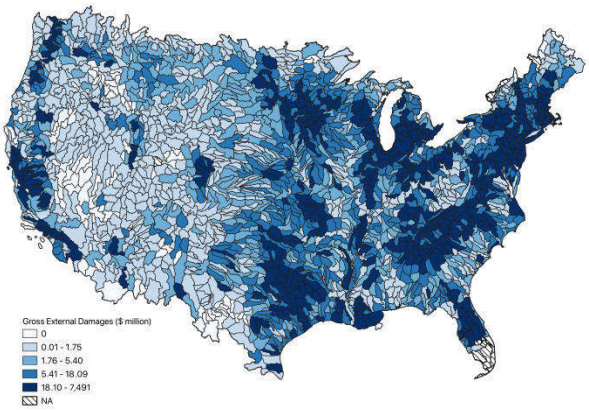
(d) Housing Price Elasticity

Notes: Panels a) and c) display baseline water-based recreation day estimates by county and assessed values of homes within 300m of a lake > 0.1 km^2. Panels b) and d) display estimates of recreation and housing price elasticities with respect to water quality. Elasticity estimates and corresponding 95% confidence intervals are shown for Total Phosphorus (light blue diamonds), Secchi depth for waterfront homes (dark blue circles), Secchi depth for non-waterfront homes (dark blue squares) and other water quality parameters (gray triangles).

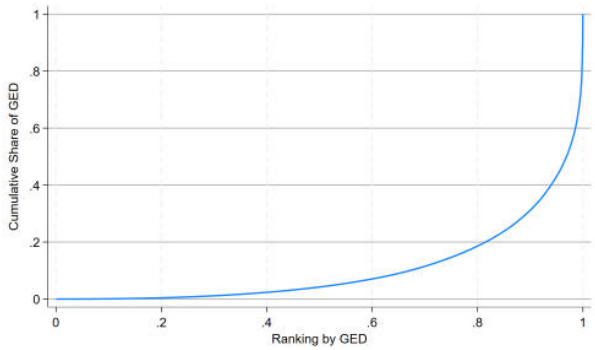
Figure 4: Marginal Damages and Gross External Damages by Watershed



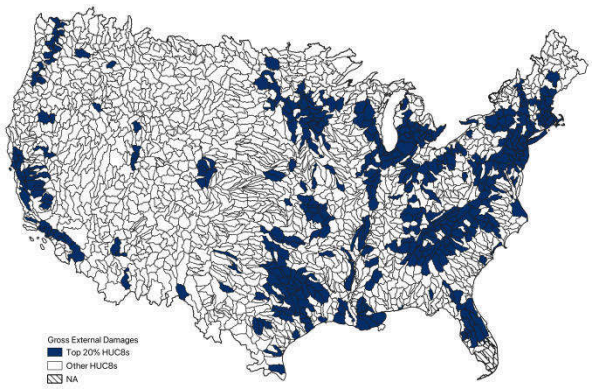
(a) Average Marginal Damages ($/kg)



(b) Gross External Damages ($M)



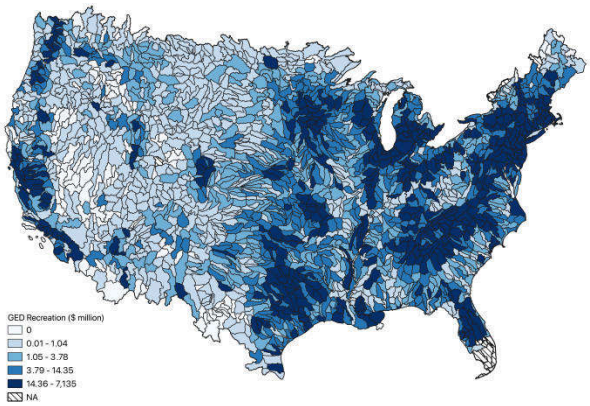
(c) Distribution of GED



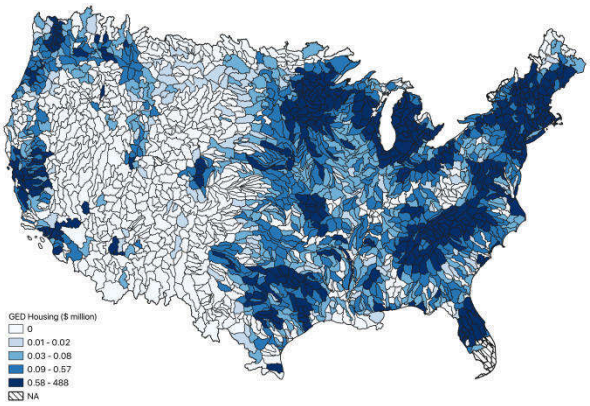
(d) GED Hot Spots

Notes: Panel a) displays average marginal damages by watershed. Panel b) displays corresponding Gross External Damages (GED). Darker shades of blue represent larger damages. Watersheds where damages are not estimated are displayed with hatched lines. Panel c) displays the cumulative share of GED (y-axis) by the ranking of watersheds according to GED (x-axis). Equal distribution of damages would follow a 45-degree line. Panel d) displays the location of the top 20% of watersheds by GED.

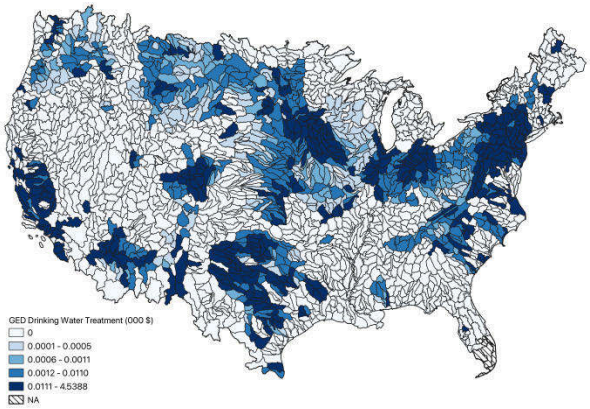
Figure 5: Gross External Damages by Use by Watershed



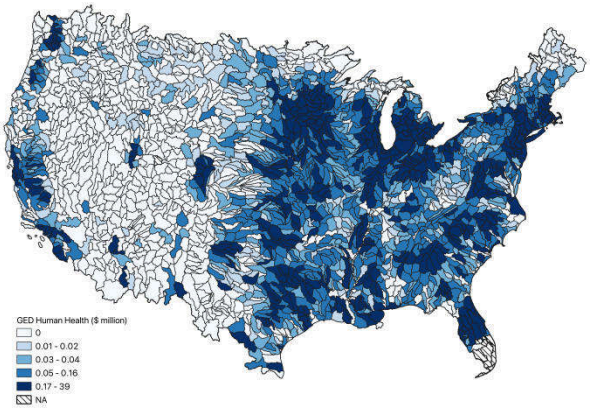
(a) Recreation



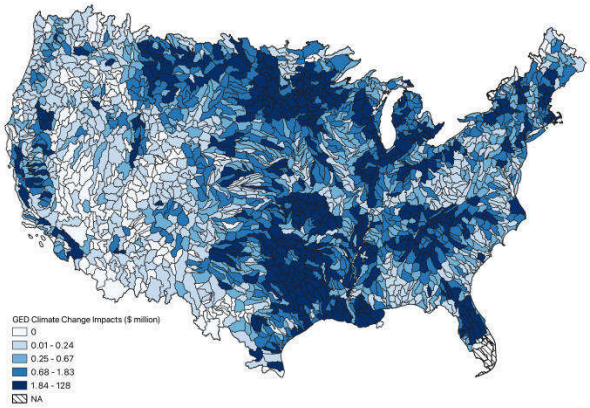
(b) Housing



(c) Drinking Water Treatment



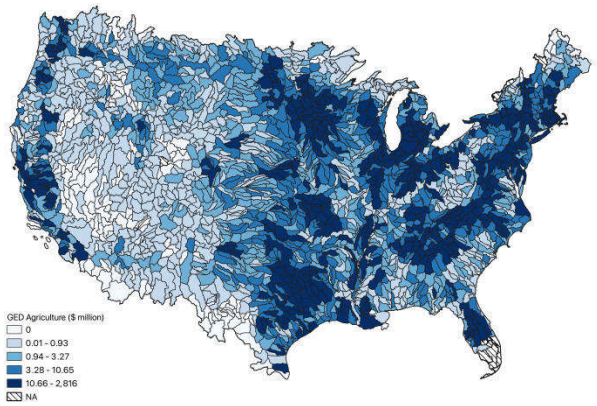
(d) Human Health



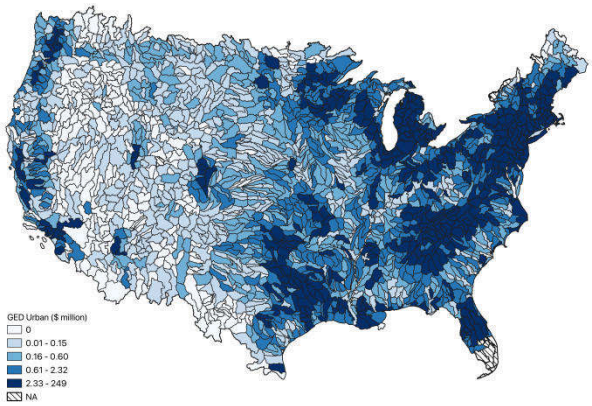
(e) Climate Change Impacts

Notes: This figure plots Gross External Damages (GED) by watershed and by use category ($M per year). Panel a) corresponds to water-based recreation damages, Panel b) displays housing damages, Panel c) captures drinking water treatment costs, Panel d) reflects human health costs and Panel e) reflects climate change impacts. Darker shades of blue represent larger damages. Watersheds where damages are not estimated are displayed with hatched lines.

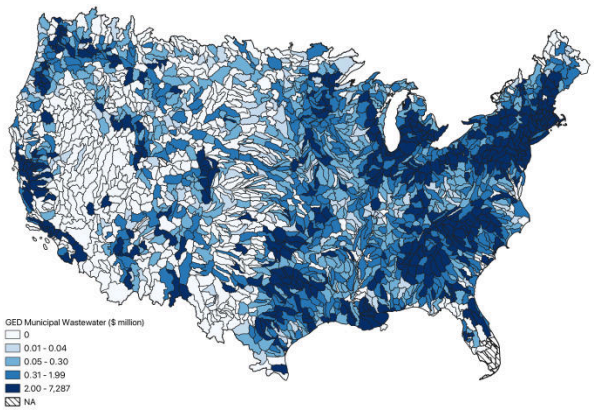
Figure 6: Gross External Damages by Source



(a) Agriculture



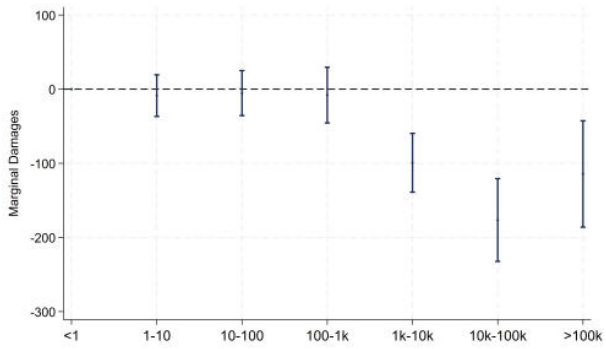
(b) Urban Non-Point Sources



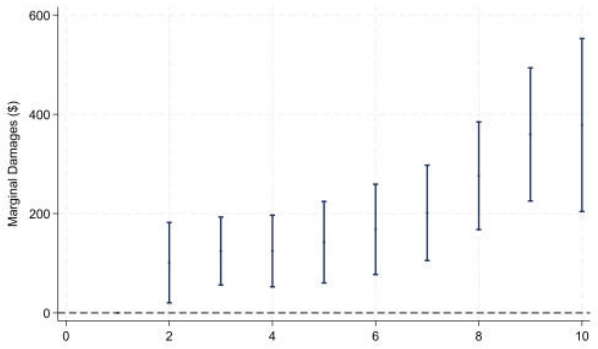
(c) Municipal Wastewater

Notes: This figure plots Gross External Damages by watershed by source ($M per year). Panel a) corresponds to damages from Agricultural non-point sources, Panel b) displays damages from Urban non-point sources, Panel c) provides estimates of damages from Municipal Wastewater. Darker shades of blue represent larger damages. Watersheds where damages are not estimated are displayed with hatched lines.

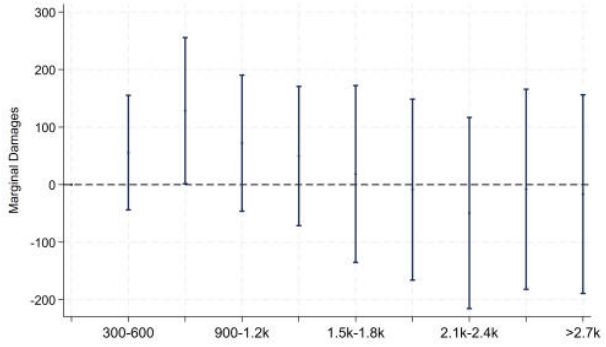
Figure 7: Variation in Marginal Damages by Hydrologic and Geographic Features



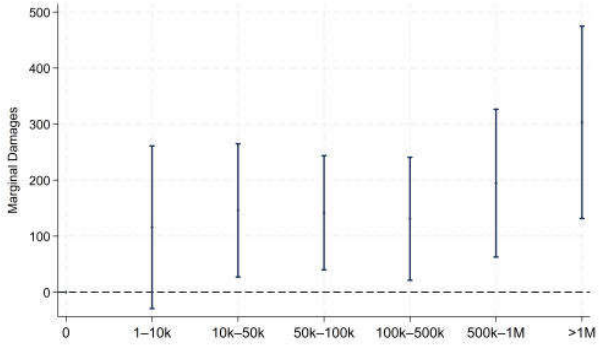
(a) River Flow



(b) Stream Level



(c) Distance to Terminal Point



(d) Population within 25 miles

Notes: This figure displays how marginal damages ($) vary by hydrologic and geographic features. Each figure plots coefficient estimates from regressions of marginal damages on a hydrologic/geologic characteristic, controlling for HUC2 and state fixed effects. Standard errors are clustered at the HUC8 level.

Appendix for:

Gross External Damages of Water Pollution in the United States

David A. Keiser, Daniel J. Phaneuf, Catherine L. Kling, Yongjie Ji, Tihitina Andarge, Mohammad Mainul Hoque, Nilesh Shinde, and Yau-Huo (Jimmy) Shr

A Conceptual Model

We start with a quasi-linear utility function for household h that is separable in numeraire consumption z_h and utility from local waterbody amenities, recreation trips to the J waterbodies, infant health, and local public goods:

$$U\left(A(d_{hn}, c_n), X_{h1}(c_1), \ldots, X_{hJ}(c_J), B_h(c_n), \zeta_n\right) + z_h. \tag{A1}$$

We model the residential choice set as a continuum from which households select their neighborhood by choosing values for local water quality (c_n) and local public goods (ζ_n). They determine their location within a neighborhood by choosing a distance to the local waterbody (d_{hn}). Households maximize utility by selecting these values and trips to the J recreation sites according to the budget constraint

$$Y_h = P(d_{hn}, c_n, \zeta_n) + \tau_n(c_n) + tc \cdot d_{hn} \cdot X_{hn} + \sum_{j \neq n} \left(tc \cdot d_{nj} \cdot X_{hj}\right), \tag{A2}$$

where Y_h is annual income, tc is the per mile cost of traveling to a recreation site, and d_{nj} is the exogenous distance from neighborhood n to waterbody j for all $j \neq n$. Households pay an annual tariff $\tau_n(c_n)$ to a local community water system (CWS) that is exogenously set to allow cost recovery for the CWS.

The first order conditions for utility maximization provide the conceptual basis for measuring the amenity and recreation costs of changes in ambient water quality. The conditions for local public goods (ζ_n) and local water quality (c_n) are

$$\frac{\partial P(\cdot)}{\partial \zeta_n} = \frac{\partial U(\cdot)}{\partial \zeta_n} \tag{A3}$$

and

$$\frac{\partial P(\cdot)}{\partial c_n} + tc \cdot d_{hn} \cdot \frac{\partial X_{hn}}{\partial c_n} - \tau_n'(c_n) = \frac{\partial U(\cdot)}{\partial A}\frac{\partial A}{\partial c_n} + \frac{\partial U(\cdot)}{\partial X_n}\frac{\partial X_{hn}}{\partial c_n} + \frac{\partial U(\cdot)}{\partial B_h}B_h'(c_n), \tag{A4}$$

respectively. To simply our presentation we assume there is no behavioral response operating through τ_n and B_h and so A4 can be reduced to

$$\frac{\partial P(\cdot)}{\partial c_n} + tc \cdot d_{hn} \cdot \frac{\partial X_{hn}}{\partial c_n} = \frac{\partial U(\cdot)}{\partial A}\frac{\partial A}{\partial c_n} + \frac{\partial U(\cdot)}{\partial X_n}\frac{\partial X_{hn}}{\partial c_n}. \tag{A5}$$

Equation A5 shows that households are willing to pay for better water quality through two channels, for two reasons. Holding d_{hn} fixed, they are willing to pay higher rent and spend marginally more on recreation trips (the left-hand side) in return for better amenity services and higher utility from recreation (the right-hand side). Furthermore, the first-order condition for X_{hn} is $\frac{\partial U(\cdot)}{\partial X_n} = tc \cdot d_{hn}$, and so A5 can be broken out as

$$\begin{aligned} -\frac{\partial P(\cdot)}{\partial c_n} &= -\frac{\partial U(\cdot)}{\partial A}\frac{\partial A}{\partial c_n} \\ -tc \cdot d_{hn} \cdot \frac{\partial X_{hn}}{\partial c_n} &= -\frac{\partial U(\cdot)}{\partial X_n}\frac{\partial X_{hn}}{\partial c_n}. \end{aligned} \tag{A6}$$

This shows that in equilibrium and conditional on d_{hn}, the rent price gradient for water quality reveals the amenity mechanism marginal willingness to pay for a change in c_n, the change in recreation behavior reveals the recreation mechanism marginal willingness to pay, and equation A5 shows these that values are additive.

The first order condition for d_{hn} is

$$-\frac{\partial P(\cdot)}{\partial d_{hn}} = tc \cdot X_{hn} - \frac{\partial U(\cdot)}{\partial A}\frac{\partial A}{\partial d_{hn}}. \tag{A7}$$

Conditional on water quality c_n, this expression shows that households pay for (reduced) distance through one channel (home prices) for two reasons: to consume amenity services and to reduce the cost of accessing recreation services at the local waterbody. We are interested in the value of changes in water quality conditional on location and so this expression does not enter our empirical damage calculations.

Finally, the first-order conditions for non-local trips are

$$\frac{\partial U(\cdot)}{\partial X_j} = tc \cdot d_{nj}, \; \forall j \neq n, \tag{A8}$$

which give rise to $J-1$ demand equations for visits to the non-local waterbodies.

In our simple model, we have abstracted from behavioral responses to the CWS tariff and the infant health externality. We assume the water utility sets its tariff to recover costs and that a higher pollution level increases these costs, which are passed on to households. The infant health mechanism operates through in utero contact with local water pollution c_n from drinking water and dermal or ambient exposure. As discussed in Appendix C.4, our empirical representation of infant health costs uses an emerging understanding of birthweight and cyanobacteria exposure that is unlikely to affect household location decisions. With these

simplifications, the CWS component in our model operates as a pure income effect and the birthweight component generates additive damages.

Equations A5 and A6, and assumptions on CWS cost passthrough and the infant health externality, provide the basis for our empirical measurement of local marginal damages from a change in water quality in the landscape, holding location fixed. Specifically,

$$\begin{aligned} \mathrm{MD}_{hn}(c_1, \ldots, c_J) &= \tau_n'(c_n) - \frac{\partial U}{\partial A}\frac{\partial A(d_{hn}, c_n)}{\partial c_n} - \sum_{j=1}^{J} \frac{\partial U}{\partial X_j}\frac{\partial X_{hj}(c_j)}{\partial c_j} - \frac{\partial U}{\partial B}B_h'(c_n) \\ &= \tau_n'(c_n) - \frac{\partial P(\cdot)}{\partial c_n} - \sum_{j=1}^{J} RDV_{hj} \cdot \frac{\partial X_{hj}(\cdot)}{\partial c_j} - \frac{\partial U}{\partial B}B_h'(c_n), \end{aligned} \tag{A9}$$

where RDV_{hj} is a unit value for trips to site j that we discuss in Section 4.1 and Appendix C.1.5. We estimate marginal damages by assembling empirical representations of the right-hand side terms in A9.

B Hydrology

An important hydrologic component of our IAM is built off the USGS's SPARROW model. The modeling framework is described in Smith et al. (1997) and Schwarz et al. (2006). Preston et al. (2009) provide a non-technical overview. SPRARROW is a hybrid model that combines physical and statistical elements to characterize the spatial distribution of nutrient pollution loading and ambient water quality outcomes. The focus is on long-term (steady state) averages based on hydrologic features of the landscape, land use patterns, and major point and nonpoint sources of emissions. It abstracts from intra-annual variation in pollution outcomes that arise from idiosyncratic weather patterns, seasonal land use changes, and other short-term shocks. We use five regional SPARROW models that collectively represent the conterminous U.S. to (a) quantify TN and TP emissions (loads) into watersheds; (b) characterize baseline concentrations of TN and TP across the landscape; and (c) estimate counterfactual changes in downstream concentrations from marginal changes in loading. In this Appendix we describe the SPARROW model and how we use the model structure for these tasks.

The SPARROW framework uses a river or lake segment and its catchment area – the surrounding land area that drains into the river segment – as the unit of analysis. River and lake segments and their catchments are identified from the NHDPlusV2 at a fine level of spatial resolution.[28] River and lake segments can represent sections of small streams with perennial or intermittent flow, large permanent rivers, or impoundments (lakes and reservoirs) and impoundment outlets. For modeling purposes, river and lake segments are defined as lines connecting upstream and downstream nodes (points). A river segment i is hydrologically connected to upstream and downstream segments. An adjacent upstream segment $i-1$ flows into segment i, while an adjacent downstream segment $i+1$ receives flow from segment i. The SPARROW model maintains an ordering of river and lake segments in the direction of flow from a headwater (a segment with no upstream adjacent segment) to a receiving water (a segment with no downstream adjacent segment). It enforces mass balance conditions so that the quantity of emissions released at a point in space is tracked throughout the stream network as the sum of losses (decay) during constituent transport

[28] Throughout the paper, we describe most rivers, streams, and other waterbodies used to route pollution as river or lake "segments." In our hydrology data and models, these individual river, streams, and lake segments are known as "comids" and serve as unique identifiers in the data. Our routing program uses comids for rivers, streams, lakes, and other waterbodies to connect upstream areas to downstream areas. HydroEcon uses the NHDPlusV2 Medium Resolution Model, which is also the model that forms the river and stream network for the USGS SPARROW regional models.

downstream and loading into receiving waters.

The statistical component of SPARROW uses stream flow and pollution concentration data from a monitoring network to calibrate the relationship between pollution mass flowing through a river or lake segment and pollution sources and land use characteristics. An important pre-processing step uses the network of stream flow gauges and pollution monitors to predict long-term average stream flow (in cubic feet per second) and long-term average flux of TN and TP (in kg per year) leaving each segment in the model. SPARROW then uses nonlinear regression and a specific parametric relationship to estimate the parameters θ in the function

$$F_i = g(F_{i-1}, S_{i1}, ..., S_{iN}, Z_i; \theta), \tag{B1}$$

where F_i is the flux of pollution (e.g., kg per year of TP) leaving segment i, F_{i-1} is the flux of pollution leaving the adjacent upstream segment (or segments), and S_{in} is one of N local pollution sources entering segment i from its catchment or being directly emitted into segment i. The term Z_i is a collection of variables describing the stream and landscape characteristics, including long run average flow, that affect delivery of pollution locally into the segment and the attenuation processes acting on flux as it travels along the segment pathway.

There are three features of the empirical SPARROW model that are important for HydroEcon. First, predictions at the segment level for long-term average flux of TN and TP and long-term average stream flow allow us to compute long term average baseline concentrations of TN and TP across the landscape. Second, S_{in} can represent directly observed emissions (e.g., point source emissions from wastewater treatment plants) or it can be a proxy for unobserved emissions (e.g., emissions from cropland acres in a catchment). Including both types of source variables allows the model to be relatively comprehensive in its coverage of major pollution sources. In addition, the functional form and parameter estimates allow F_i to be apportioned across contributions from the upstream flux and contributions from local pollution sources entering the segment. This in turn allows us to assign annual nutrient loads at the segment level for the major point and nonpoint sources of pollution.

Third, the empirical specification captures constituent loss from local sources S_{in} and upstream flux F_{i-1} as pollution mass travels along the stream network. River and lake segment specific variables such as stream flow, average temperatures, soil characteristics, conservation practices, and withdrawals for irrigation and other purposes determine the

proportion of upstream flux and local loads that are passed to the downstream segment. This key SPARROW capability allows us to observe the local load of TN and TP entering a river or lake segment (incremental yield into segment i) and the amount of the local load that is delivered to the receiving water (delivered yield from segment i).

We use the ratio of delivered yield to incremental yield to predict the changes in downstream concentrations that follow a marginal change in nutrient load. We define a delivery ratio $DR_i \leq 1$ as the proportion of the total local load of a nutrient (TN or TP) that is delivered to a terminal segment. There is variation in delivery ratios across the landscape due to variation in landscape attributes and distance to the downstream terminal segment. In general, we expect delivery ratios to be smaller for distant upstream segments and larger for downstream segments that are near the terminal segment. In practice this means that

$$DR_{i-1} \leq DR_i \leq DR_{i+1} \tag{B2}$$

for upstream and downstream adjacent segments that are similar in their physical characteristics. We use this to define a segment-specific decay rate

$$decay_{i+1}^{i} = \frac{DR_{i+1} - DR_i}{DR_{i+1}} \tag{B3}$$

If equation B2 holds, the decay rate will be a positive number in the unit interval.[29] It measures the proportion of incremental load to river or lake segment i that is lost on its way downstream to segment $i+1$.

We use the decay rates to predict downstream changes in concentration that follow from a place specific change in load. Define $\triangle L_i$ as an incremental addition to long term average load for a nutrient in segment i. The resulting change in long term average nutrient flux at downstream segment $i+k$ is

$$\triangle F_{i+k} = \triangle L_i \prod_{j=0}^{k-1} (1 - decay_{i+j+1}^{i+j}). \tag{B4}$$

[29] In our application, this restriction holds for over 98% of our river segments. In the rare instances where a delivery ratio in the i segment is less than the delivery ratio in the $i-1$ segment, we set it equal to the $i-1$ segment. We also assume no decay occurs in headwater streams.

For each downstream segment $i+k$ we compute the updated long term average flux as $F_{i+k} + \triangle F_{i+k}$ and compute the updated nutrient concentration using the baseline long term average flow.

For our empirical implementation we use five regional SPARROW models that collectively represent the conterminous U.S. The models are the result of a major effort completed in 2019 that implemented state of the art SPARROW methodology and used monitoring data for stream flow and nutrient pollution concentrations collected between 2000 and 2014. The effort was completed in stages, with Saad et al. (2019) describing the preprocessing steps that were used to estimate long-term average stream flow and nutrient loads at monitoring sites across the country. Robertson and Saad (2019), Ator (2019), Wise (2019), Hoos and Roland II (2019), and Wise et al. (2019) then describe models for the Midwest, Northeast, Pacific, Southeast, and Southwest regions, respectively. We access results for each model from the USGS SPARROW Mapper website.[30] The output files provide river or lake segment specific predictions for long-term average TN and TP flux, stream flow, incremental and delivered yield for TN and TP, and individual pollution sources.

Figure 1 and Appendix Figure B.1 display the spatial distribution of loads and concentrations for TN and TP at the watershed level. Watershed level loads are the sum of incremental (local) loads across all segments in the watershed. Concentrations are computed as the flow-weighted average of segment concentrations (calculated as the ratio of flux and stream flow) across all segments in a watershed. Unless stated otherwise, we winsorize concentrations at the 1st and 99th percentile within each SPARROW region. We adjust loads only for our marginal damage calculations, but do not winsorize them separately for GED estimates. Our marginal damages calculations rely on simulating changes in concentrations from a pulse in upstream loads. When we follow these loads downstream, we add the marginal change in loads to the river segment specific load to estimate changes in concentrations. For those river segments where we winsorize concentrations, we adjust the annual load estimate for that river segment to reflect the winsorized concentration.

Table B.1 provides summary statistics on the hydrologic network and corresponding water quality measurements for each of the five SPARROW models. For some measures, we report the corresponding statistics for NHDPlusV2, SPARROW, and HydroEcon since our model uses the NHDPlusV2 as its hydrologic base and then links to SPARROW for information on decay, concentrations, and loads. The table shows that our model captures a high level of granularity - HydroEcon uses approximately 2.5 million river segments and each river

[30] See USGS (2020).

segment is, on average, less than 2 km in length. The Midwest SPARROW model comprises the largest area, encompasses the most states, and accounts for approximately half of the river segments. HydroEcon uses most of the river segments in NHDPlusV2 and SPARROW; however, some river segments are dropped through the following process. First, we use the National Seamless Database from the NHDPlusV2 as our routing network. We keep only river segment pairs where the direction is defined as having a flowing connection between these pairs (this corresponds to direction value of 709 which is a flowing connection). This restriction excludes most coastal boundaries. We then exclude divergences from this list. This leaves us with 2,664,564 unique river segments as reported in Table B.1. HydroEcon uses all watersheds from NHDPlusV2 except one, HUC8 12110203. Nearly all river segments in this watershed are classified as coastal. The one exception, comid 1587295, crosses another HUC8 watershed, 12110205, and is primarily found within it. Thus, we assign comid 1587295 to this non-coastal watershed. We also report the TP and TN concentrations for the winsorized sample, as well as the full sample. Due to some extremely and unrealistically large outliers, especially in the Pacific and Midwest models, our winsorization process yields smaller average concentrations for TP and TN.

Table B.2 provides additional summary statistics for HydroEcon. First, we report the number of distinct lakes that are used within the recreation, housing, health and climate change damages modules since these damage estimates are highly dependent on the presence and water quality of lakes. Our recreation and housing modules use approximately 7 to 17% of all lakes within the NHDPlusV2 and reflect those lakes that are larger than 0.1 km^2. Our human health and climate change modules include many more smaller lakes, including all lakes greater than 0.01 km^2. This captures most (94%) of the waterbodies designated as lakes within the NHDPlusV2. Figure 2 displays the distribution of these lakes across the U.S. and Figure B.2 displays the count of lakes by county for recreation (Panel A), property value (Panel B), and the human health and climate change (Panel C) modules. For purposes of reporting, when lakes overlap multiple states, we assign lakes to states based on the centroid of each lake.

In Table B.2, we also report the TP and TN loads by Census region that correspond to loads reported in Table 3. These statistics show that the Midwest contributes the largest share of TP and TN loads, primarily due to agriculture's large contribution. However, municipal wastewater and urban non-point sources also contribute significant portions, especially in the South. For purposes of reporting of loads and GED, when watersheds overlap states, we assign watersheds to states based on the share of the watershed that is within a state.

Table B.1: Summary of Regional SPARROW Models

Category	Northeast	Southeast	Midwest	Southwest	Pacific	All
States	ME, NY, NJ, PA, MD, VA, VT, NH, DE, MA, RI, WV, CT, DC	AL, FL, GA, NC, SC, VA, TN, MS	CO, WY, MT, OK, MI, AR, TX, WV, KY, VA, OH, MO, IA, NE, KS, IL, NM, LA, MN, WI, TN, SD, ND, IN, PA, MD, NC, MS, NY, AL, GA	TX, ID, UT, NV, CO, AZ, NM, LA, CA, WY, OR	CA, OR, MT, ID, WA, WY, UT, NV	-
Avg. Length of River Segment (km)	1.52	1.46	1.91	2.56	1.88	1.93
No. of River Segments						
NHDPlusV2	197,332	323,647	1,378,929	440,497	342,453	2,689,508
SPARROW	197,576	323,770	1,379,904	440,985	342,454	2,684,689
HydroEcon	190,812	316,752	1,373,610	438,810	338,846	2,664,564
Number of HUC8s						
NHDPlusV2	159	193	1,006	440	319	2,113
SPARROW	159	193	1,006	440	319	2,106
HydroEcon	159	193	1,006	439	319	2,112
Avg. TP Concentration (mg/l)						
Winsorized	0.073	0.117	0.329	0.153	0.169	0.235
Full Sample	0.08	0.125	4.396	0.255	$1.14 \cdot 10^{19}$	$1.47 \cdot 10^{18}$
Avg. TN Concentration (mg/l)						
Winsorized	1.126	1.101	2.332	0.705	1.051	1.665
Full Sample	1.236	1.108	2.598	0.741	$2.10 \cdot 10^{98}$	$2.67 \cdot 10^{97}$

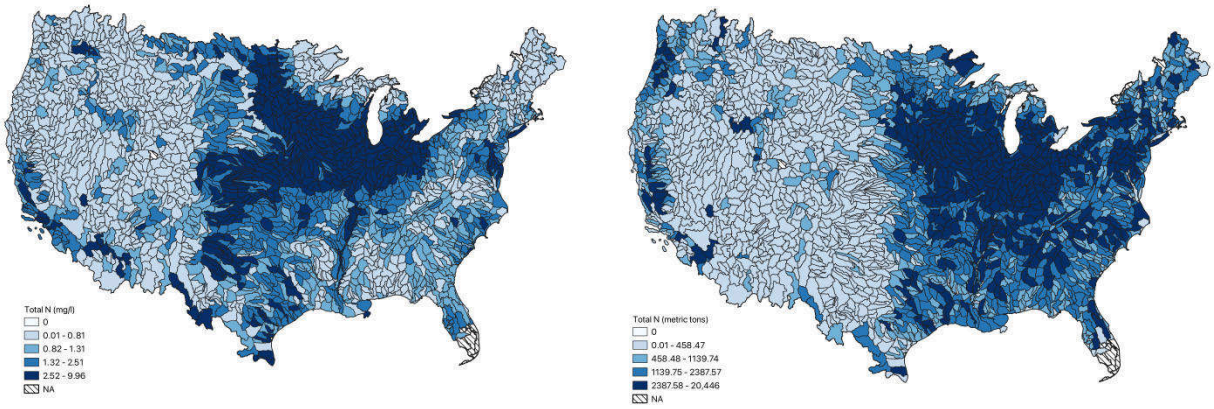
Notes: This table displays summary statistics for the NHDPlusV2, SPARROW, and HydroEcon across the five primary regional SPARROW models.

Table B.2: Summary of HydroEcon by Census Region

	Northeast	**South**	**Midwest**	**West**	**All**
No. of Lakes					
NHDPlusV2	27,695	140,641	150,629	60,112	379,077
Recreation (> 0.1 km^2)	6,899	20,655	27,072	10,012	64,638
Housing (> 0.1 km^2)	4,455	8,452	12,345	1832	27,084
Human Health and Climate Change (> 0.01 km^2)	25,385	133,032	142,285	55,284	355,986
TP Loads (thousand metric tons/yr)					
Non-point Sources					
Agriculture	6.2	74.3	106.2	13.8	200.4
Urban	4.0	17.5	13.3	2.5	37.3
Point Sources					
Municipal Wastewater	10.5	21.1	20.7	13.1	65.3
Other	0	2.9	0	1.4	4.4
TP Loads per Year	20.6	115.8	140.2	30.8	307.4
TN Loads (thousand metric tons/yr)					
Non-point Sources					
Agriculture	97.5	348.1	1,097.4	89.0	1,631.9
Urban	29.7	41.5	30.0	17.4	118.6
Point Sources					
Municipal Wastewater	76.5	135.3	77.2	52.1	341.2
Other	0	8.91	0	7.9	16.81
TN Loads per Year	203.6	533.7	1,204.6	166.5	2,108.4

Notes: This table displays summary statistics for the NHDPlusV2 and HydroEcon across Census regions. Other sources of Total Phosphorus loads include emissions from mining and inter-basin transfer. Other sources of total nitrogen loads include inter-basin transfers.

Figure B.1: Baseline Total Nitrogen Concentrations and Loads

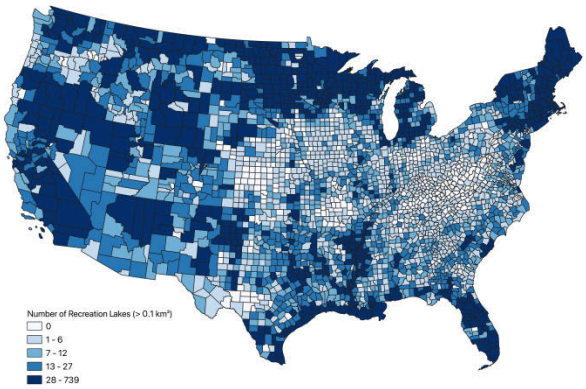


(a) Total Nitrogen Concentration (mg/l)

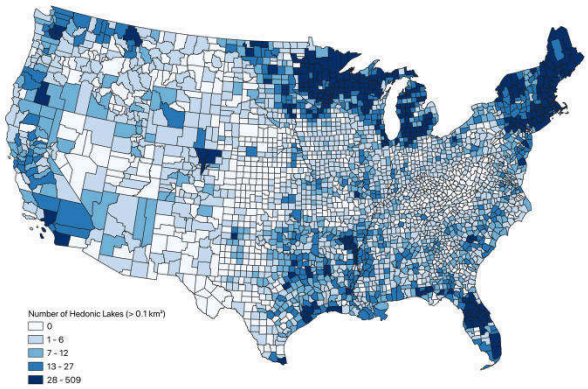
(b) Total Nitrogen Loads (metric tons/yr)

Notes: This figure displays baseline, long-term annual average concentrations and loads of Total Nitrogen by watershed. Darker shades of blue represent higher concentrations and greater loads.

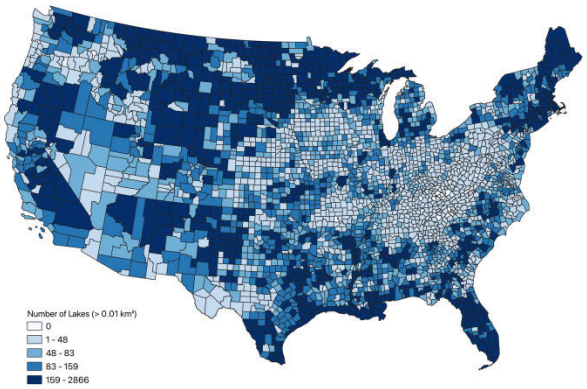
Figure B.2: Count of U.S. Lakes By County By Module



(a) Lakes > 0.1 km^2 (Recreation Module)



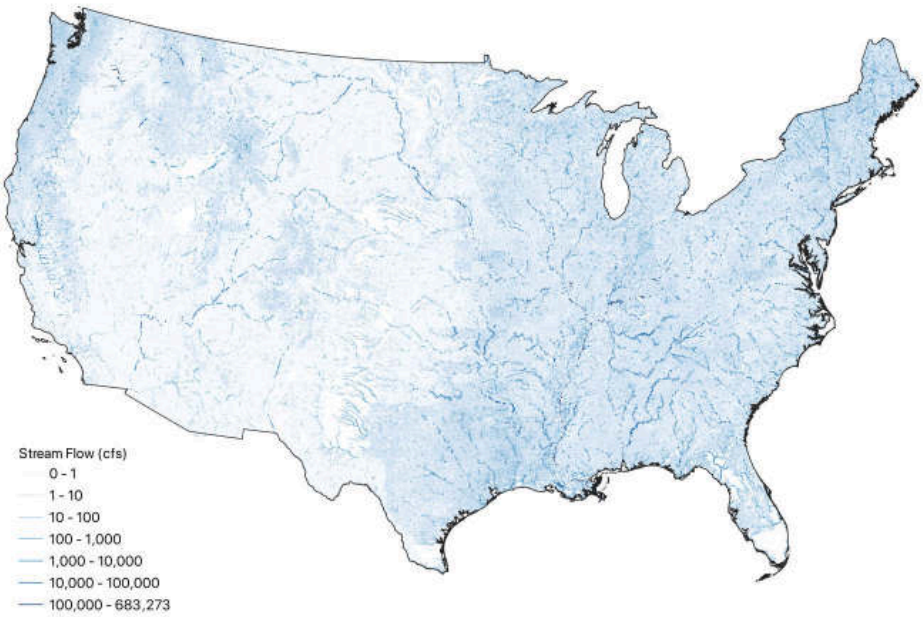
(b) Lakes > 0.1 km^2 (Hedonic Module)



(c) Lakes > 0.01 km^2 (Human Health and Climate Change Modules)

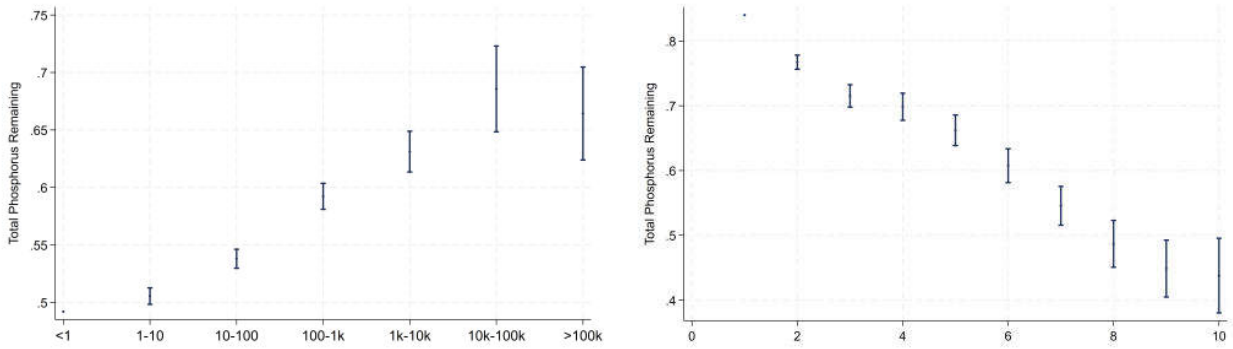
Notes: This figure displays lakes greater than 0.01 km^2 used in the human health and climate change modules (Panel A) and lakes greater than 0.1 km^2 used in the recreation (Panel B) and hedonic (Panel C) modules.

Figure B.3: Volume of Water Flow by River Segment



Notes: This figure displays baseline water flow by river segment. Darker shades of blue represent higher levels of flow and lighter shades represent smaller levels.

Figure B.4: Downstream Transport of Total Phosphorus by Hydrologic Factors



(a) Flow (cfs)

(b) Stream Level

Notes: These figures plot estimates how much Total Phosphorus (TP) is transported downstream as a function of various hydrologic factors. The dependent variable is the fraction of TP that remains at the furthest outlet when added to a starting river segment. Regressions control for HUC2 fixed effects. Coefficient estimates adjusted to reflect the mean decay in the reference category. Standard errors clustered at the HUC8 level. Bars display 95% confidence intervals.

C Damage Modules

C.1 Recreation

Our recreation module provides an empirical representation of the recreation component of total marginal damage from a pulse in upstream nutrient load, given in equation 3 by:

$$\triangle RV = -\sum_{h=1}^{H}\sum_{j=1}^{J}\frac{\partial U}{\partial X_j}\frac{\partial X_{hj}(c_j)}{\partial c_j}\triangle c_j, \tag{C1}$$

where j indexes lakes, X_{hj} is trips by person h to lake j, c_j is the nutrient concentration in lake j, and $\frac{\partial U}{\partial X_j}$ is the value of a recreation trip. Specifically, the module predicts the landscape wide economic damage from changes in TP concentration in lakes, which arise from an upstream change in phosphorus emissions. The key empirical components include a dose-response function relating a change in individual water-based recreation trip taking to a change in TP, the value of a lost recreation day, and the baseline and changed levels of TP concentration in the choice set of lakes. Equations 5-13 detail the calculations that we make to estimate $\triangle RV$. The following sub-sections detail the important inputs to these calculations.

C.1.1 Baseline Recreation Data

We calibrate baseline recreation trips, $X_{ct}(WQ^0_{ct})$, in equation 6 using survey responses to the National Survey of Recreation and the Environment (NSRE) (USDA 2009). These data reflect surveys conducted between 1999 and 2009. We use the NSRE for three reasons. First, the survey reflects one of the most comprehensive datasets on water-based recreation across the U.S. It was maintained by the Forest Service for many decades, beginning in the 1960s. The last iteration of the survey was completed in 2010, and the survey years 1999 to 2009 reflect the most recent data available to us. Second, this time period for baseline recreation roughly corresponds to the time period that our water quality data represent. This means that baseline recreation reflects baseline water quality conditions in our model. Finally, the NSRE was the recreation dataset used in Keiser (2019). This study strongly influences our preferred water quality elasticity estimate as we detail in the next sub-section.

To estimate $X_{ct}(WQ^0_{ct})$, we use information on approximately 74,000 individuals that

completed the NSRE and answered questions about water-based recreation activities, including swimming in natural bodies of water, warm and coldwater fishing, boating with a motor, and visits to watersides that are not beaches for other recreation purposes. We choose these activities since they correspond to the same categories studied by Keiser (2019). He shows that these activities are responsive to changes in phosphorus concentrations, while most other activities that serve as a falsification tests are not. We also choose these activities to approximate, as closely as possible, the number of visits that occur to freshwater resources, though it is likely they also reflect some coastal activities. Our measure of freshwater fishing (warm and coldwater) almost surely reflects freshwater fishing since the survey distinguishes them from saltwater fishing. However, it is possible that swimming, boating (with a motor) and visits to watersides that are not beaches could reflect coastal activities. Our version of the NSRE data distinguishes between swimming in natural bodies of water from swimming in a pool. It also distinguishes visits to a beach from visits to a waterside that is not a beach. We do not use visits to a pool or visits to a beach. Our estimates are likely a conservative estimate of overall water-based recreation activities because they do not include boating (without a motor) or activities that may be less directly impacted by water quality such as camping, hiking, picnicking, viewing wildlife and other activities not captured by the categories above.

The survey links respondents to the county in which they live. We use these data to construct county-level estimates of water-based recreation participation rates and an annual average number of recreation days per capita. Some survey iterations only asked about participation in these activities and do not provide information about the number of recreation days. For instance, approximately 65% of respondents report participating in the water-based recreation activities that we list above. Of the respondents that participated, we observe corresponding recreation days for approximately 65% of this sub-sample. We use these responses to calculate a conditional per capita average number of recreation days for those that participate in water-based recreation activities. We then calculate an unconditional average number of water-based recreation days for each county by multiplying this conditional average by the county-level participation rate.

Using the steps above, we observe participation rates for every county within the conterminous U.S. and recreation days for 91% of counties. Over the 10 year NSRE period that we observe, each county has an average of 24 respondents. To account for differences in the number of responses per county, we take additional steps to smooth our county-level participation and recreation day estimates. We calculate weighted average participation and recreation day estimates for a county by averaging a county's own estimate with neighbor-

ing counties, using the number of responses in each county as weights. This approach also allows us to impute recreation participation and day estimates for the counties where we do not observe recreation day estimates from the NSRE. We make one additional adjustment. Emporia City (FIPS code 51595) sits entirely within Greensville County (FIPS code 51081). We assign Greensville's estimate of water-based recreation days to Emporia given Emporia would otherwise be the only county within the U.S. with zero water-based recreation days. Our smoothing process has very little impact on our estimate of county-level participation rates - they remain effectively unchanged at 65%. Our estimate of water-based recreation days declines from 15.55 days on average per county in the raw data to 12.92 days after our smoothing procedure. The county is the finest spatial scale that we can observe in the NSRE data. However, we are able to model differences in choice sets and water quality at an even finer scale. To leverage these other data for our marginal damage estimates, we set baseline recreation estimates at the census tract level equal to the corresponding county-level estimates in which they reside.

C.1.2 Recreation Elasticity Estimates

To select the range of elasticities used in our recreation module, we proceeded in two steps. First, we re-analyzed Keiser (2019) using the baseline water quality data in our model. We then compared the elasticity estimate from this exercise to estimates from the recreation demand literature. To make this comparison, we derive the steps necessary to recover trip elasticities from discrete choice models. We believe these derivations serve as a novel contribution.

Keiser (2019)'s analysis covered a large portion of the US, but excluded much of the Southwest due to data limitations. Given the newly produced SPARROW models, which cover the entire continental US, we re-ran Keiser (2019) using these water quality data. Table C.1 provides results for an instrumental variables model (Panel A) and a simple OLS model (Panel B). For each approach, we present results across four specifications. Columns (1) and (2) present results using TP concentrations from lakes. Columns (3) and (4) present results using TP concentrations from rivers. Columns (1) and (3) reproduce the log-linear model from Keiser (2019). Given our focus on elasticities, we also estimate log-log models in Columns (2) and (4). We use the preferred instrumental variable from Keiser (2019), which defines upstream concentrations as those that utilize non-adjacent counties that are up to 200 miles upstream from a local county. Across all specifications, we find that the instrumental variables estimates are larger than the corresponding OLS estimates, which is

consistent with the findings from Keiser (2019). The instrumental variables strategy utilizes plausibly exogenous upstream concentrations as an instrument for local concentrations. The Kleibergen-Paap rk Wald F statistics shows that this instrument is strong. We direct the reader to Keiser (2019) for a deeper discussion on the potential validity of this approach, including an analysis of measurement error, falsification tests, and various ways of defining upstream concentrations. Given the re-analysis of Keiser (2019), we next sought to understand how these elasticity estimates compared to other published studies, as well as the prior estimates from Keiser (2019). We reviewed published recreation demand studies with applications that were U.S. based and featured analyses that compared water recreation behavior to continuous measures of water quality. We used these studies to assemble trip demand elasticities with respect to a range of water quality indicators. Most modern recreation demand research uses discrete choice models to estimate site choice or trip demand models. Here we describe the computational approaches we used to compute elasticities for different types of discrete choice models.

Many applications model the choice of recreation site conditional on taking a trip. In these models the demand for a visit to a site is the probability that a respondent selects the site on a trip taking occasion. We refer to these as trip allocation models. Let Pr_j denote the probability that a sampled person visits site j from J alternatives. Let q_j denote a generic environmental quality indicator at site j. The elasticity of trip demand with respect to q_j is

$$\epsilon_{j,q_j} = \frac{q_j}{Pr_j}\frac{\partial Pr_j}{\partial q_j}. \tag{C2}$$

For the conditional logit model, this elasticity has the form (see Train (2009), pp. 58-59)

$$\epsilon_{j,q_j} = (1 - Pr_j) \cdot \beta \cdot q_j, \tag{C3}$$

where β is the coefficient on q_j in a linear indirect utility function. For trip allocation models, we calibrate elasticities using the average quality $\overline{q}$ and $Pr_j = 1/J$.

Discrete choice models are also used to estimate trip demand models. In this case there are $J + 1$ choice alternatives consisting of J recreation sites and a no-trip option $j = 0$. We refer to these as repeated discrete choice models. If there are T choice occasions the total number of trips a person takes in the study period is

$$trips = (1 - Pr_0) \cdot T, \tag{C4}$$

where Pr_0 is the probability of not taking a trip on a choice occasion. Similarly, trips to site j during the study period is

$$trips_j = Pr_j \cdot T. \tag{C5}$$

The elasticity of $trips_j$ with respect to q_j is given by equation C3.

The elasticity of total trips with respect to quality at one of the sites j is

$$\epsilon_{q_j} = -T\frac{\partial Pr_0}{\partial q_j}\frac{q_j}{(1-Pr_0)T}. \tag{C6}$$

For the conditional logit version of the repeated discrete choice model, the elasticity is (see Train (2009), p. 58-59)

$$\epsilon_{q_j} = \beta\frac{Pr_0}{1-Pr_0}Pr_j \cdot q_j. \tag{C7}$$

The change in total trips from a one percent change in quality at all J sites is

$$\sum_{j=1}^{J}\epsilon_{q_j} = \beta\frac{Pr_0}{1-Pr_0}\underbrace{\sum_{j=1}^{J}Pr_j \cdot q_j}_{\tilde{q}}. \tag{C8}$$

Note that $\tilde{q}$ is the probability weighted sum of quality at the J sites, where probabilities add up to $1-Pr_0$. To calibrate the elasticity of total trips with respect to a landscape wide one percent change in quality we replace q_j with $\overline{q}$ so that equation C8 can be rewritten as

$$\sum_{j=1}^{J}\epsilon_{q_j} \approx \beta \cdot Pr_0 \cdot \overline{q}. \tag{C9}$$

Pr_0 is calibrated by observing the value for T and the average total trips taken by respondents.

Tables C.2 to C.4 display elasticities pulled from the published literature along with our reanalysis of Keiser (2019) and Keiser (2019). Figure 3, Panel B provides a corresponding visual of these elasticities. Here we note that behavioral responses to all the site quality variables we examined are inelastic. The range of estimates for TP and Secchi depth are especially relevant for our analysis. The median TP elasticity is 0.18. Likewise, the median Secchi depth elasticity is 0.19.

We expect behavior to be more responsive to a visible attribute like water clarity than to TP, since the latter contributes to visible water quality outcomes but is itself invisible, and so the magnitude order is intuitive. However, as mentioned previously, only Keiser (2019) uses a research design that generates plausibly causal estimates for water quality. Furthermore, given that Keiser (2019) finds evidence for significant measurement error in

water quality data, we expect some attenuation bias in most of the estimates presented in the table, including those for Secchi depth. Our analysis of the physical relationship between Secchi depth and TP (see Appendix C.2 below) suggests that a 1 percent change in TP is associated with a 0.50 percent change in Secchi depth. Thus, we speculate that a consistent estimate of the Secchi depth elasticity would be twice as large as its TP counterpart. Figure 3 shows that half of the median secchi depth elasticity is 0.09 and the median TP elasticity is 0.18. However, the Secchi depth elasticity from Keeler et al. (2015) and the estimate of elasticity of TP using USGS data from Keiser (2019) are both somewhat outliers from the other estimates. If we do not consider these two studies, the median TP elasticity is 0.16 and the median Secchi depth elasticity is 0.20. Thus, it appears that the elasticity with respect to TP is likely somewhere between 0.10 and 0.20. Given this reasoning along with the results from Keiser (2019) and our reanalysis of Keiser (2019), we use 0.15 as our main value for the elasticity of recreation trips with respect to TP and test the sensitivity of our findings to different elasticity values.

C.1.3 Choice Set

Defining the choice set of water bodies is always an important aspect of recreation demand models. We make three key decisions in defining these choice sets and then test the sensitivity of our estimates to these decisions. First, we focus on lakes. This focus is consistent with many freshwater-based recreation demand studies, including most studies from which we draw our elasticity estimates.

Second, we choose to focus on lakes larger than 0.1 km^2. A long running recreation demand project in the state of Iowa tracks recreation behavior at lakes ranging from 0.04 km^2 to 77 km^2, with an average size of 2.7 km^2 (Egan et al. 2009). A report from this effort in 2009 suggests that the most frequently visited lakes are approximately 20 km^2 (Saylorville Dam), 15 km^2 (Clear Lake), and 16 km^2 (West Okoboji Lake), whereas the least visited lakes are approximately 0.06 km^2 (Slip Buff Lake), 0.13 km^2 (Pierce Creek Lake), and 0.10 km^2 (Windmill Lake) (Evans et al. 2011).[1] We chose 0.1 km^2 given that recreation activities more frequently occur at larger lakes.

Third, we use a distance of 100 miles from the center of a census tract to define the lakes that fall within each representative individual's choice set. Keiser (2019) cites a distance of 30 miles for most recreation trips based on a 2009 survey from the National Household

[1] We retrieved lake sizes from the Iowa Lakes Project (Iowa State University 2022).

Travel Survey. Keiser and Shapiro (2019) calculate a straight-line distance of 25 miles for capitalizing water quality benefits using data from the Nationwide Personal Transportation Survey. This survey reported recreation travel distances in the 1980s and 1990s. Viscusi et al. (2008) utilize a distance of 100 miles for measuring water quality improvements based on an analysis done by the National Center for Environmental Economics at USEPA. This analysis suggested that most water-based recreation trips occur within a 100 mile radius of a person's home. The authors cite statistics on various water-based recreation activities that were calculated from the 1996 version of the NSRE. These activities included fishing, boating, and swimming. Most trips (∼80%) were within a 100 mile radius of the respondent's home. We use 100 miles as our main distance estimate given that this third study examined travel distances for specific water-based recreation and acquired these distances from an earlier version of our recreation data. We explore the sensitivity of our damage estimates to shorter and longer distances.

C.1.4 Marginal Value of Water Quality

Our reference quantity is water pollution, and we so need the marginal value (implicit price) of nutrient pollution. For the recreation mechanism we use the canonical travel cost model whereby recreation travel is a vehicle through which we estimate the demand for water quality. Two assumptions – known as weak complementarity and the Willig condition – allow us to derive the inverse demand for water quality by estimating how water quality shifts the demand for trips (see Phaneuf and Requate (2016), chapter 15 for details). We illustrate this using the following example. Let the demand for recreation trips be given by

$$lnX = \beta_0 + \beta_1 TC + \beta_2 Y + \theta WQ, \tag{C10}$$

where X, TC, and Y are trips to available lakes, travel cost, and income, respectively, and WQ is water quality. Symbols are demand parameters, with $\beta_1 < 0$. Note that baseline consumer surplus is

$$CS_0 = \int_{TC_0}^{\infty} exp(\beta_0 + \beta_1 TC + \beta_2 Y + \theta WQ) dTC = -\frac{exp(\beta_0 + \beta_1 TC + \beta_2 Y + \theta WQ)}{\beta_1} = -\frac{X_0}{\beta_1}. \tag{C11}$$

For this specification, $-\frac{1}{\beta_1}$ is the average consumer surplus per trip (the recreation day value).

For the functional form in equation C10 the indirect utility function is given by (see Phaneuf and Requate (2016) p. 433):

$$V = \frac{exp(-\beta_2 Y)}{\beta_2} + \frac{exp(\beta_0 + \beta_1 TC + \theta WQ)}{\beta_1}. \tag{C12}$$

We can derive the inverse demand for water quality using an approach that is akin to Roy's identity (see Phaneuf and Requate (2016), p. 409):

$$\frac{\partial V/\partial WQ}{\partial V/\partial Y} = -\frac{exp(\beta_0 + \beta_1 TC + \beta_2 Y + \theta WQ)\theta}{\beta_1} = -\frac{X\theta}{\beta_1}. \tag{C13}$$

Note that for the demand equation in C10, $X\theta$ is the marginal change in trips from a change in WQ and so the marginal willingness to pay for water quality is a dose-response ($X\theta$) times the per trip average consumer surplus ($-1/\beta_1$).

This example allows us to emphasize that the applied objective is to characterize the inverse demand for water quality and use it to assign a marginal value for nutrient quantity changes. The canonical travel cost model provides a way to use the demand for recreation trips to identify the marginal value a good (water quality) that is not traded in a market.

C.1.5 Value of a Recreation Day

To value a lost recreation day, we use RDV = \$78 (2022 dollars) based on the College of Forestry, Oregon State University (1981-2010). This database, updated through 2016, combines 3,129 estimates of the value of a recreation day from 421 documents / studies in the U.S. and Canada spanning the period 1958-2016. The database provides estimates of net willingness to pay or consumer surplus in per person-day units for 21 primary recreation activities, differentiated by broad geographic scale.

To construct our RDV, we limit our studies to the United States and those published after 2000. This sequentially removes 17 and then 253 studies respectively. We then calculate the median value for each study by activity given that some studies report more than one estimate per activity. The median of these median values across all activities is \$77 and the mean of these median values is \$110 (see Appendix Figure C.2).

These estimates do not necessarily represent water-based recreation activities. Thus, we analyze the records in this database and construct an indicator for water-based recreation.

We focus on activities that are most likely freshwater activities and align with our estimates of water-based recreation from the NSRE. We first restrict estimates to those with the primary activity designated as Freshwater Fishing, Non-motorized Boating, Motorized Boating, and Swimming. We remove one study that captures coastal swimming values and one study that captures coastal motorized boating. We include one study that describes Sturgeon viewing. We also include several studies that designate the activity as "General" or "Other" based on the study name and/or site that strongly suggests water-based activities that occur at rivers, lakes, creeks, falls, reservoirs, or streams. We remove one study in Lake Tahoe that focuses on snow sports. After these restrictions, we are left with 55 studies that report 414 estimates of the value of water-based recreation activities. We then calculate the median value of a day by activity category within a study. This leaves us with 60 observations. 37 observations reflect freshwater fishing (median value \$85), 6 represent nonmotorized boating (median value \$69), 3 are for motorized boating (median value \$130), 2 are for swimming (median value \$25), 1 is for wildlife viewing (Sturgeon Viewing, \$57), 10 are for general recreation (median value \$60), and 1 is for other recreation (\$7). Given our limited sample sizes by activity, we use the median value (\$78) across all of these observations. The corresponding mean of the median is (\$108). Panel B of Appendix Figure C.2 displays the corresponding histogram.

As a robustness check on our RDV estimate, we conducted an additional review of more recent water-based recreation demand studies that were not included in the Oregon State University database. These studies use contemporary discrete choice econometric models. Many studies use the conditional logit and related frameworks to estimate random utility maximum (RUM) models to recover the conditional indirect and expected maximum utility from taking a trip to a recreation site. This framework provides a convenient way to back out the RDV from estimated parameters. To complete this subsection, we first sketch the necessary logic for recovering the RDV from RUM models and then summarize an alternative measure of the RDV based on our review.

A typical RUM model specification defines the conditional indirect utility for person i on a choice occasion t from an alternative j using

$$V_{ijt} = \delta_j + \beta TC_{ij} + \epsilon_{ijt}, \tag{C14}$$

where there are $j = 1, \ldots, J$ recreation sites available, TC_{ij} is the travel cost for site j, and ϵ_{ijt} is distributed type I extreme value. The parameter $-\beta$ is referred to as the marginal

utility of income. For trip demand models using a repeated discrete choice framework, there is also an outside option indexed $j = 0$ with $TC_{i0} = 0$, and on choice occasion t the person decides if they will visit a site or opt out from visiting.

Under the distributional assumption the expected maximum utility on a choice occasion is (see Train (2009), p.55):

$$E(max\{V_{ijt}\}) = ln\left(\sum_{k=0}^{J} exp(\delta_k + \beta TC_{ik})\right) + K, \tag{C15}$$

where K is a constant. The willingness to pay per choice occasion to prevent the removal of site j from the choice set is the monetized difference in expected maximum utilities:

$$WTP_{ijt} = -\beta^{-1}\left(ln\left(\sum_{k=0}^{J} exp(\delta_k + \beta TC_{ik})\right) - ln\left(\sum_{k\neq j}^{J} exp(\delta_k + \beta TC_{ik})\right)\right), \tag{C16}$$

This simplifies to (see McConnell (1995), p. 265).

$$WTP_{ijt} = -\beta^{-1} ln\left(1 - Pr_{ijt}\right) \approx -\beta^{-1} Pr_{ijt}, \tag{C17}$$

where Pr_{ijt} is the probability that the person selects site j on choice occasion t and the approximation holds if the probability is small. If T is the number of choice occasions and $T \cdot Pr_{ijt}$ is the number of trips the person takes to site j, then the *total* willingness to pay to prevent the loss of site j is $-\beta^{-1} \cdot Pr_{ijt} \cdot T$ and the *per trip* willingness to pay for site j is

$$RDV_{ij} = -\beta^{-1} \cdot Pr_{ijt} \cdot \frac{T}{PR_{ijt} \cdot T} = -\frac{1}{\beta}. \tag{C18}$$

This logic motivates a review of RUM model studies focused on water recreation to recover the inverse of the marginal utility of income as the average value of a trip to a site in the choice set. Tables C.5 to C.6 display the recreation day values, computed as the inverse of the marginal utility of income, pulled from the published literature on recreation studies of water destinations that adhere to modern standards such as including a full set of alternative specific constants. Based on this survey, RDV*s* inferred from recent studies range from \$15.35 to \$225.55, with an average of \$94.83 and a median of \$78.53. These figures are quite close to the per trip values pulled from the Oregon State University database and give us additional confidence in our decision to use \$78 per day as our baseline estimate of the RDV.

C.1.6 Damage Estimates

For census tract population estimates, pop_{ct}, we use the 2019 American Community Survey (ACS). We use these population estimates to estimate damages from a change in TP concentration according to

$$\triangle RV = \sum_{ct=1}^{C} \triangle RV_{ct} \cdot pop_{ct}, \tag{C19}$$

where C is the number of census tracts in the conterminous U.S.

Table C.1: Update to Keiser (2019)

	(1)	(2)	(3)	(4)
Panel A: Instrumental Variables Models				
Total Phosphorus (mg/l)	-0.323***	-	-0.411***	-
	(0.108)		(0.130)	
Ln(Total Phosphorus (mg/l))	-	-0.137***	-	-0.176***
		(0.0395)		(0.0505)
Resource Type	Lakes	Lakes	Rivers	Rivers
Observations	44,153	44,153	44,153	44,153
Kleibergen-Paap	30.044	114.091	41.469	93.595
rk Wald F Stat	(16.38)	(16.38)	(16.38)	(16.38)
Panel B: OLS Models				
Total Phosphorus (mg/l)	-0.0773***	-	-0.118***	-
	(0.0297)		(0.0349)	
Ln(Total Phosphorus (mg/l))	-	-0.0637**	-	-0.0821***
		(0.0137)		(0.0150)
Resource Type	Lakes	Lakes	Rivers	Rivers
Observations	44,153	44,153	44,153	44,153

Notes: The dependent variable is the natural log of the number of water-based recreation days + 1. All regressions include state fixed effects and controls for metropolitan and micropolitan statistical areas, coastal and Great Lakes shorelines, 30-year climate normals for temperature and precipitation and temperature squared, indicators for race, gender, age, income, education, and predominant land cover type. Standard errors are clustered at the county level. Panel A presents results from a instrumental variables regression. Panel B presents results from a simple OLS regression. Kleibergen-Paap rk Wald F Stat, Stock-Yogo weak identification critical value for 10% maximal instrumental variable size in parentheses. Mean Total Phosphorus values for the estimating sample are 0.2430226 mg/l for lakes and 0.2712859 mg/l for rivers. *** Significant at the 1 percent level; ** Significant at the 5 percent level. * Significant at the 10 percent level.

Table C.2: Recreation Trip Elasticities with Respect to Water Quality

Study	**Resource Type**	**Water Quality Variable**	**Context**	**Implied Elasticity**
Abidoye et al. (2012b)	Lakes	Total Phosphorus	Repeated discrete choice model for 2002 visits to Iowa lakes. Model with alternative specific constants. Parameter estimate drawn from table 5, model A.	-0.12 [-0.01 -0.21]
Egan et al. (2009)	Lakes	Secchi Depth	Repeated discrete choice model for 2002 visits to Iowa lakes. Model without alternative specific constants. Parameter estimate drawn from table 7, model C.	0.18 [0.11 0.24]
Egan et al. (2009)	Lakes	Total Phosphorus	Same as above.	-0.21 [-0.07 -0.36]
Egan et al. (2009)	Lakes	Log(Total Phosphorus)	Same as above. Parameter estimate drawn from table 7, model A.	-0.25 [-0.08 -0.41]
Ji et al. (2020b)	Lakes	Secchi Depth	Repeated discrete choice model for 2009 visits to Iowa lakes. Model with alternative specific constants. Estimates drawn from table 3, column 5.	0.21 [0.06 0.35]
Keeler et al. (2015)	Lakes	Secchi Depth	Log-linear regression of summertime lake visits on lake-level water clarity measured using Secchi depth. Visits to nearly 1,100 lakes in MN and IA proxied using time and place referenced photographs uploaded to Flickr. Estimate drawn from table 1.	0.03 [0.01 0.06]

Notes: This table summarizes elasticity estimates from the literature. 95 percent confidence intervals in square brackets.

Table C.3: Recreation Trip Elasticities with Respect to Water Quality (Continued)

Study	Resource Type	Water Quality Variable	Context	Implied Elasticity
Keiser (2019)	Lakes	Total Phosphorus	Log-linear regression of individual recreation visits to freshwater resources on county-level average phosphorus concentrations from 2007 EPA National Lakes Assessment. IV design uses upstream concentrations as instruments. National scale analysis using NSRE to measure trips, with southwestern states excluded. Estimate drawn from table 2, column 2.	-0.21 [-0.09 -0.33]
Keiser (2019)	Lakes	Total Phosphorus	Same as above but using 2002 SPARROW model for county level phosphorus concentrations. Estimate drawn from table 3, column 2.	-0.39 [-0.18 -0.61]
Update to Keiser (2019)	Lakes	Total Phosphorus	Internal replication of Keiser (2019) research design using the 2012 SPARROW model estimates for lake water quality and including all U.S. states. See table C.1, column 1.	-0.08 [-0.03 -0.13]
Update to Keiser (2019)	Lakes	Total Phosphorus	Internal replication of Keiser (2019) research design using the 2012 SPARROW model estimates for lake water quality and including all U.S. states. Log-log specification. See table C.1, column 2.	-0.14 [-0.06 -0.21]
Update to Keiser (2019)	Rivers	Total Phosphorus	Internal replication of Keiser (2019) research design using the 2012 SPARROW model estimates for river water quality and including all U.S. states. See table C.1, column 3.	-0.11 [-0.04 -0.18]
Update to Keiser (2019)	Rivers	Total Phosphorus	Internal replication of Keiser (2019) research design using the 2012 SPARROW model estimates for river water quality and including all U.S. states. Log-log specification. See table C.1, column 4.	-0.18 [-0.08 -0.28]

Notes: This table summarizes elasticity estimates from the literature. 95 percent confidence intervals in square brackets.

Table C.4: Recreation Trip Elasticities with Respect to Water Quality (Continued)

Study	Resource Type	Water Quality Variable	Context	Implied Elasticity
Kuwayama et al. (2022)	Saltwater Coastal Angling Sites	Dissolved Oxygen	Trip allocation model for angling trips to sites in Tampa Bay, FL. Trips are observed between 1998 and 2014. Estimate drawn from table 2, column 1.	0.46 [0.35 0.57]
Merrill et al. (2024)	Saltwater Coastal Beaches	Bacteria Counts	Trip allocation model for 4,812 coastal New England beaches. Trip-taking population is households within 100 miles of Cape Code, MA. Estimates drawn from table 3, column 3.	-0.23 [-0.12 -0.35]
Merrill et al. (2024)	Saltwater Coastal Beaches	Secchi Depth	Same as above.	0.26 [0.09 0.43]
Nelson et al. (2023)	Lakes	Secchi Depth	Log-linear regression of summertime lake visits on lake-level water clarity measured using Secchi depth. Visits to 51,000 U.S. lakes proxied using time and place referenced photographs uploaded to Flickr. Estimate drawn from table 4, column 3.	0.19 [0.01 0.37]
Phaneuf et al. (2009)	Lakes	Chlorophyll a	Linear demand model for 2002 visits to Iowa lakes. Individual visits to each of 129 lakes were modeled as a function of travel cost and chlorophyll a. Alternative specific constants included.	-0.15 [-0.06 -0.23]
Wolf et al. (2019)	Lake Erie	Algae Counts	Trip allocation model for beach visitors and anglers to 106 locations along Lake Erie. Sample represents Ohio residents living within 50 miles of the lake. Estimate drawn from table 4, column 1.	-0.15 [-0.03 -0.28]
Wolf et al. (2019)	Lake Erie	E. coli counts	Same as above.	-0.11 [-0.01 -0.21]

Notes: This table summarizes elasticity estimates from the literature. 95 percent confidence intervals in square brackets.

Table C.5: Recreation Use Day Value from Recent RUM Studies

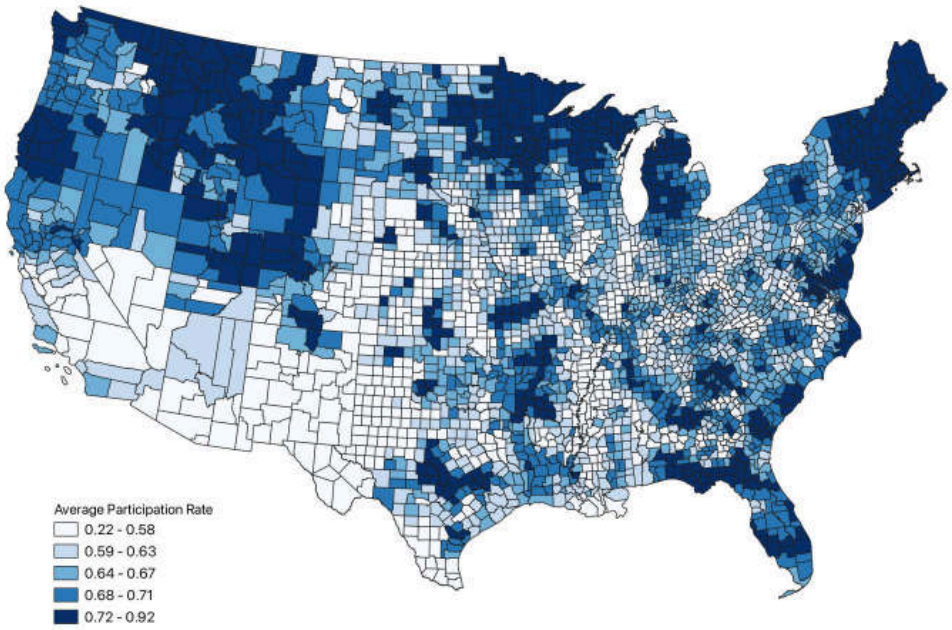
Study	**Resource Type**	**Study Region(s)**	**Context**	**Implied RUDV($)**
Abidoye and Herriges (2012)	Lakes	IA	Bayesian Repeated discrete choice model for 2002 visits to Iowa lakes. Model with alternative specific constants. Parameter estimate drawn from table 3.	117.90
Abidoye et al. (2012a)	Lakes	IA	Repeated discrete choice model for 2002 visits to Iowa lakes. Model with alternative specific constants. Parameter estimate drawn from table 4	120.54
Ji et al. (2020a)	Lakes	IA	Repeated discrete choice model for five years of visits to Iowa lakes. Model with alternative specific constants. Estimates drawn from table 3.	55.28
Yi and Herriges (2017)	Lakes	IA	Repeated discrete choice model for 2004 , 2005 (actual and expected) visits to Iowa lakes. Model with alternative specific constants. Estimates drawn from tables 2 and 3.	87.89
Lupi et al. (2022)	Great Lakes	MI	Repeated discrete choice model for 2012 visits to Great Lakes beaches in Michigan. Model with alternative specific constants. Estimates drawn from table 2.	74.70
Melstrom et al. (2015)	Rivers, Streams	MI	Discrete choice model for 2008-2010 angling visits to rivers and streams in Michigan. Model with alternative specific constants. Estimate drawn from table 4.	43.87
Jayasekera et al. (2024)	Reservoirs	NE	Discrete choice model for 2019 visits to reservoirs in Omaha metropolitan region. Model with alternative specific constants. Parameter estimates drawn from table 2.	168.63

Table C.6: Recreation Use Day Value from Recent RUM Studies (continued)

Study	**Resource Type**	**Study Region(s)**	**Context**	**Implied RUDV($)**
Melstrom and Jayasekera (2017)	Lakes,Rivers	OK	Discrete choice model for 2014 visits to Oklahoma lakes and rivers. Model with alternative specific constants. Parameter estimate drawn from table 2.	64.00
Melstrom and Reeling (2024)	Reservoirs	OK	Zonal RUM model for 2014 visits to Oklahoma public reservoirs. Model with alternative specific constants. Parameter estimates drawn from table 6.	56.32
Cha and Melstrom (2018)	Lakes,Rivers	OK,AR,KS,MO	Discrete choice model for 2015 visits to Oklahoma lakes and rivers. Model with alternative specific constants. Parameter estimates drawn from table 4.	35.29
Smith and Lamborn (2024)	Lakes, Rivers	UT	Discrete choice model for 2021-2022 season to lake, rivers, reserviors in Utah. Model with alternative specific constants. Estimates draw from table 2.	180.01
Timmins and Murdock (2007)	Lakes, Rivers	WI	Discrete choice model for 1998 visits to lakes and rivers in Wisconsin. Model with alternative specific constants. Estimates drawn from tables 4 and 7.	15.35
W. Zhang et al. (2024)	Lakes, Rivers, Coastal	Contiguous US	Discrete choice based model for 2019 cellphone inferred visits to water recreational sites in US. Model with alternative specific constants. Parameter estimates draw from tables 2 and 3.	225.55
Lloyd-Smith and Zawojska (2025)	Lakes, Rivers	Alberta	Discrete choice model using administrative campground visit data to on-water provincial parks 2013-2022. Model with alternative specific constants. Estimates drawn from table 2.	82.37

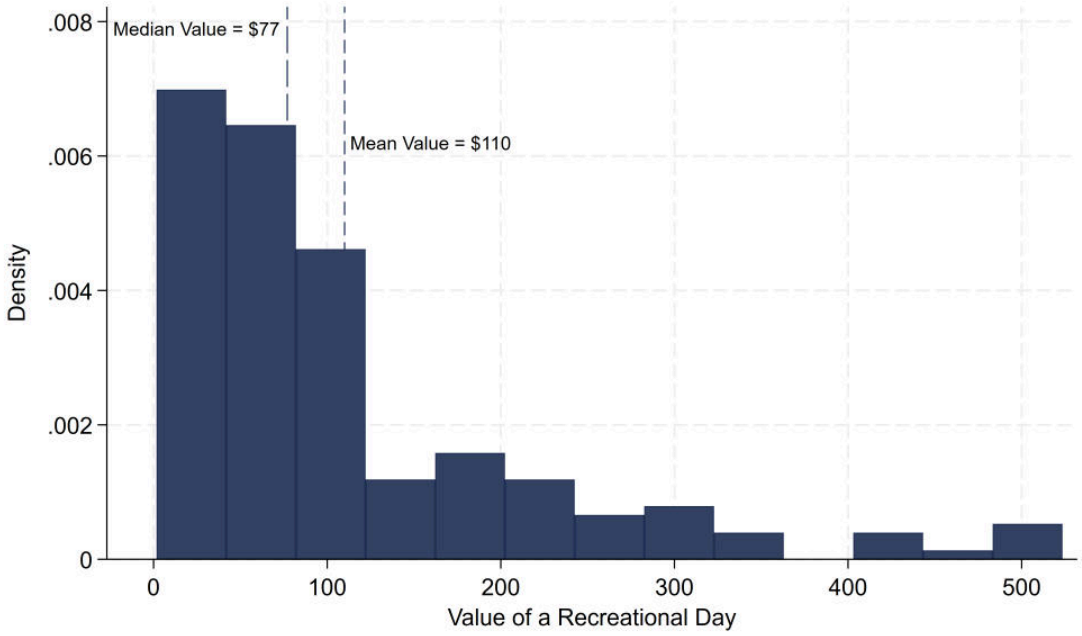
Notes: In some papers, more than one travel cost coefficient is reported. In such cases, we collected all reported coefficients, computed the corresponding recreation use day values, and then averaged them after adjusting prices to 2022 dollars.

Figure C.1: Baseline Water-Based Recreation - Participation Rates

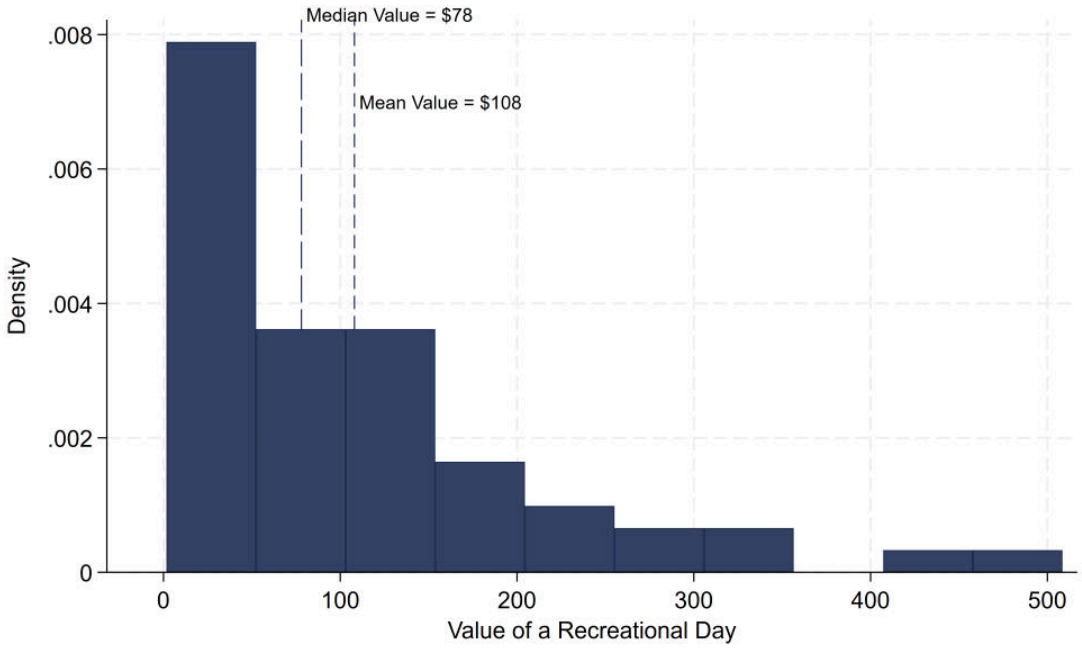


Notes: This figure displays baseline water-based recreation participation rates. Darker shades of blue represent higher participation rates.

Figure C.2: Value of a Recreation Day



(a) Values of a Recreation Day



(b) Values of a Water-Based Recreation Day

Notes: Panels A and B plot histograms of the day value ($2022) for all forms of recreation (Panel A) and water-based recreation (Panel B) for studies that use data collected in the U.S. since the year 2000.

C.2 Property Values

Our property value module provides an empirical representation of the amenity component of total marginal damage from a pulse in upstream nutrient load, given in equation 3 by:

$$\triangle HV = -\sum_{n=1}^{N}\sum_{h=1}^{H_n}\frac{\partial U}{\partial A}\frac{\partial A(d_{hn}, c_n)}{\partial c_n}\triangle c_n \tag{C20}$$

where n indexes neighborhoods, H_n is the number of households in neighborhood n affected by the amenity change, c_n is the nutrient concentration in the local waterbody, and the utility terms denote the marginal willingness to pay by household h for a change in amenity services operating through a change in local water quality. Equation A5 shows that, in equilibrium, the marginal willingness to pay is equal to the hedonic price gradient:

$$\frac{\partial U}{\partial A}\frac{\partial A(d_{hn}, c_n)}{\partial c_n} = \frac{\partial P(\cdot)}{\partial c_n} \tag{C21}$$

Equation 14 details the calculation and components used to estimate equation C20 and C21. The sub-sections below describe these components.

C.2.1 Baseline Housing Value Data

We use the Core Logic Basic Property assessment database to inventory single-family residential homes that are within 300 m of inland lakes larger than 0.1 km^2. We begin by selecting the most recent assessment for single-family residential homes in the conterminous U.S. that occurred in the 2017-2021 window. This generated 92.13 million unique records with valid location and assessed values. We matched these properties to their nearest inland lake larger than 0.1 km^2. There were 45,335 lakes with properties closer than 3 km from shore. We then identified properties with distances equal to or smaller than 300 m to the nearest lake. This resulted in 3.51 million homes near 31,597 lakes. The total assessed value for these properties is \$961.41 billion (2021 dollars). This divides out as V^1 = \$378.85 billion for homes closer than 100 m from shore and V^2 = \$582.55 billion for homes in the 100 m to 300 m distance bin. Figure 3 provides a county-level map of these estimates for the U.S.

C.2.2 Housing Elasticity Estimates

To select the range of elasticities used in our property value module, we reviewed published studies with applications that were U.S. based and focused on freshwater lakes or reservoirs. Appendix Tables C.8 to C.10 shows point estimates and 95 percent confidence intervals for 19 elasticities from eight original studies published between 2016 and 2024. Nine elasticities are for water clarity (measured via Secchi depth) at different distance-from-shore reference points and are the focus of our summary. Elasticities for other water quality parameters are included for context. The table also lists a meta-analysis from Guignet et al. (2022), who reviewed 36 documents (26 in peer reviewed journals) appearing between 1979 and 2017. We show the authors' point estimates and confidence intervals for water clarity.

Including the Guignet et al. (2022) entry, the median water clarity elasticity in Tables C.8 to C.10 for properties less than 100 m from the shoreline is 0.18. The median water clarity elasticity for properties between 100 m and 500 m from the shoreline is 0.05. Figure 3, Panel D provides a corresponding coefficient plot of these estimates.

Based on our review we selected baseline elasticities of 0.18 for properties closer than 100 m to the lakefront and 0.04 for properties in the 100 m to 300 m distance band. We assume that home prices more than 300 m from the lakefront are not affected by water quality changes. We have selected these baselines because they are close to the averages noted above and they correspond to the findings in Mamun et al. (2023), which is the most comprehensive U.S.-wide analysis among the available studies. Mamun et al. (2023) conducted a national-scale hedonic property analysis to measure the relationship between home values and water quality in inland lakes in the conterminous U.S. They assemble water quality data on over 1,600 freshwater lakes from the USEPA Water Quality Portal and the LAGOS-NE database (Soranno et al. 2017). Data on water clarity (measured as Secchi depth) are matched to residential home sales drawn from the Zillow ZTRAX database, occurring between 2000 and 2019 and within 2 km of freshwater lakes. The estimation sample includes nearly 750,000 sales. Identification relies on a fixed effects design that exploits within-lake variation in water quality over time. Mamun et al. (2023) find that a 1 percent increase in water clarity leads to a $\eta_1 = 0.18$ percent increase in sale price for properties within 100 m of the lakeshore. For properties between 100 m and 300 m from shore, a 1 percent increase in clarity increases price by $\eta_2 = 0.04$ percent.

C.2.3 Set of Impacted Lakes

We selected a lake size threshold of 0.1 km^2 as our baseline set of lakes based on the distribution of lake sizes included in Mamun et al. (2023). Specifically, among the 1,632 lakes included in their analysis, the average surface area is 9.84 km^2. The median surface area is 0.67 km^2, the 10th and 90th percentiles are 0.10 and 10.75 km^2, respectively, and the minimum lakes size is 0.04 km^2. The distribution of lake size in the property sales data, however, is skewed further right by the higher volume of sales on larger lakes. Mamun et al. (2023) report a sales-weighted average lake size of 39.83 km^2 with a median of 1.12 km^2. A comparison of the unweighted and sales weighted lake size distribution led to our selection of 0.1 km^2 as our size threshold for lakes in the property values module.

C.2.4 Relationship between Total Phosphorus and Secchi Depth

To measure amenity damages from nutrient pollution, we need to translate changes in nutrient concentration into changes in Secchi depth (water clarity). We use data from all four waves of the NLAs, conducted in 2007, 2012, 2017, and 2022, to establish this relationship (USEPA 2007, 2012, 2017, 2022). The 2007 NLA selected 1,028 lakes representative of approximately 50,000 natural and manmade lakes larger than 0.04 km^2. The 2012, 2017, and 2022 NLAs respectively selected 1,038, 1,005, and 981 lakes, each representative of approximately 110,000, 222,000, and 268,000 natural and manmade lakes larger than 0.01 km^2. A range of water quality parameters was measured for each lake in the two cross-sections, including Secchi depth and TP concentrations. For some lakes, multiple measurements were taken at different times during the sample year. To quantify the relationship between Secchi depth and TP, we pool the four cross-sections and use a log-log specification given by:

$$ln(SD_{ijr}) = v_{0r} + v_r ln(TP_{ijr}) + \epsilon_{ij} \tag{C22}$$

where SD_{ij} is measure i of Secchi depth at lake j in eco-region r and TP_{ij} is TP concentration. The key parameter is v_r, which measures the percentage change in Secchi depth from a 1 percent change in TP. Appendix Table C.7 displays regression results for three models. Column (1) shows an average elasticity of Secchi depth with respect to phosphorus of -0.47. This estimate is negative and statistically significant. We expect this negative relationship because as TP concentrations go up, the clarity of the water should decrease. The parameter remains relatively unchanged when we control for ecological region (Column 2). In Column (3), we estimate nine values of this parameter, each corresponding to an ecological region. The parameter estimates range from 0.39 to 0.50. In Column (4), we estimate the national average elasticity with a model including lake and survey year fixed effects.

C.2.5 Annualization Rate

We use findings from Bishop et al. (2025) to select an annualization rate for translating a change in asset value into an annual measure. Bishop et al. (2025) estimate the time and space-varying annual cost of homeownership and express this cost as a proportion of the asset value, labeled the user cost rate. In their analysis, the user cost rate for a given homeowner depends on variables like the loan-to-value ratio, the rate of return on capital, the mortgage rate, property taxes, the rate of depreciation, expected capital gains, and features of the tax code. Bishop et al. (2025) assemble data and perform analysis to compute the user cost rates for the years 2012 to 2019 for all Public Use Microdata Areas (PUMAs) in the U.S. We use the nationwide average user cost rate for 2019 of $\alpha = 0.0448$ for our model.

C.2.6 Damage Estimates

These hedonic components allow us to compute the amenity mechanism damages from a change in nutrient concentration in downstream river segments following an emissions pulse. We focus on damages from changes in phosphorus concentration. Denote $\triangle c_j$ as the change in phosphorus concentration at a lake j that is affected by an upstream change in load. The damages from the set of lakes J_k that are affected by the load change is:

$$\triangle HV = \sum_{j=1}^{J_k} \alpha \cdot \frac{(V_j^1 \cdot \eta_1 + V_j^2 \cdot \eta_2)}{100} \cdot v_j \cdot \%\triangle c_j, \tag{C23}$$

where α, η_1, η_2, and v_j are defined above, V_j^1 is the total assessed value of homes within 100 m of lake j, and V_j^2 is the total assessed value of homes in the 100 m to 300 m distance band of lake j. We assign the baseline phosphorus concentration for lake j to be equal to the average concentration in the watershed (HUC8) it is contained in. Likewise, the new concentration for lake j following an upstream pulse is assigned based on the new watershed average concentration.

Table C.7: Secchi Depth-Phosphorus Elasticity

	(1)	(2)	(3)	(4)
Ln(P)	-0.4651***	-0.4383***		-0.2093***
	(0.0122)	(0.0139)		(0.0292)
Ln(P) × CPL			-0.4494***	
			(0.0352)	
Ln(P) × NAP			-0.5034***	
			(0.0382)	
Ln(P) × NPL			-0.3871***	
			(0.0430)	
Ln(P) × SAP			-0.4156***	
			(0.0395)	
Ln(P) × SPL			-0.4230***	
			(0.0477)	
Ln(P) × TPL			-0.4313***	
			(0.0337)	
Ln(P) × UMW			-0.4138***	
			(0.0437)	
Ln(P) × WMT			-0.4619***	
			(0.0312)	
Ln(P) × XER			-0.4599***	
			(0.0608)	
CPL		-0.8240***	-0.8562***	
		(0.0548)	(0.1106)	
NAP		0.2821***	0.0378	
		(0.0544)	(0.1966)	
NPL		0.4357***	0.5640***	
		(0.0665)	(0.1546)	
SAP		0.2643***	0.3810**	
		(0.0556)	(0.1921)	
SPL		0.0527	0.1224	
		(0.0662)	(0.1795)	
TPL		0.3119***	0.3614**	
		(0.0532)	(0.1481)	
UMW		0.4069***	0.5309***	
		(0.0522)	(0.2032)	
WMT		0.6451***	0.5843***	
		(0.0521)	(0.1711)	
XER		0.5753***	0.5444**	
		(0.0728)	(0.2383)	
Observations	4,750	4,750	4,750	2,808
R-squared	0.292	0.321	0.322	0.906
Eco-Region-Specific Elasticity	No	No	Yes	No
Eco-Region-Specific Constant	No	Yes	Yes	No
Lake and Year Fixed Effects	No	No	No	Yes
Number of Unique Lakes	2,786	2,786	2,786	1,120

Notes: The dependent variable is the natural log of Secchi depth. The coefficients show the national (Columns 1, 2, and 4) and eco-region-specific (Column 3) elasticities between Secchi depth and phosphorus concentration and eco-region-specific constants (Columns 2 and 3). CPL: Coastal Plains, NAP: Northern Appalachians, NPL: Northern Plains, SAP: Southern Appalachians, SPL: Southern Plains, TPL: Temperate Plains, UMW: Upper Midwest, WMT: Western Mountains, XER: Xeric. CPL is the baseline eco-region constant term. *** Significant at the 1% level; ** Significant at the 5% level.

Table C.8: Housing Price Elasticities with Respect to Water Quality

Study	**Resource Type**	**Water Quality Variable**	**Distance from Shore**	**Context**	**Implied Elasticity**
Guignet et al. (2022)	Various	Secchi Depth	Waterfront	Meta-analysis of elasticities from 18 studies published prior to 2017. Summary drawn from table 1, column 4.	0.11 [0.10 0.12]
Guignet et al. (2022)	Various	Secchi Depth	Non-waterfront	As above, based on 6 studies.	0.026 [0.017 0.034]
Mamun et al. (2023)	Lakes	Secchi Depth	<100m	Nationwide model with 1,600 lakes. Sample is properties 2000 m and closer to shore. Census tract by year fixed effects design. Average lake size 40 km^2. Minimum lake size 0.04 km^2. Estimate drawn from table 2, column 1.	0.18 [0.12 0.23]
Mamun et al. (2023)	Lakes	Secchi Depth	100-300 m	As above.	0.05 [0.01 0.09]
Mamun et al. (2023)	Lakes	Chlorophyll a	<100m	As above. Estimate drawn from table 2, column 2.	-0.12 [-0.06 -0.17]
Mamun et al. (2023)	Lakes	Chlorophyll a	100-300 m	As above. Estimate drawn from table 2, column 2.	-0.04 [-0.02 -0.06]
Moore et al. (2020)	Lakes	Secchi Depth	Lakefront	Nationwide model with 113 lakes. Sample is properties 100 m and closer to shore. IV design. Average lake size 40 km^2. Minimum lake size 0.01 km^2. Estimate drawn from table 2, column 4a.	0.21 [0.13 0.29]
Swedberg et al. (2024)	Lakes	Secchi Depth	Lakefront	Nationwide regression to examine heterogeneity across space. Reported elasticity is for lakefront properties in MN. Sample include properties closer than 2500 m from shore. Census tract by year fixed effects design. Average lake size 10 km^2. Estimate drawn from table 3.	0.26 [0.17 0.34]

Notes: This table summarizes elasticity estimates from the literature. 95 percent confidence intervals in square brackets.

Table C.9: Housing Price Elasticities with Respect to Water Quality (Continued)

Study	**Resource Type**	**Water Quality Variable**	**Distance from Shore**	**Context**	**Implied Elasticity**
Swedberg et al. (2024)	Lakes	Secchi Depth	150 m	As above.	0.061 [0.00 0.13]
Walsh and Milon (2016)	Lakes	Total Phosphorus	Lakefront	Orange County, FL, model with 76 lakes. Sample includes properties 1000 m and closer to shore. Estimates based on a spatial lag model. Average lake size 1.65 km^2. Minimum lake size 0.01 km^2. Estimate drawn from table 3, TP column for average lake size.	-0.12 [-0.06 -0.17]
Walsh and Milon (2016)	Lakes	Chlorophyll a	Lakefront	As above. Estimate drawn from table 3, CHLA column for average lake size.	-0.06 [-0.04 -0.09]
Walsh and Milon (2016)	Lakes	Total Nitrogen	Lakefront	As above. Estimate drawn from table 3, TN column for average lake size.	-0.25 [-0.12 -0.37]
Weng et al. (2020)	Lakes	Secchi Depth	Lakefront	Dane County, WI, model for single lake. No distance threshold for sample. Single lake (Lake Mendota) is 40 km^2. School district fixed effects design. Estimate drawn from table 2, Secchi depth quality variable.	0.13 [0.12 0.14]
Weng et al. (2020)	Lakes	Chlorophyll a	Lakefront	As above. Estimate drawn from table 2, Chl-a quality variable.	-0.057 [-0.049 -0.065]
Wolf and Kemp (2021)	Lakes	Secchi Depth	140 m avg.	Vilas and Oneida County model with 104 lakes. Sample is properties 500 m and closer to shore. Census block group fixed effects design. Average lakes size 6 km^2. Minimum lake size is 0.08 km^2. Estimates drawn from table 4, model 5, Secchi disc quality measurement.	0.11 [0.07 0.14]

Notes: This table summarizes elasticity estimates from the literature. 95 percent confidence intervals in square brackets.

Table C.10: Housing Price Elasticities with Respect to Water Quality (Continued)

Study	Resource Type	Water Quality Variable	Distance from Shore	Context	Implied Elasticity
J. Zhang et al. (2022)	Lakes	Cyanobacteria Blooms	<100 m	Upper Midwest ecoregion model with 2019 lakes. Sample is properties <10 km from nearest lake. Census block group and state-by-sale-year fixed effects design. Quality variable is satellite measured frequency of harmful algal blooms. Estimate drawn from table 4, UM ecoregion.	-0.08 [-0.07 -0.09]
J. Zhang et al. (2022)	Lakes	Cyanobacteria Blooms	<400 m	As above, for the Northeastern ecoregion. Estimate drawn from table 4, NE ecoregion.	-0.075 [-0.068 -0.082]
J. Zhang et al. (2022)	Lakes	Cyanobacteria Blooms	<300 m	As above, for the Southern ecoregion. Estimate drawn from table 4, S ecoregion.	-0.10 [-0.09 -0.11]
J. Zhang et al. (2022)	Lakes	Cyanobacteria Blooms	<400 m	As above, for the Southern ecoregion. Estimate drawn from table 4, SE ecoregion.	-0.05 [-0.005 -0.09]

Notes: This table summarizes elasticity estimates from the literature. 95 percent confidence intervals in square brackets.

C.3 Drinking Water

Our drinking water treatment module identifies CWSs that likely treat for nitrogen and estimates corresponding treatment costs that arise from changes in nitrogen concentrations. Our estimates closely follow the work of Mosheim and Ribaudo (2017) given the study's national scope and detailed nitrogen abatement data. The authors obtained data from 48 CWSs that reported data on the levels of nitrogen in raw and finished drinking water, variable costs of treatment, and characteristics of drinking water systems. The authors also obtained a more limited dataset on baseline variable costs and characteristics of an additional 264 systems, though these did not include information on nitrogen abatement. These data reflect a survey performed by the American Water Works in 1996 and were the most recent data available for use in their study. Using these data, Mosheim and Ribaudo (2017) estimate an elasticity of variable cost with respect to total nitrogen removal of 0.03 for small utilities (those with fewer than 100,000 customers) and 0.004 for large utilities (those with 100,000 customers or more). Given the importance of returns to scale in drinking water treatment, we utilize this differentiation along system size in our model. The authors report other specifications that explore the elasticity of costs with respect to finer system size classifications based on water production scale (small systems with elasticity of treatment costs equal to 0.05, medium systems with an elasticity of 0.006, and large systems with an elasticity of 0.003), network density (which they describe displays an "undetermined pattern"), and baseline nitrogen concentrations (varying from 0.002 for low nitrogen levels, 0.005 for medium levels, and 0.05 for high levels).[2]

To estimate changes in drinking water treatment costs, we first obtained data on the universe of CWSs from USEPA. We use these data to identify intake locations and the set of systems that draw their water from surface waters, groundwater under the influence of surface water or those that purchase water from systems that source from either of the first two categories. These data provide information on system characteristics, including the HUC12 of the source water intake location, count of the population served, activity status (active vs. inactive), primary water source type (groundwater vs. surface water), reporting and violation records, and the use of certain treatment technologies. Starting with a universe of 49,443 active CWSs, we observe intake source HUC12s for 41,084 CWSs. We first restrict our data to those CWSs that draw at least some of their water from surface water sources. Some CWSs draw from both ground and surface water sources. We retain systems where the primary source is classified as "Groundwater under the influence of surface

[2]As far as we are aware, the authors do not report size thresholds to distinguish medium sized CWSs versus small and large.

water" or "Surface water." We also retain systems that purchase water from other systems if the purchased water is sourced from surface water or ground water under the influence of surface water. Below we refer to this grouping as those that have "surface water" sources. This process leaves us with 6,263 CWSs with HUC12 intake location (another 5,318 do not have intake location data). We then link these CWSs to baseline TN concentration by HUC8, obtained from the SPARROW model as described above. However, some CWSs may draw raw water from multiple HUC12s. For instance, 1,918 of the above-mentioned 6,263 systems report multiple intake-source HUC12s. In the absence of information on the relative contribution of each source, we assign equal weights to all intake sources for those systems.

Next, we identify which systems likely treat for nitrogen. This information is not widely shared with USEPA, so we take several steps to identify these systems. First, we identify CWSs where long-run average baseline source water nitrogen concentration is 5 mg/l or above. Although the EPA drinking water threshold for nitrates is 10 mg/l, we assume this level may lead to instances where intake levels exceed the regulatory threshold. For context, Mosheim and Ribaudo (2017) reported that nitrogen (N) concentrations in raw water for systems treating for N averaged 2.87 mg/l, with values ranging from 0.04 mg/l to 57 mg/l. Of our surface water systems, 313 are located in 257 HUC12s (98 HUC8s) where the baseline level of N is 5 mg/l or above. Next, we supplement this list with CWSs that have treatment technologies that are capable of treating for nitrogen. This includes 250 additional surface water systems that have Reverse Osmosis, Electrodialysis, Ion Exchange, and/or Distillation. Although these technologies can remove a range of contaminants, we include them since they can also remove nitrates (Pennino et al. 2017).

We add to this list of treatment systems 180 additional surface water CWSs that are located in HUC12s with a previous history of MCL-N violations. To create this list of HUC12s we examine any MCL-N violation (involving Nitrate, Nitrite or Nitrate-Nitrite) for any CWS or non-CWS regardless of where it sources its water. We obtained these data on N violations from USEPA's SDWIS system from the first quarter of 2018 through the second quarter of 2022.

Using these steps, we arrive at a list of CWSs that likely treat for nitrogen. This list includes surface water CWSs that have (i) elevated long-run average N concentrations in their intake source watershed, (ii) use a treatment technology capable of removing nitrogen, or (iii) draw water from a HUC12 where a MCL-N violation has occurred in the recent past. There are 743 water systems with intake sources located across 662 HUC12s, which are nested within 321 HUC8 watersheds, that meet one or more of the above criteria. The total population served by these systems is approximately 36.8 million, with 79 percent of

the systems serving 1,000 or more people, and 11 percent of the systems serving at least 100,000 people each.

Equation 15 shows that our estimate of drinking water treatment costs requires three components. These include baseline levels of nitrogen in the source water, estimates of baseline variable costs, and an elasticity parameter that relates changes in total nitrogen levels in source water to changes in variable costs. Variable costs and the elasticity parameter both vary by large and small system size, while variable costs also differ across USEPA regions. As described in the paper, we model changes in baseline nitrogen concentration levels within the HUC12 from which a CWS draws water. Second, we utilize the reported elasticities from Mosheim and Ribaudo (2017) and obtain system size and USEPA region classifications from USEPA's SDWIS data. Finally, we calibrate variable cost estimates for large and small systems and by USEPA region using data from Mosheim and Ribaudo (2017). Using data from the full sample of 312 CWSs in the paper, we first deflate variable costs to 2022 dollars, as done in other parts of this study. We assume Mosheim and Ribaudo (2017) report variable costs in 1995 dollars, as reported elsewhere in their paper, but were unable to confirm this base year for variable costs with the authors.

C.4 Health

Our infant health module provides an empirical representation of the birthweight component of total marginal damage from a pulse in upstream nutrient load, given in equation 3 by:

$$\triangle BV = -\sum_{h=1}^{H}\sum_{n=1}^{N}\frac{\partial U}{\partial B}B'_h(c_n)\triangle c_n, \tag{C24}$$

where n indexes neighborhoods and H is the number of households in the landscape. We proxy infant health using birthweight and so the key empirical components include a dose response function linking a change in nutrient pollution to a change birthweight ($B'_h(c_n)\triangle c_n$) and an estimate of the external cost of low birthweight ($\partial U/\partial B$). Equations 16-18 describe assumptions and calculations we use to measure $\triangle BV$. The following provides additional details supporting our assumptions.

C.4.1 Microcystin and Human Health

Several types of cyanobacteria produce toxins in freshwater systems that impact human health. Among these, species belonging to the genus *Microcystis* are dominant in lakes, where they produce the hepatoxic chemical microcystin (Melaram et al. 2024). Exposure to microcystin has been linked to acute and chronic human health conditions. Most of the research on acute reactions has focused on contact through recreation (Stewart et al. 2006; Backer et al. 2010) and drinking water (Wu et al. 2021), though reported clinical cases attributable to exposure through these mechanisms are relatively rare (Otten and Paerl 2015). Research on non-acute effects has emphasized liver and gastrointestinal (Lad et al. 2022), neurological (Batterman et al. 2025), and reproductive (S. Zhang et al. 2021) impacts. We focus on infant birthweight because there is toxicological (Zhao et al. 2020) and epidemiological (Jones 2019) evidence establishing the plausibility and size of birthweight impacts in humans, and births near freshwater resources are frequent enough that in utero exposure to microcystin may constitute an important component of social costs of nutrient pollution in surface water.

Exposure to microcystin is through ingestion from drinking water or during recreation, dermal contact during recreation, and inhaling aerosolized particles. We expect that in utero exposure results from a combination of these, and that the degree of exposure depends on distance from a source lake, especially regarding inhalation. In reviewing human health and aerosolized harmful algal blooms, Lim et al. (2023) report that lake spray aerosolized

particles have been observed more than 25 km inland (May et al. 2018); they also note that microcystin in atmospheric particles is stable and can be transported inland without degradation (Plaas and Paerl 2021). Other research has shown that inhaled microcystin generates a higher level of toxicity than when it is orally ingested (Olson et al. 2020). Finally, Backer et al. (2010) and Wood and Dietrich (2011) have measured microcystin in ambient air samples near freshwater lakes in California and New Zealand, respectively.

Based on this review, we assume in our baseline analysis that in utero exposure to microcystin occurs up to 1 mile from the shoreline of lakes larger than 0.01 km^2. This is a conservative assumption that is supported by three other studies. First, in studying airborne toxins from manufacturing plants, Currie et al. (2015) find that ambient concentrations are detected up to one mile from an emission source, and that birthweight is affected by plants operating up to 1 mile away. Second, Batterman et al. (2025) show that individuals with amyotrophic lateral sclerosis (ALS) who live within 5 km of waterbodies with high cyanobacteria concentrations have shorter survivability. Finally, most residents of the two microcystin treated counties in Jones (2019) live within 25 miles of Gull Lake.

C.4.2 Nutrients and Microcystin

To measure the birthweight cost of nutrient pollution, we need to translate changes in nutrient pollution into changes in microcystin concentration. We use a protocol like what was described in Section C.2.4 for TP and Secchi depth, whereby we use data from the four waves of the USEPA NLAs to estimate the relationship

$$\ln(MC_{ij}) = \phi_{0r} + \phi_r \ln(TP_{ij}) + \epsilon_{ij} \tag{C25}$$

where MC_{ij} is the concentration of microcystin in lake i during wave j and TP_{ij} is TP concentration. The detection limit for microcystin in the NLA data is 0.1 µg/l. For samples below the detection limit we randomly assign an outcome between 0 and 0.1 prior to estimation. The key parameter is ϕ_r, which measures the percentage change in microcystin from a 1 percent change in TP. Appendix Table C.11 displays regression results for three models. Column 1 show an average elasticity of 0.45, which reduces to 0.40 when we add ecoregion fixed effects (Column 2). The positive elasticity confirms that microcystin concentration is higher in lakes with higher TP concentration. This relationship is supported by limnological research on causes of cyanobacteria blooms in freshwater resources (M. Beaulieu et al. 2013; Brehob et al. 2024; Buley et al. 2022; Yuan and Pollard 2015). In Column 3 we estimate nine elasticity values, each corresponding to an ecological region. The estimates range from

0.24 to 0.62. Column 3 provides the elasticities used to compute equation 16 for each lake. Column 4 shows that our elasticity estimates is still significant when including lake and year fixed effects in our model. As with our TP and Secchi results, this parameter is smaller than our baseline estimate; the sample size is also significantly smaller for the set of lakes where we observe multiple rounds of sampling.

C.4.3 Baseline Birth and Low Birthweight Incidence

We assemble information on annual births occurring within 1 mile of lakes and the county-level incidence of low birthweight from University of Wisconsin Population Health Institute's County Health Rankings & Roadmaps (CHR&R) project. Statistics on birth and low birthweight outcomes are published annually based on rolling averages of previous years' National Center for Health Statistics data. The 2023 report summarizes natality data for the years spanning 2014-2020.[3] For the county-level incidence of low birthweight we use fraction of births over the seven years that were below 2,500 g. For annual county-level births we use the total births during 2014-2020 divided by seven. To assign baseline annual births we first compute the proportion of each county's population that lives within 1 mile of a lake larger than 0.01 km^2. We then assume that the proportion of births in the county within 1 mile of a lake is equal to the proportion of the population that lives within 1 mile. Figure C.3 shows the distribution of births near lakes in our model and baseline low birthweight incidence. These data provide the variables LB_j and B_{jk} for k=1 mile in equation 17.

C.4.4 Economic Cost of Low Birthweight

There is a large literature in economics and public health establishing a robust relationship between birthweight and economic and health outcomes in adulthood. Examples include Behrman and Rosenzweig (2004), Almond et al. (2005), Black et al. (2007), and Bharadwaj et al. (2018). A common research design is to use twins fixed effects, whereby the difference in birthweight within twin pairs is compared to later in life earnings and health outcomes. Bharadwaj et al. (2018) is unique among the high-quality research designs in the range of outcomes studied, including earnings across the lifecycle. For a sample of twins born in Sweden between 1933 and 1958, the authors examine the effect of birthweight on permanent income and permanent labor income, defined as the average taken across ages 35 to 45. In Table 4, Column 4, Panel B they report that a 10 percent increase in birth weight increases

[3]For a description of the data, see University of Wisconsin Population Health Institute (2023).

permanent labor income by 1.25 percent. To translate this into an economic cost of low birthweight, we need an estimate of the average present discounted value of lifetime wage earnings for workers in the U.S. Guvenen et al. (2022) study trends in lifetime earnings, defined as average annual earnings across 31 working years spanning ages 25 to 55. In Table C.12, the authors report mean annual earnings of $42,330 (2013 USD) and $6,300 in non-wage compensation for workers who were 25 in 1983 (the youngest cohort in their data), or $59,300 in 2022 dollars. Assuming forty working years and discounting at 3 percent, we arrive at $1.43 million as the discounted sum of annual earnings for a typical worker.

For medical costs, Russell et al. (2007) analyze the 2001 National Inpatient Sample from the Healthcare Cost and Utilization Project. Table 1 shows that the median length of hospital stay for a low birthweight infant is 4 days, compared to 2 days for uncomplicated newborns. The mean and median costs for low birthweight infants are $15,00 and $3,000, respectively in 2001 dollars. The corresponding figures for uncomplicated newborns are $600 and $500. We use the difference in medians and convert to 2022 USD to arrive at the median additional health costs from low birthweight as $4,200.

C.4.5 Damage Estimates

Based on the findings in Bharadwaj et al. (2018), we assume a birthweight outcome of 2,500 g reduces lifetime discounted earnings by 3.25 percent, relative to 3,400 g. This implies lost earnings of $46,500 per low birthweight outcome. Combining this with our estimate of additional health care costs lead to VBW=$50,700. The birthweight mechanism damages from an upstream phosphorus pulse affecting a set of downstream lakes J_k is

$$\triangle BV = VBW \sum_{j=1}^{J_k} 100^{-1} \left(\underbrace{\epsilon_{LB} \cdot \phi_j \cdot \%\triangle c_j}_{\%\triangle LB_j} \right) \cdot \left(\underbrace{LB_j \cdot B_{j1}}_{LBW_{0j}} \right), \tag{C26}$$

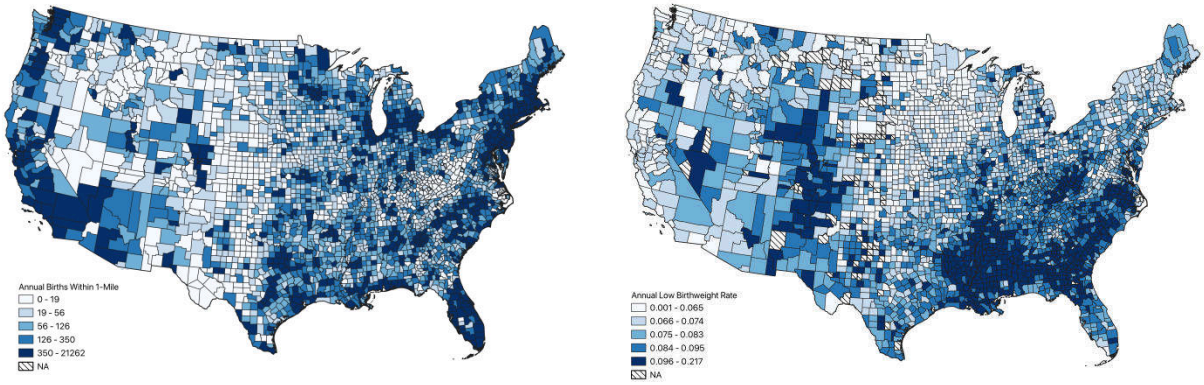
where $\%\triangle LB_j$ is the percentage change in the incidence of low birthweight at lake j from the nutrient pulse, LBW_{0j} is the baseline number of low weight births in within 1 mile of lake j, and the other terms are defined in Section 4.4.

Table C.11: Microcystin-Phosphorus Elasticity

	(1)	(2)	(3)	(4)
Ln(P)	0.4536***	0.4030***		0.2841***
	(0.0179)	(0.0199)		(0.0600)
Ln(P) × CPL			0.2930***	
			(0.0536)	
Ln(P) × NAP			0.2548***	
			(0.0608)	
Ln(P) × NPL			0.5194***	
			(0.0716)	
Ln(P) × SAP			0.2403***	
			(0.0557)	
Ln(P) × SPL			0.5406***	
			(0.0635)	
Ln(P) × TPL			0.6202***	
			(0.0617)	
Ln(P) × UMW			0.5314***	
			(0.0654)	
Ln(P) × WMT			0.2963***	
			(0.0403)	
Ln(P) × XER			0.2714***	
			(0.0590)	
CPL		-1.2208***	-1.5419***	
		(0.0548)	(0.1106)	
NAP		0.2821***	0.0378	
		(0.0544)	(0.1966)	
NPL		0.4357***	0.5640***	
		(0.0665)	(0.1546)	
SAP		0.2643***	0.3810**	
		(0.0556)	(0.1921)	
SPL		0.0527	0.1224	
		(0.0662)	(0.1795)	
TPL		0.3119***	0.3614**	
		(0.0532)	(0.1481)	
UMW		0.4069***	0.5309***	
		(0.0522)	(0.2032)	
WMT		0.6451***	0.5843***	
		(0.0521)	(0.1711)	
XER		0.5753***	0.5444**	
		(0.0728)	(0.2383)	
Observations	4,784	4,784	4,784	2,860
R-squared	0.149	0.186	0.198	0.645
Eco-Region-Specific Elasticity	No	No	Yes	No
Eco-Region-Specific Constant	No	Yes	Yes	No
Lake and Year Fixed Effects	No	No	No	Yes
Number of Unique Lakes	2,790	2,790	2,790	1,138

Notes: The dependent variable is the natural log of microcystin concentration (µg/l). The coefficients show the national (Columns 1, 2, and 4) and eco-region-specific (Column 3) elasticities between microcystin and Total Phosphorus concentration and eco-region-specific constants (Columns 2 and 3). CPL: Coastal Plains, NAP: Northern Appalachians, NPL: Northern Plains, SAP: Southern Appalachians, SPL: Southern Plains, TPL: Temperate Plains, UMW: Upper Midwest, WMT: Western Mountains, XER: Xeric. CPL is the baseline eco-region constant term. *** Significant at the 1% level; ** Significant at the 5% level.

Figure C.3: Annual Births Within 1-Mile Radius and Low Birthweight Rate by County



(a) Annual Births Within 1-Mile Radius

(b) Annual Low Birthweight Rate

Notes: Panel (a) displays the estimated average annual number of births occurring within a 1-mile radius for each county. Darker shades indicate higher numbers of births. Panel (b) shows the average annual low birthweight rate by county, defined as the share of live births weighing less than 2,500 grams. Darker shades represent higher rates. Counties with missing data are shown with hatched lines.

C.5 Climate Change

Our main text focuses on the damages associated with methane given that it is responsible for nearly all climate change related damages. Here we describe how we also estimate damages for carbon dioxide and nitrous oxide.

Utilizing historical data from spatially varying lentic ecosystems, DelSontro et al. (2018) and J. J. Beaulieu et al. (2019) studied the relationship between nutrient concentration, lake surface area, and emissions of major greenhouse gases. In this study, we include all lakes larger than 0.01 km^2, except the Great Lakes and Great Salt Lake, to quantify GHG emissions resulting from nutrient–driven eutrophication.

To predict emissions of CO_2 from total phosphorus concentration in lake j, DelSontro et al. (2018) report the following best-fit model:

$$\begin{aligned}\log_{10}(\mathrm{CO}_{2j}+43) = 2.447 - 0.034\log_{10}(\text{Surface Area}_j) + 0.08\log_{10}(\mathrm{TP}_j) \\ - 0.072\log_{10}(\text{Surface Area}_j)\log_{10}(\mathrm{TP}_j),\end{aligned} \tag{C27}$$

where the units for CO_2 are mg per m^2 of surface area per day. As equation C27 suggests, due to internal biophysical processes that vary with lake area, a lentic system can act as both a sink and a source (DelSontro et al. 2018). The prediction of nitrous oxide (N_2O) follows a similar structure as for methane:

$$\log_{10}(\mathrm{N_2O}_j+0.25) = -0.505 + 0.030\log_{10}(\text{Surface Area}_j) + 0.104\log_{10}(\mathrm{Chla}_j), \tag{C28}$$

where the unit of N_2O is mg per m^2 of surface area per day. We use equation C27 and the lake surface area to predict how an increase in phosphorus concentration, Δc_j, translates into additional CO_2 emission from lake j. For nitrous oxide (N_2O), we use equation 19, lake surface areas estimates, and (C28) to predict how an increase in phosphorus concentration, Δc_j, generates additional annual N_2O emissions from lake j.

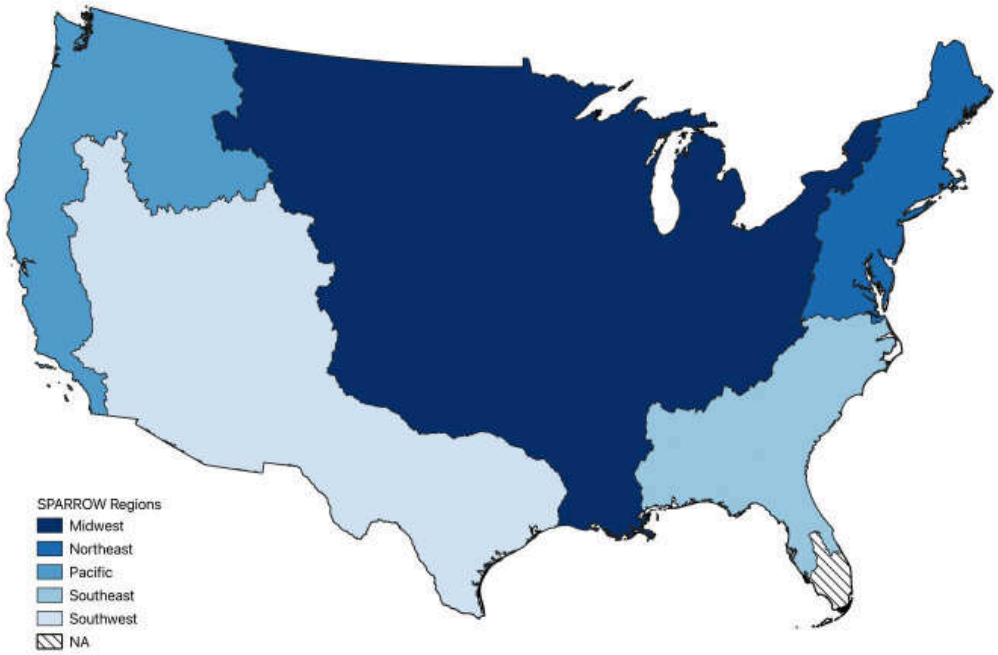
To monetize the damages from the emission changes, we use the social cost of one metric ton of carbon dioxide and nitrous oxide, based on USEPA (2022b). Using a 2% discount rate, we assign $SCC = \$200$ and $SCN = \$56,590$ to calculate the annual social cost from a change in CO_2 and N_2O emissions in lake j based on:

$$\Delta CC_{j,\mathrm{CO_2}} = \Delta CO_2(\Delta c_j)\cdot SCC,$$

$$\Delta CC_{j,\mathrm{N_2O}} = \Delta N_2O(\Delta c_j)\cdot SCN.$$

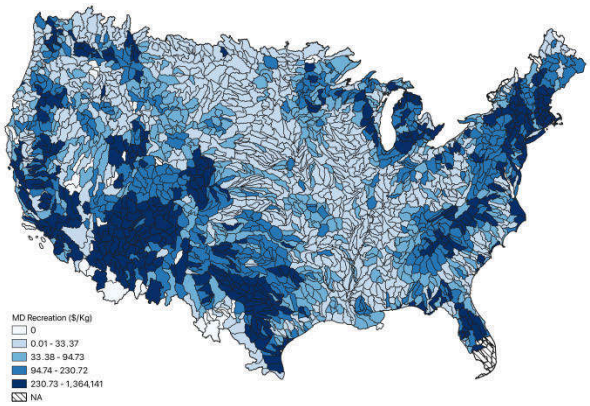
C.6 Additional Maps

Figure C.4: SPARROW Regions Used for Watershed Modeling

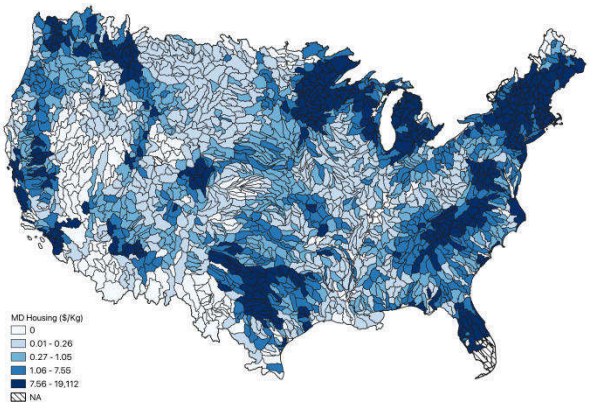


Notes: This map shows the SPARROW modeling regions across the contiguous United States. Each region represents a grouping of watersheds used by the USGS to estimate nutrient loading and water quality damages. Areas in hatched lines (e.g., Southern Florida) are excluded from the regional models.

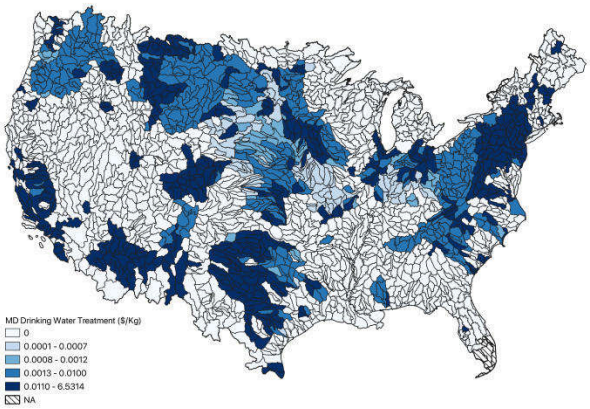
Figure C.5: Average Marginal Damages by Use by Watershed



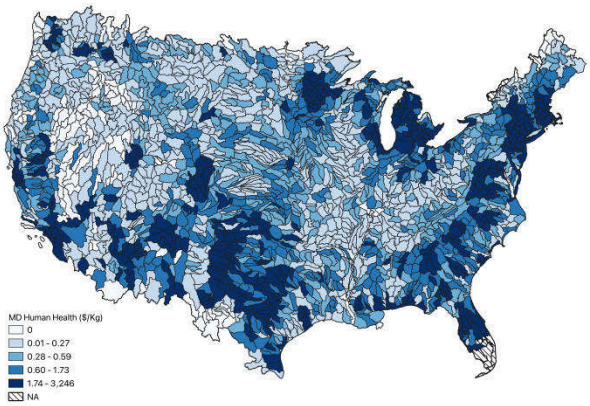
(a) Recreation



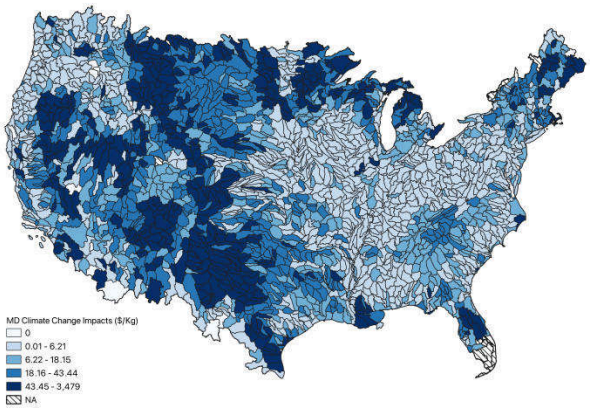
(b) Housing



(c) Drinking Water Treatment



(d) Human Health



(e) Climate Change Impacts

Notes: This figure plots average marginal damages by watershed by use category. Panel a) corresponds to water-based recreation damages, Panel b) displays housing damages, Panel c) captures drinking water treatment costs, Panel d) depicts damages related to human health and Panel e) reflects climate change impacts. Darker shades of blue represent larger damages. Watersheds where damages are not estimated are displayed with hatched lines.

C.7 Additional Results

Table C.12: Variation in Marginal Damages

	(1)	(2)	(3)	(4)
Mean stream flow (cfs)	-0.0006***	-0.0003**	-0.0003*	-0.0002
	(0.0002)	(0.0002)	(0.0002)	(0.0001)
Stream level	50.0466***	23.1754***	27.7243***	29.6619***
	(10.4651)	(8.1337)	(6.4171)	(6.9720)
Distance from start to end (km)	-0.0698***	0.0533***	-0.0084	0.0393
	(0.0098)	(0.0116)	(0.0094)	(0.0265)
Local Population (within 25 miles)	0.0002***	0.0002***	0.0002***	0.0002***
	(6.89e-06)	(0.00001)	(8.85e-06)	(0.00001)
Constant	56.6644	-14.7544	66.3103*	-16.8499
	(45.8644)	(41.3270)	(36.1439)	(48.1850)
Observations	2,454,514	2,454,514	2,452,356	2,452,356
Fixed Effects	No	HUC2	State	HUC2 and State

Notes: This table displays results from regressions of marginal damages ($) by river segment on several hydrologic and geographic features. Standard errors are clustered at the HUC8 level. *** Significant at the 1 percent level; ** Significant at the 5 percent level; * Significant at the 10 percent level.

Appendix References

Abidoye, Babatunde O., and Joseph A. Herriges. 2012. "Model Uncertainty in Characterizing Recreation Demand." *Environmental and Resource Economics* 53 (2): 251–277.

Abidoye, Babatunde O., Joseph A. Herriges, and Justin L. Tobias. 2012a. "Controlling for Observed and Unobserved Site Characteristics in RUM models of Recreation Demand." *American Journal of Agricultural Economics* 94 (5): 1070–1093.

———. 2012b. "Controlling for Observed and Unobserved Site Characteristics in RUM models of Recreation Demand." *American Journal of Agricultural Economics* 94(5):1070–1093.

Almond, Douglas, Kenneth Y. Chay, and David S. Lee. 2005. "The Costs of Low Birth Weight." *The Quarterly Journal of Economics* 120 (3): 1031–1083.

Ator, Scott W. 2019. "Spatially Referenced Models of Streamflow and Nitrogen, Phosphorus, and Suspended Sediment Loads in Streams of the Northeastern United States." U.S. Geological Survey Scientific Investigations Report No. 2019-5118.

Backer, Lorraine C., Sandra V. McNeel, Terry Barber, Barbara Kirkpatrick, Christopher Williams, Mitch Irvin, Yue Zhou, Trisha B. Johnson, Kate Nierenberg, Mark Aubel, Rebecca LePrell, Andrew Chapman, Amanda Foss, Susan Corum, Vincent R. Hill, Stephanie M. Kieszak, and Yung-Sung Cheng. 2010. "Recreational Exposure to Microcystins During Algal Blooms in Two California Lakes." *Toxicon* 55 (5): 909–921.

Batterman, Stuart A., Md Kamrul Islam, Dae Gyu Jang, Eva L. Feldman, and Stephen A. Goutman. 2025. "Life Course Exposure to Cyanobacteria and Amyotrophic Lateral Sclerosis Survival." *International Journal of Environmental Research and Public Health* 22 (5): 763.

Beaulieu, Jake J, Tonya DelSontro, and John A Downing. 2019. "Eutrophication Will Increase Methane Emissions from Lakes and Impoundments during the 21st Century." *Nature communications* 10 (1): 1375.

Beaulieu, Marieke, Frances Pick, and Irene Gregory-Eaves. 2013. "Nutrients and Water Temperature Are Significant Predictors of Cyanobacterial Biomass in a 1147 Lakes Data Set." *Limnology and Oceanography* 58 (5): 1736–1746.

Behrman, Jere R., and Mark R. Rosenzweig. 2004. "Returns to Birthweight." *The Review of Economics and Statistics* 86 (2): 586–601.

Bharadwaj, Prashant, Petter Lundborg, and Dan-Olof Rooth. 2018. "Birth Weight in the Long Run." *Journal of Human Resources* 53 (1): 189–231.

Bishop, Kelly C., Jakob Dowling, Nicolai V. Kuminoff, and Alvin Murphy. 2025. "Tax Policy and the Heterogeneous Costs of Homeownership." *Accepted at Journal of Urban Economics.*

Black, Sandra E., Paul J. Devereux, and Kjell G. Salvanes. 2007. "From the Cradle to the Labor Market? The Effect of Birth Weight on Adult Outcomes." *The Quarterly Journal of Economics* 122 (1): 409–439.

Brehob, Meredith M., Michael J. Pennino, Amalia M. Handler, Jana E. Compton, Sylvia S. Lee, and Robert D. Sabo. 2024. "Estimates of Lake Nitrogen, Phosphorus, and Chlorophyll-*a* Concentrations to Characterize Harmful Algal Bloom Risk Across the United States." *Earth's Future* 12 (8): e2024EF004493.

Buley, Riley P., Matthew F. Gladfelter, Edna G. Fernandez-Figueroa, and Alan E. Wilson. 2022. "Can Correlational Analyses Help Determine the Drivers of Microcystin Occurrence in Freshwater Ecosystems? A Meta-Analysis of Microcystin and Associated Water Quality Parameters." *Environmental Monitoring and Assessment* 194 (7): 493.

Cha, Wonkyu, and Richard T. Melstrom. 2018. "Catch-and-release Regulations and Paddlefish Angler Preferences." *Journal of Environmental Management* 214:1–8.

College of Forestry, Oregon State University. 1981-2010. "Recreational Use Values Database." Accessed September 10, 2013. http://recvaluation.forestry.oregonstate.edu/.

Currie, Janet, Lucas Davis, Michael Greenstone, and Reed Walker. 2015. "Environmental Health Risks and Housing Values: Evidence from 1,600 Toxic Plant Openings and Closings." *American Economic Review* 105(2):678–709.

DelSontro, Tonya, Jake J Beaulieu, and John A Downing. 2018. "Greenhouse Gas Emissions from Lakes and Impoundments: Upscaling in the Face of Global Change." *Limnology and Oceanography Letters* 3 (3): 64–75.

Egan, Kevin J., Joseph A. Herriges, Catherine L. Kling, and John A. Downing. 2009. "Valuing Water Quality as a Function of Water Quality Measures." *American Journal of Agricultural Economics* 91(1):106–123.

Evans, Keith S., Joseph A. Herriges, and Catherine L. Kling. 2011. *Iowa Lakes Survey 2009: A Report to the Iowa Department of Natural Resources.* Technical report.

Guignet, Dennis, Matthew T. Heberling, Michael Papenfus, and Olivia Griot. 2022. "Property Values, Water Quality, and Benefit Transfer: A Nationwide Meta-Analysis." *Land Economics* 98(2):198–218.

Guvenen, Fatih, Greg Kaplan, Jae Song, and Justin Weidner. 2022. "Lifetime Earnings in the United States over Six Decades." *American Economic Journal: Applied Economics* 14 (4): 446–479.

Hoos, Anne B., and Victor L. Roland II. 2019. "Spatially Referenced Models of Streamflow and Nitrogen, Phosphorus, and Suspended Sediment Loads in Streams of the Southeastern United States." U.S. Geological Survey Scientific Investigations Report No. 2019-5135.

Iowa State University. 2022. *Iowa Lakes Project.* Accessed June 5, 2025. https://lakes.card.iastate.edu/interactive-maps.

Jayasekera, D. Harshanee, Richard T. Melstrom, and Kevin L. Pope. 2024. "Economic Losses to Inland Recreational Fisheries from Harmful Algal Blooms." *Journal of Environmental Management* 372:123238.

Ji, Yongjie, David A. Keiser, and Catherine L. Kling. 2020a. "Temporal Reliability of Welfare Estimates from Revealed Preferences." *Journal of the Association of Environmental and Resource Economists* 7 (4): 659–686.

———. 2020b. "Temporal Reliability of Welfare Estimates from Revealed Preferences." *Journal of the Association of Environmental and Resource Economists* 7(4):659–686.

Jones, Benjamin A. 2019. "Infant Health Impacts of Freshwater Algal Blooms: Evidence from an Invasive Species Natural Experiment." *Journal of Environmental Economics and Management* 96:36–59.

Keeler, Bonnie L., Spencer A. Wood, Stephen Polasky, Catherine Kling, Christopher T. Filstrup, and John A. Downing. 2015. "Recreational Demand for Clean Water: Evidence from Geotagged Photographs by Visitors to Lakes." *Frontiers in Ecology and the Environment* 13(2):76–81.

Keiser, David A. 2019. "The Missing Benefits of Clean Water and the Role of Mismeasured Pollution Data." *Journal of the Association of Environmental and Resource Economists* 6(4):669–707.

Keiser, David A., and Joseph S. Shapiro. 2019. "Consequences of the Clean Water Act and the Demand for Water Quality." *Quarterly Journal of Economics* 134(1):349–396.

Kuwayama, Yusuke, Sheila Olmstead, and Jiameng Zheng. 2022. "A More Comprehensive Estimate of the Value of Water Quality." *Journal of Public Economics* 207:104600.

Lad, Apurva, Joshua D. Breidenbach, Robin C. Su, Jordan Murray, Rebecca Kuang, Alison Mascarenhas, John Najjar, Shivani Patel, Prajwal Hegde, Mirella Youssef, Jason Breuler, Andrew L. Kleinhenz, Andrew P. Ault, Judy A. Westrick, Nikolai N. Modyanov, David J. Kennedy, and Steven T. Haller. 2022. "As We Drink and Breathe: Adverse Health Effects of Microcystins and Other Harmful Algal Bloom Toxins in the Liver, Gut, Lungs and Beyond." *Life* 12 (3): 418.

Lim, Chris C., Jeonggyo Yoon, Kelly Reynolds, Lynn B. Gerald, Andrew P. Ault, Seulkee Heo, and Michelle L. Bell. 2023. "Harmful Algal Bloom Aerosols and Human Health." *eBioMedicine* 93:104604.

Lloyd-Smith, Patrick, and Ewa Zawojska. 2025. "How Stable and Predictable Are Welfare Estimates Using Recreation Demand Models?" *American Journal of Agricultural Economics* 107 (3): 846–868.

Lupi, Frank, Roger H. Von Haefen, and Li Cheng. 2022. "Distributional Effects of Entry Fees and Taxation for Financing Public Beaches." *Land Economics* 98 (3): 509–19.

Mamun, Saleh, Adriana Castillo-Castillo, Kristen Swedberg, Jiarui Zhang, Kevin J. Boyle, Diego Cardoso, Catherine L. Kling, Christoph Nolte, Michael Papenfus, Daniel Phaneuf, and Stephen Polasky. 2023. "Valuing Water Quality in the United States Using a

National Dataset on Property Values." *Proceedings of the National Academy of Sciences* 120 (15): e2210417120.

May, Nathaniel W., Matthew J. Gunsch, Nicole E. Olson, Amy L. Bondy, Rachel M. Kirpes, Steven B. Bertman, Swarup China, Alexander Laskin, Philip K. Hopke, Andrew P. Ault, and Kerri A. Pratt. 2018. "Unexpected Contributions of Sea Spray and Lake Spray Aerosol to Inland Particulate Matter." *Environmental Science & Technology Letters* 5 (7): 405–412.

McConnell, Kenneth E. 1995. "Consumer Surplus from Discrete Choice Models." *Journal of Environmental Economics and Management* 29 (3): 263–270.

Melaram, Rajesh, Amanda Rose Newton, Anna Lee, Scott Herber, Anthony El-Khouri, and Jennifer Chafin. 2024. "A Review of Microcystin and Nodularin Toxins Derived from Freshwater Cyanobacterial Harmful Algal Blooms and Their Impact on Human Health." *Toxicology and Environmental Health Sciences* 16 (3): 233–241.

Melstrom, Richard T, and Carson Reeling. 2024. "A Zonal RUM Model to Value Recreation Sites with Aggregate Visitation Data." *Land Economics* 100 (4): 589–605.

Melstrom, Richard T., and Deshamithra HW. Jayasekera. 2017. "Two-stage Estimation to Control for Unobservables in a Recreation Demand Model with Unvisited Sites." *Land Economics* 93 (2): 328–341.

Melstrom, Richard T., Frank Lupi, Peter C. Esselman, and R. Jan Stevenson. 2015. "Valuing Recreational Fishing Quality at Rivers and Streams." *Water Resources Research* 51 (1): 140–150.

Merrill, Nathaniel, Marisa Mazzotta, Kate Mulvaney, Joshua Sawyer, Julia Twichell, Sarina Atkinson, Darryl Keith, and Laura Erban. 2024. "The Value of Water Quality for Coastal Recreation in New England, USA." *Agricultural and Resource Economics Review* 53(3):492–514.

Moore, Michael R., Jonathan P. Doubek, Hui Xu, and Bradley J. Cardinale. 2020. "Hedonic Price Estimates of Lake Water Quality: Valued Attribute, Instrumental Variables, and Ecological-Economic Benefits." *Ecological Economics* 176:106692.

Mosheim, Roberto, and Marc Ribaudo. 2017. "Costs of Nitrogen Runoff for Rural Water Utilities: A Shadow Cost Approach." *Land Economics* 93 (1): 12–39.

Nelson, Erik, Maggie Rogers, Spencer A. Wood, Jesse Chung, and Bonnie Keeler. 2023. "Data-driven Predictions of Summertime Visits to Lakes Across 17 US States." *Ecosphere* 14(4):e4457.

Olson, Nicole E., Madeline E. Cooke, Jia H. Shi, Johnna A. Birbeck, Judy A. Westrick, and Andrew P. Ault. 2020. "Harmful Algal Bloom Toxins in Aerosol Generated from Inland Lake Water." *Environmental Science & Technology* 54 (8): 4769–4780.

Otten, Timothy G., and Hans W. Paerl. 2015. "Health Effects of Toxic Cyanobacteria in U.S. Drinking and Recreational Waters: Our Current Understanding and Proposed Direction." *Current Environmental Health Reports* 2 (1): 75–84.

Pennino, Michael J., Jana E. Compton, and Scott G. Leibowitz. 2017. "Trends in Drinking Water Nitrate Violations Across the United States." *Environmental Science & Technology* 51 (22): 13450–13460.

Phaneuf, Daniel J., Jared C. Carbone, and Joseph A. Herriges. 2009. "Non-price Equilibria for Non-Marketed Goods." *Journal of Environmental Economics and Management* 57(1):45–64.

Phaneuf, Daniel J., and Till Requate. 2016. *A Course in Environmental Economics: Theory, Policy, and Practice.* Cambridge University Press.

Plaas, Haley E., and Hans W. Paerl. 2021. "Toxic Cyanobacteria: A Growing Threat to Water and Air Quality." *Environmental Science & Technology* 55 (1): 44–64.

Preston, Stephen D., Richard B. Alexander, Michael D. Woodside, and Pixie A. Hamilton. 2009. "SPARROW Modeling - Enhancing Understanding of the Nation's Water Quality." U.S. Geological Survey Fact Sheet 2009 - 3019.

Robertson, Dale M., and David A. Saad. 2019. "Spatially Referenced Models of Streamflow and Nitrogen, Phosphorus, and Suspended Sediment Loads in Streams of the Midwestern United States." U.S. Geological Survey Scientific Investigations Report No. 2019-5114.

Russell, Rebecca B., Nancy S. Green, Claudia A. Steiner, Susan Meikle, Jennifer L. Howse, Karalee Poschman, Todd Dias, Lisa Potetz, Michael J. Davidoff, Karla Damus, and Joann R. Petrini. 2007. "Cost of Hospitalization for Preterm and Low Birth Weight Infants in the United States." *Pediatrics* 120 (1): e1–e9.

Saad, David A., Gregory E. Schwarz, Denise M. Argue, David W. Anning, Scott A. Ator, Anne B. Hoos, Stephen D. Preston, Dale M. Robertson, and Daniel Wise. 2019. "Estimates of Long-Term Mean Daily Streamflow and Annual Nutrient and Suspended-sediment Loads Considered for Use in Regional SPARROW Models of the Conterminous United States, 2012 Base Year." U.S. Geological Survey Scientific Investigations Report No. 2019-5069.

Schwarz, Gregory E., A.B. Hoos, R.B. Alexander, and R.A. Smith. 2006. "The SPARROW Surface Water-quality Model: Theory, Application and User Documentation." U.S. Geological Survey Techniques and Methods Book 6, Section B, Chapter 3.

Smith, Jordan W., and Chase C. Lamborn. 2024. "The Influence of Population Growth and Weather on the Value of Recreational Angling Trips within Utah (USA)." *Fisheries Research* 274:106976.

Smith, Richard A., Gregory E. Schwarz, and Richard B. Alexander. 1997. "Regional Interpretation of Water-Quality Monitoring Data." *Water Resources Research* 33 (12): 2781–2798.

Soranno, Patricia A., Linda C. Bacon, Michael Beauchene, Karen E. Bednar, Edward G. Bissell, Claire K. Boudreau, Marvin G. Boyer, and et al. 2017. "LAGOS-NE: A Multi-Scaled Geospatial and Temporal Database of Lake Ecological Context and Water Quality for Thousands of US Lakes." *GigaScience* 6(12):gix101.

Stewart, Ian, Penelope M. Webb, Philip J. Schluter, Lora E. Fleming, John W. Burns, Miroslav Gantar, Lorraine C. Backer, and Glen R. Shaw. 2006. "Epidemiology of Recreational Exposure to Freshwater Cyanobacteria – An International Prospective Cohort Study." *BMC Public Health* 6 (1): 93.

Swedberg, Kristen, Diego S. Cardoso, Adriana Castillo-Castillo, Saleh Mamun, Kevin J. Boyle, Christoph Nolte, Michael Papenfus, and Stephen Polasky. 2024. "Spatial Heterogeneity in Hedonic Price Effects for Lake Water Quality." *Land Economics* 100(1):89–108.

Timmins, Christopher, and Jennifer Murdock. 2007. "A Revealed Preference Approach to the Measurement of Congestion in Travel Cost Models." *Journal of Environmental Economics and Management* 53 (2): 230–249.

Train, Kenneth E. 2009. *Discrete Choice Methods with Simulation.* Cambridge University Press.

University of Wisconsin Population Health Institute. 2023. *County Health Rankings & Roadmaps.* https://www.countyhealthrankings.org/health-data/methodology-and-sources/data-documentation/national-data-documentation-2010-2023.

USDA. 2009. "National Survey on Recreation and the Environment." The Interagency National Survey Consortium, Coordinated by the USDA Forest Service, Recreation, Wilderness, Demographics Trends Research Group, Athens, GA, and the Human Dimensions Research Laboratory. Accessed August 29, 2013. http://nhts.ornl.gov/det/.

USEPA. 2007. "National Lakes Assessment." http://water.epa.gov/type/lakes/lakessurvey_index.cfm.

———. 2012. "National Lakes Assessment." http://water.epa.gov/type/lakes/lakessurvey_index.cfm.

———. 2017. "National Lakes Assessment." http://water.epa.gov/type/lakes/lakessurvey_index.cfm.

———. 2022. "National Lakes Assessment." https://nationallakesassessment.epa.gov/webreport/.

USGS. 2020. *SPARROW Mappers.* https://www.usgs.gov/mission-areas/water-resources/science/sparrow-mappers.

Viscusi, W. Kip, Joel Huber, and Jason Bell. 2008. "The Economic Value of Water Quality." *Environmental and Resource Economics* 41:169–187.

Walsh, Patrick J., and J. Walter Milon. 2016. "Nutrient Standards, Water Quality Indicators, and Economic Benefits from Water Quality Regulations." *Environmental and Resource Economics* 64(4):643–661.

Weng, Weizhe, Kevin J. Boyle, Kaitlin J. Farrell, Cayelan C. Carey, Kelly M. Cobourn, Hilary A. Dugan, Paul C. Hanson, Nicole K. Ward, and Kathleen C. Weathers. 2020.

"Coupling Natural and Human Models in the Context of a Lake Ecosystem: Lake Mendota, Wisconsin, USA." *Ecological Economics* 169:106556.

Wise, Daniel R. 2019. "Spatially Referenced Models of Streamflow and Nitrogen, Phosphorus, and Suspended Sediment Loads in Streams of the Pacific Region of the United States." U.S. Geological Survey Scientific Investigations Report No. 2019-5112.

Wise, Daniel R., David W. Anning, and Olivia L. Miller. 2019. "Spatially Referenced Models of Streamflow and Nitrogen, Phosphorus, and Suspended Sediment Transport in Streams of the Southwestern United States." U.S. Geological Survey Scientific Investigations Report No. 2019-5106.

Wolf, David, Wei Chen, Sathya Gopalakrishnan, Timothy Haab, and H Allen Klaiber. 2019. "The Impacts of Harmful Algal Blooms and E. Coli on Recreational Behavior in Lake Erie." *Land Economics* 95(4):455–472.

Wolf, David, and Thomas Kemp. 2021. "Convergent Validity of Satellite and Secchi Disk Measures of Water Clarity in Hedonic Models." *Land Economics* 97(1):39–58.

Wood, S. A., and D. R. Dietrich. 2011. "Quantitative Assessment of Aerosolized Cyanobacterial Toxins at Two New Zealand Lakes." *Journal of Environmental Monitoring* 13 (6): 1617–1624.

Wu, Jianyong, Elizabeth D. Hilborn, Blake A. Schaeffer, Erin Urquhart, Megan M. Coffer, Cynthia J. Lin, and Andrey I. Egorov. 2021. "Acute Health Effects Associated with Satellite-Determined Cyanobacterial Blooms in a Drinking Water Source in Massachusetts." *Environmental Health* 20 (1): 83.

Yi, DongGyu, and Joseph A. Herriges. 2017. "Convergent Validity and the Time Consistency of Preferences: Evidence from the Iowa Lakes Recreation Demand Project." *Land Economics* 93 (2): 269–291.

Yuan, Lester L., and Amina I. Pollard. 2015. "Deriving Nutrient Targets to Prevent Excessive Cyanobacterial Densities in U.S. Lakes and Reservoirs." *Freshwater Biology* 60 (9): 1901–1916.

Zhang, Jiarui, Daniel J. Phaneuf, and Blake A. Schaeffer. 2022. "Property Values and Cyanobacterial Algal Blooms: Evidence from Satellite Monitoring of Inland Lakes." *Ecological Economics* 199:107481.

Zhang, Shiyu, Xingde Du, Haohao Liu, Michael D. Losiewic, Xinghai Chen, Ya Ma, Rui Wang, Zhihui Tian, Linjia Shi, Hongxiang Guo, and Huizhen Zhang. 2021. "The Latest Advances in the Reproductive Toxicity of Microcystin-LR." *Environmental Research* 192:110254.

Zhang, Wendong, Xibo Wan, Wenran Fan, and Yongjie Ji. 2024. "Uncovering Disparities in Water-based Outdoor Recreation using Cell Phone Mobility Data." *Environmental Research Letters* 19 (11): 114057.

Zhao, Sujuan, Shengzheng Zhong, Fang Wang, Honghui Wang, Dexiang Xu, and Guangyu Li. 2020. "Microcystin-LR Exposure Decreased the Fetal Weight of Mice by Distur-

bance of Placental Development and ROS-Mediated Endoplasmic Reticulum Stress in the Placenta." *Environmental Pollution* 256:113362.